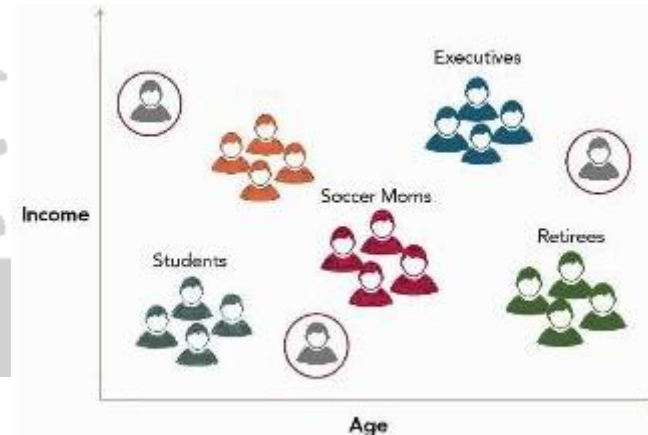




COLLABORATIVE FILTERING CLUSTERING

Recommendation Engines

Customer Segmentation

Reminder

- Fuqua CheckIn
- Deliverable 2 on the final project: this weekend
 - Do not try to increase – data set size; OSR2
- Elevator pitch in Class 10 (Monday Oct 5):
 - What problem are you studying?
 - Why you study this problem (why this problem matters to **you**)?
 - One thing you learned so far (you did not expect this before working on the project)
 - No need to tell “the whole story” – that’s for the last class
 - No slides, just a speech of up to 2 minutes from each team

Recap of where we are

Class 1: Network analytics

- Descriptive analytics about network

Classes 2-3: Regression

- Linear and logistic regression for predictive analytics

Classes 4-5: Tree-based methods

- CART for regression and classification, random forest and ensemble methods

Class 6: Model selection

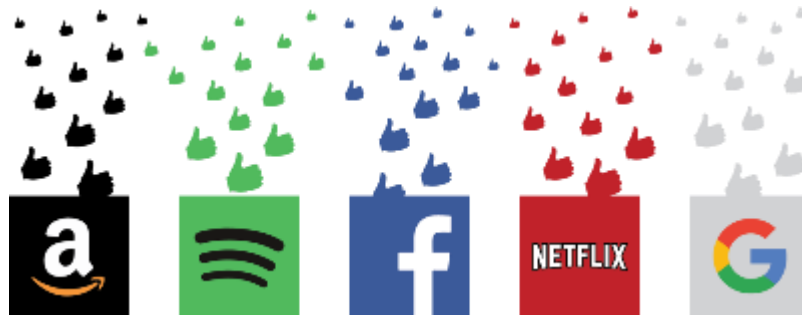
- What features to include in the predictive analytics model?

Class 7 (today)

- Beyond tabular data with one column of dependent variable

Looking forward

- Neural networks
- Optimization



RECOMMENDER SYSTEMS AND COLLABORATIVE FILTERING

Personalized Predictions from Collective Behavior

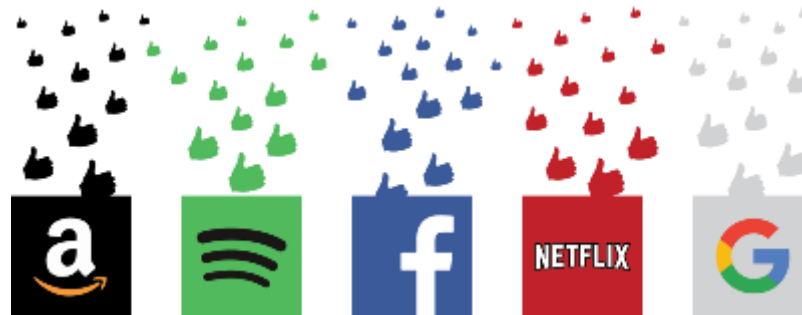
Agenda

- Motivating recommender systems
 - Concrete example based on the Netflix Prize
 - Traditional approach: content-based features
- Collaborative Filtering
 - Neighborhood-based Approach
 - Model-based approach using user archetypes
- Ensemble Learning
 - Combine collaborative filtering + content features

Recommender Systems

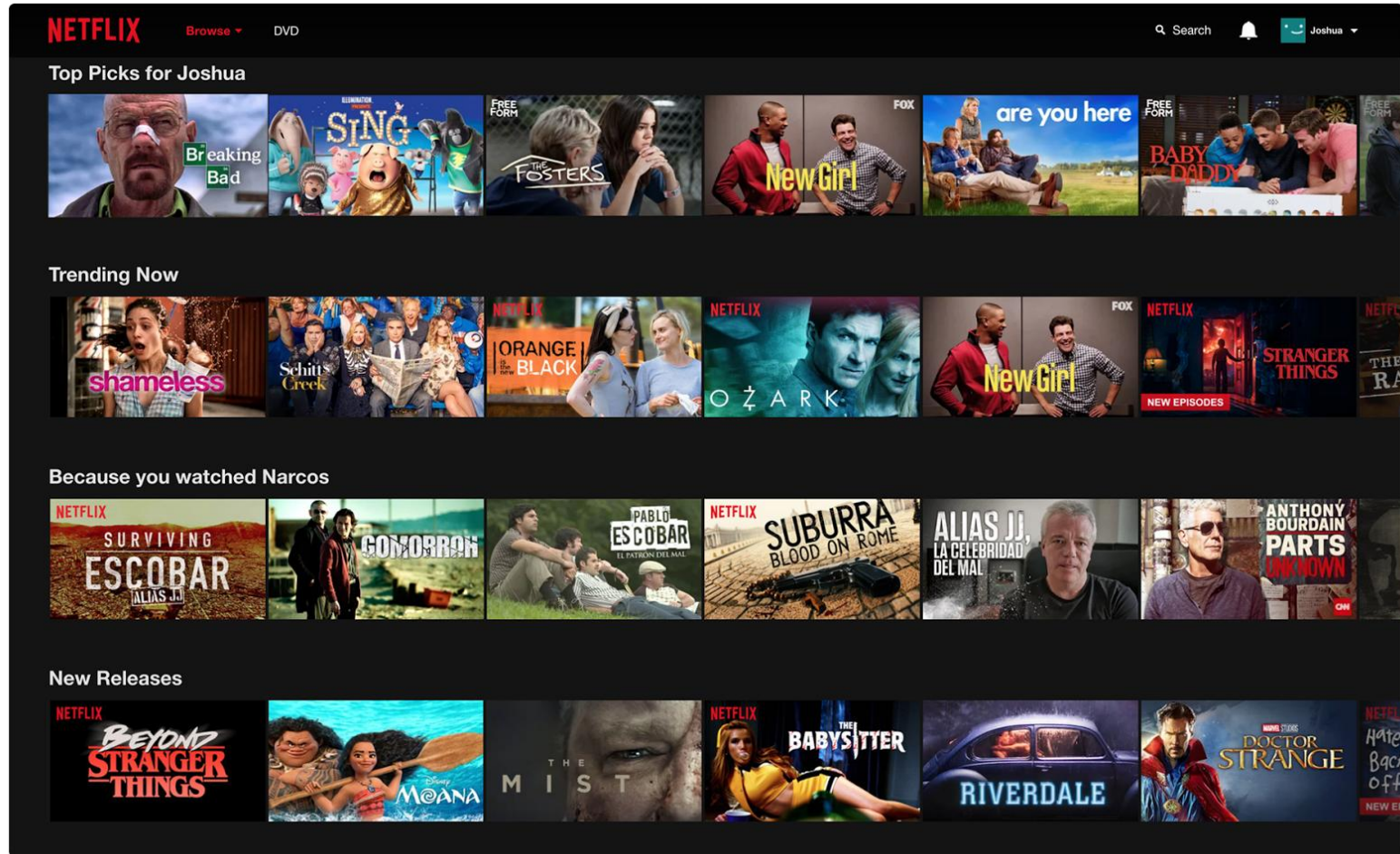
The Netflix Prize

What is a recommender system?



- A recommender system is an algorithm that predicts what a user might like based on their past preferences and interactions.
 - **Entertainment.** Movie (Netflix), Video (YouTube, TikTok), Music (Spotify)
 - **Ecommerce.** Amazon, eBay and others
 - **Social Media.** Facebook news feed, friend suggestions
 - **Online Dating.** Dating apps like eHarmony
- **Main Challenge:** Predict how much a user will like an item
 - Recommend items the user is most likely to engage with.

Netflix Recommendations

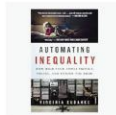


Amazon Recommendations

amazon.com/Black-Box-Society-Algorithms-Information/dp/0674970845



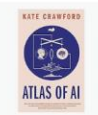
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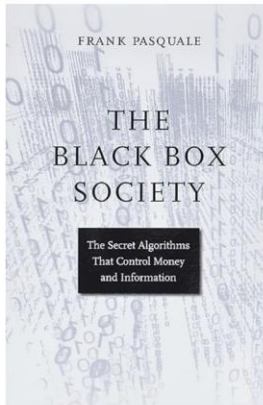
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The Black Box Society: The Secret Algorithms That Control Money and Information Reprint Edition

by Frank Pasquale (Author)

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Every day, corporations are connecting the dots about our personal behavior—silently scrutinizing clues left behind by our work habits and Internet use. The data compiled and portraits created are incredibly detailed, to the point of being invasive. But who connects the dots about what firms are doing with this information? *The Black Box Society* argues that we all need to be able to do so—and to set limits on how big data affects our lives.

Hidden algorithms can make (or ruin) reputations, decide the destiny of entrepreneurs, or even devastate an entire economy. Shrouded in secrecy and complexity, decisions at major Silicon Valley and Wall Street firms were long assumed to be neutral and technical. But leaks, whistleblowers, and legal disputes have shed new light on automated judgment. Self-serving and reckless behavior is surprisingly common, and easy to hide in code protected by legal and real secrecy. Even after billions of dollars of fines have been levied, underfunded regulators may have only scratched the surface of this troubling behavior.

Frank Pasquale exposes how powerful interests abuse secrecy for profit and explains ways to rein them in. Demanding
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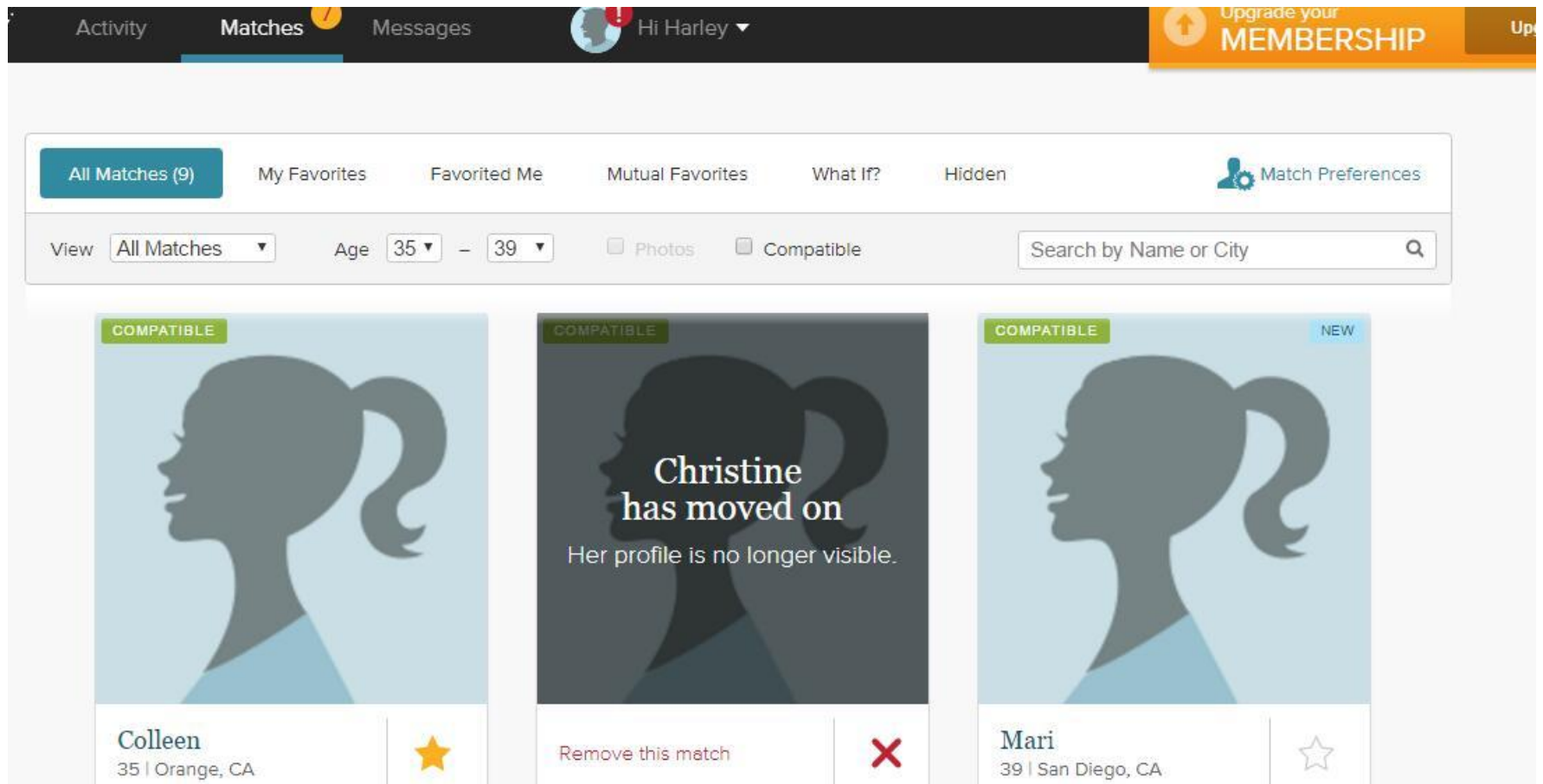
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eHarmony Recommendations



The Netflix Prize



- In October 2006, Netflix ran a contest inviting anyone to develop a recommendation method that could beat Netflix's own algorithm called Cinematch
 - This competition took three years
 - The winning team won Cinematch by over 10%
- Cinematch predicts user ratings of unseen movies (rating 1-5).
- \$1,000,000 prize to the team that could best beat Cinematch.
 - Catalyzed research innovation in recommender systems
 - Fostered collaboration between academic researchers and industry practitioners
 - Crowdsourcing innovation with over 44,000 submissions in 180 countries

Netflix Prize Dataset

- The Netflix dataset from 2006:
 - 480,189 users
 - 17,700 movies
 - 8,499,345,300 overall user-movie pairings possible
- Netflix dataset is very “sparse”:
 - On average, each user rated 209 of the 17,700 movies
 - Only 100,480,507 user ratings of movies (1.2%)
 - Need to estimate the remaining 8,398,864,793 user-movie ratings (98.8%)
- Netflix can also utilize some “metadata”:
 - movie metadata: actors, director, genre, year released, etc., and
 - user metadata: age, gender, income, etc.

MovieLens Dataset

- Netflix dataset is no longer available—due to a privacy-based lawsuit from users
- We will work instead with the MovieLens dataset
 - 6,040 users
 - 3,900 movies
 - 1,000,209 ratings of movies
- On average, each user rated 166 of the 3,900 movies (4.2%)
- The goal is to predict ratings of the other 95.8% user-movie pairings, about 22,555,791 “missing” user ratings

MovieLens Data

n : # of observations ($n = 1,000,209$)

	rating	userID	movieID	date
0	5.0	1	1193	2001-01-01 06:12:40
1	3.0	1	661	2001-01-01 06:35:09
2	3.0	1	914	2001-01-01 06:32:48
3	4.0	1	3408	2001-01-01 06:04:35
4	5.0	1	2355	2001-01-07 07:38:11
...	...	...	...	...
1000204	1.0	6040	1091	2000-04-26 10:35:41
1000205	5.0	6040	1094	2000-04-26 07:21:27
1000206	5.0	6040	562	2000-04-26 07:19:06
1000207	4.0	6040	1096	2000-04-26 10:20:48
1000208	4.0	6040	1097	2000-04-26 10:19:29

Tabular Form

User ID	Movie ID				
	1	2	3	...	3900
1	5				4
2		3	1		
3	1				2
...					
6040			3		5

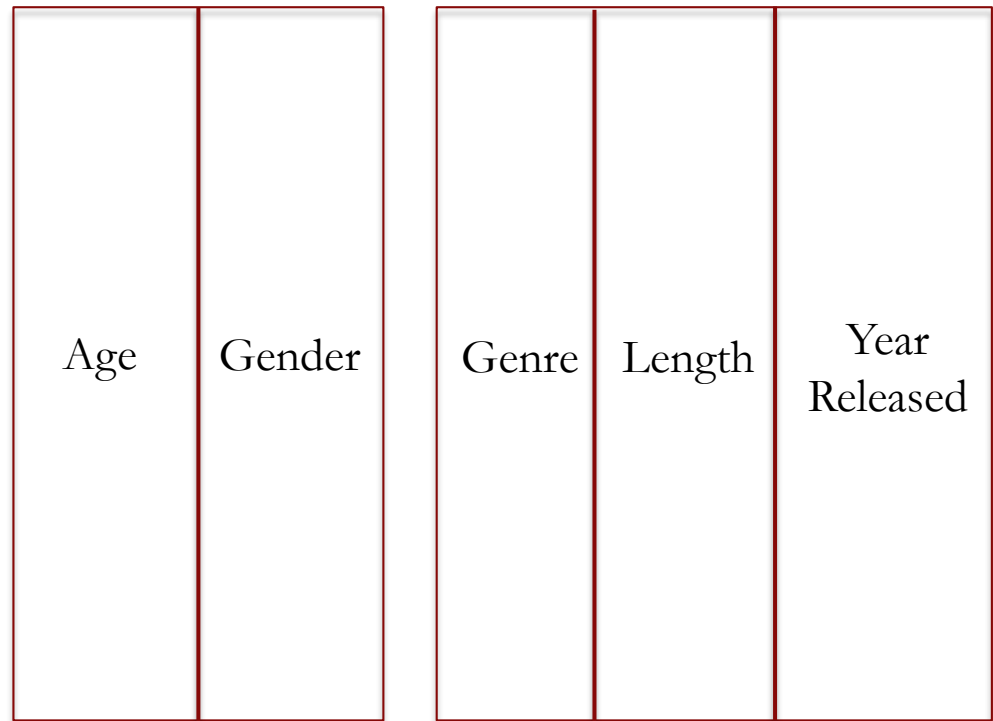
Matrix Form

Our goal is to fill in the blanks

Traditional Approach: Content-Based Recommenders

- Gather features of users and characteristics movies.
 - Learn what types of users like what types of movies?

	rating	userID	movieID
0	5.0	1	1193
1	3.0	1	661
2	3.0	1	914
3	4.0	1	3408
4	5.0	1	2355
...	...	...	...
1000204	1.0	6040	1091
1000205	5.0	6040	1094
1000206	5.0	6040	562
1000207	4.0	6040	1096
1000208	4.0	6040	1097
1000209 rows × 4 columns			



User Features

Movie Features

Limitations

- Rely heavily on manually curated user features and movie characteristics.
 - Users have varied preferences that cannot be “written down”.
 - Movies have features that also are not easily characterized.
- Missing Piece: Leveraging similar users (collaboration)
 - Users who have similar rating history are likely to be similar in preferences.
 - Our algorithm should learn from similar users, without explicitly knowing “why” users are similar.
- Let’s develop some intuition for this collaborative part.

Some Intuition

To avoid confusion between user ID's and movie ID's, I'll use names for the users and letters (A-Z) for the movies.

User	Movie																										
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	...
John	5		3		1		2		4																		...
Isabel	5		3		1		2		?																		...
Charles											2		1		2		1		1								...
Paul											3		3		2		3		?								...
Rahul											4		5		2		5		3								...
Jose																					1	5	2	2	4		...
Caroline																					5	1	4	4	?		...
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

- What might we guess for the entries with “?”

Some Intuition, continued

User	Movie																											
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	...	
John																	1		5									...
Isabel																	4		2									...
Charles											2		2				1		5									...
Paul											3		3				3		?									...
Rahul											4		?				5		1									...
Jose											1		1				2		4									...
Caroline											5		5															...
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

- What might we guess for the entries with “?”

Where's the structure?

User	Movie																											
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	...	
John				5					1				3			4												...
Isabel														4				4						1		2		...
Charles			3				2				2				5													...
Paul					1				1										4		5							...
Rahul												2												4				...
Jose		5				3											2			2							1	...
Caroline								4							1							4						...
Stephanie	3				2					2	2							3							5			...
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

The previous slides were artificial of course: there were visibly obvious ways to impute the missing ratings using easily identifiable users or movies.

How do we find such structure more generally?

Collaborative Filtering

- Neighborhood-Based Approach
- Model-Based Approach (focus)

Collaborative Filtering

- **Collaborative filtering** approaches leverage only the set of user-movie ratings to make predictions for missing entries.
 - No content features are used, so no domain knowledge is needed.
 - Content features can be added to augment collaborative filtering (more on this later)

User	Movie				
	A	B	C	D	...
Alex	5			4	...
Barb		3	1		...
Cassie	1			2	...
Dana			3	5	...
...	...	...	...	...	...



User	Movie				
	A	B	C	D	...
Alex	5	4.2	2.5	4	...
Barb	1.4	3	1	1.9	...
Cassie	1	3.5	2.2	2	...
Dana	4.4	2.4	3	5	...
...	...	...	...	...	...

Collaborative Filtering Approaches

- **Core Principle:** Similar users should like similar movies.
 - Leverages collective behavior of the entire community of users.
- Approach 1: Neighborhood-Based
 - Use past ratings to define “similar” users. Predict a weighted average based on these similar users.
- Approach 2: Model-Based
 - Define user archetypes to capture “types” of (similar) users.
 - Model each user’ as a weighted combination of user archetypes
- This lecture will focus on the model-based approach, which is more widely used in practice. But let’s see how the neighborhood-based approach works. It is quite simple.

Neighborhood-Based Approach

- Suppose we are trying to predict Alex's rating for movie B.
- Using the observed ratings, calculate the correlation between Alex's ratings and each of the other users.
 - $\text{Corr}(\text{Alex}, \text{Barb}) = 1$
 - $\text{Corr}(\text{Alex}, \text{Cassie}) = 0.72$
 - $\text{Corr}(\text{Alex}, \text{Dana}) = 0$
- Predict a weighted average.
 - $(1 * 2 + 0.72 * 4 + 0 * 3) / 1.72 = 2.83$

User	Movie				
	A	B	C	D	E
Alex	5		1	3	5
Barb	5	2	1		
Cassie		4	2	1	5
Dana	3	3	3		

Relying on a lot of users with overlapping movie reviews

Model-Based Collaborative Filtering

Archetypes and user compositions

Core Concept: Archetype

- **Concept:** Archetype, a typical profile or behavior pattern.
 - Archetype 1: enjoys action, adventure, physical drama
 - Archetype 2: enjoys exploration of relationships, interpersonal drama
 - Archetype 3: emotional drama – suspense, fear, mystery, intrigue
 - ...
 - Archetype 9: enjoys chronicles of facts, events, presentation
- **Assumption:** Each person's movie taste is some weighted combination of the preferences of the archetypal user.
- Archetypes are like “clusters”, except that each user is a mixture of many archetypes instead of being assigned to one archetype.

Toy Example

Movie		Rating						
	The Dark Knight Rises	PG-13	Alex Barb Cassie Dana					
	Harry Potter and the Sorcerer's Stone	PG		Harry Potter	The Triplets of Belleville	Shrek	The Dark Knight Rises	Memento
	Shrek	PG						
	The Triplets of Belleville	PG-13						
	Memento	R						

Archetypes and Compositions



Archetype 1
“Child at Heart”



Archetype 2
“Artist”



Harry Potter



The Triplets of
Belleville



Shrek



The Dark
Knight Rises



Memento

3	5	5	1	1
1	5	1	1	5

Archetype
Ratings

-0.5	-1
+0.5	+1
+1	0
-1	+1

User Composition

Alex



Barb



Cassie



Dana



4		4	5	
	5	2	1	4
3	5	5		1
2			4	5

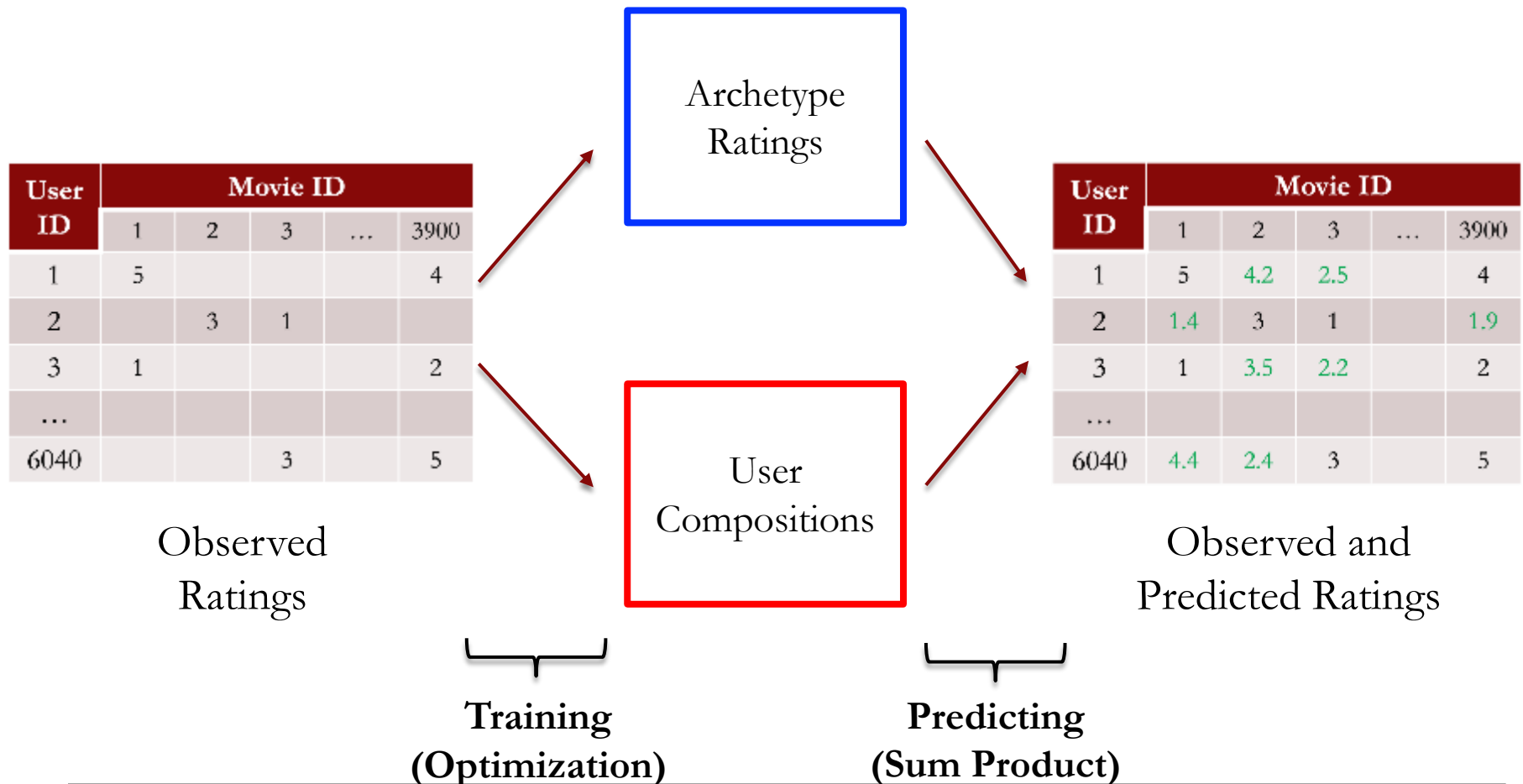
Toy Example Explanation

- 4 users and 5 movies, with 14 observed ratings and 6 missing ratings.
- Two archetypes: Child at Heart and Artist.
 - The kid at heart user likes movies that are animated or feature children
 - The artist user likes movies which are artsy and less mainstream (not Hollywood blockbusters)
- Each user is a composition of these two archetypes.
 - +1 = fits this archetype
 - - 1 = fits the opposite of this archetype
- Predictions can be made for the missing entries based on the **user compositions** and **archetype ratings**. (Details to come...)
- The **archetype ratings** and **user composition** are both discovered through the model training process. (Details to come...)

Modeling Approach

1. **Parameter.** Choose a number k representing the number of archetypes.
2. **Training.** Using the observed user-movie ratings in the matrix, solve an optimization problem to discover archetypes' ratings and user compositions.
3. **Predicting.** Compute the missing entries of the user-movie ratings matrix using sum-products.

Training and Predicting



Predicting Ratings

Sum-Product of archetype ratings and user compositions.

Archetype Users and Their Ratings

Archetype	A. Hillbilly Elegy	B. The Dig	C. Lost Girls	D. Narcos	E. The Midnight Sky	F. I Care a Lot	G. Rebecca	H. Moxie	I. The King	...
1	3	5	2	5	5	3	3	4	1	...
2	5	4	5	2	3	1	5	4	5	...
...	...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...
9	2	5	3	5	4	5	1	4	3	...

Each archetype has a rating for every movie.

- Index archetypes by $a = 1, \dots, 9$ and movies by $m = 1, \dots, 3900$
- $S_{a,m}$: supposed rating of movie m by archetype a
 - For example, in the above table: $S_{2,Narcos} = 2$ and $S_{9,Narcos} = 5$
- The blue numbers in the table are obtained through the model training process (details later)

User as a Composition of Archetypes

Each user is characterized by 9 weights across archetypes

User	Archetype								
	1	2	3	4	5	6	7	8	9
John	0.7	0.1	...	...	...	...	...	...	0.2
Isabel	0.8	1.6	...	...	...	...	...	...	-0.3
Charles	0.2	-0.6	...	...	...	...	...	...	1.1
Paul	...	...	...	...	...	...	...	...	...
Rahul	...	...	...	...	...	...	...	...	...
Jose	0.2	0.3	0.1	0.9	-0.3	-0.4	0.8	-0.8	0.7
...	...	...	...	...	...	...	...	...	...

$W_{u,a}$: the weight of user u for archetype a

For example, in the above table: $W_{Jose,2} = 0.3$ and $W_{Jose,9} = 0.7$

Jose's predicted rating of movie Q

User	Archetype								
	1	2	3	4	5	6	7	8	9
John	...	...	...	...	...	...	...	...	...
Isabel									
Charles									
Paul									
Rahul	...	...	...	...	...	...	...	...	...
Jose	.2	.3	.1	.9	-.3	-.4	.8	-.8	.7
Caroline	...	...	...	...	...	...	...	...	...
Stephanie									
...	...	...	...	...	...	...	...	...	...

Archetype	Movie																											
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	...	
1																...	3	...										...
2																...	2	...										...
3																...	4	...										...
4																...	5	...										...
5																...	3	...										...
6																...	2	...										...
7																...	1	...										...
8																...	4	...										...
9	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	3	...	...	...	...	...	...	...	...	...	...	...

$$R_{\text{Jose}, Q} = .2 \times 3 + .3 \times 2 + .1 \times 4 + .9 \times 5 - .3 \times 3 - .4 \times 2 + .8 \times 1 - .8 \times 4 + .7 \times 3 = 4.1$$

Archetype Ratings of Movies

Let us also suppose the archetype ratings of movies are to be solved by the model:

Archetype	Movie																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	...
1	S_{1A}	...	...	$S_{1,D}$	...	...	...	...	$S_{1,I}$	...	...	...	...	...	...	...	$S_{1,Q}$	...	...
2	$S_{2,A}$	...	...	$S_{2,D}$	...	...	...	...	$S_{2,I}$	...	...	...	...	...	...	...	$S_{2,Q}$	...	...
3	$S_{3,A}$	...	...	$S_{3,D}$	...	...	...	...	$S_{3,I}$	...	...	...	...	...	...	...	$S_{3,Q}$	...	...
4	$S_{4,A}$	...	...	$S_{4,D}$	...	...	...	...	$S_{4,I}$	...	...	...	...	...	...	...	$S_{4,Q}$	...	...
5	$S_{5,A}$	...	...	$S_{5,D}$	...	...	...	...	$S_{5,I}$	...	...	...	...	...	...	...	$S_{5,Q}$	...	...
6	$S_{6,A}$	...	...	$S_{6,D}$	...	...	...	...	$S_{6,I}$	...	...	...	...	...	...	...	$S_{6,Q}$	...	...
7	$S_{7,A}$	...	...	$S_{7,D}$	...	...	...	...	$S_{7,I}$	...	...	...	...	...	...	...	$S_{7,Q}$	...	...
8	$S_{8,A}$	...	...	$S_{8,D}$	...	...	...	...	$S_{8,I}$	...	...	...	...	...	...	...	$S_{8,Q}$	...	...
9	S_{9A}	...	...	$S_{9,D}$	...	...	...	...	$S_{9,I}$	...	...	...	...	...	...	...	$S_{9,Q}$	...	...

User Composition Weights

User	Archetype								
	1	2	3	4	5	6	7	8	9
John	$w_{John,1}$	$w_{John,2}$	$w_{John,3}$	$w_{John,4}$	$w_{John,5}$	$w_{John,6}$	$w_{John,7}$	$w_{John,8}$	$w_{John,9}$
Isabel	...	...	...	...	...	...	...	...	...
Charles	...	...	...	...	...	...	...	...	...
Paul	$w_{Paul,1}$	$w_{Paul,2}$	$w_{Paul,3}$	$w_{Paul,4}$	$w_{Paul,5}$	$w_{Paul,6}$	$w_{Paul,7}$	$w_{Paul,8}$	$w_{Paul,9}$
Rahul	...	...	...	...	...	...	...	...	...
Jose	$w_{Jose,1}$	$w_{Jose,2}$	$w_{Jose,3}$	$w_{Jose,4}$	$w_{Jose,5}$	$w_{Jose,6}$	$w_{Jose,7}$	$w_{Jose,8}$	$w_{Jose,9}$
Caroline	...	...	...	...	...	...	...	...	...
Stephanie									
...	...	...	...	...	...	...	...	...	...

Rating as a Sum-Product

User	Archetype								
	1	2	3	4	5	6	7	8	9
John	$w_{John,1}$	$w_{John,2}$	$w_{John,3}$	$w_{John,4}$	$w_{John,5}$	$w_{John,6}$	$w_{John,7}$	$w_{John,8}$	$w_{John,9}$
Isabel	...	...	...	...	...	...	...	...	...
Charles	...	...	...	...	...	...	...	...	...
Paul	$w_{Paul,1}$	$w_{Paul,2}$	$w_{Paul,3}$	$w_{Paul,4}$	$w_{Paul,5}$	$w_{Paul,6}$	$w_{Paul,7}$	$w_{Paul,8}$	$w_{Paul,9}$
Rahul	...	...	...	...	...	...	...	...	...
Jose	$w_{Jose,1}$	$w_{Jose,2}$	$w_{Jose,3}$	$w_{Jose,4}$	$w_{Jose,5}$	$w_{Jose,6}$	$w_{Jose,7}$	$w_{Jose,8}$	$w_{Jose,9}$
Caroline	...	...	...	...	...	...	...	...	...
Stephanie	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...

Archetype	Movie																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	...
1	$S_{1,A}$	...	...	$S_{1,D}$	...	...	...	...	$S_{1,I}$	...	...	...	...	...	...	...	$S_{1,Q}$	...	...
2	$S_{2,A}$	...	...	$S_{2,D}$	...	...	...	...	$S_{2,I}$	...	...	...	...	...	...	...	$S_{2,Q}$	...	...
3	$S_{3,A}$	...	...	$S_{3,D}$	...	...	...	...	$S_{3,I}$	...	...	...	...	...	...	...	$S_{3,Q}$	...	...
4	$S_{4,A}$	...	...	$S_{4,D}$	...	...	...	...	$S_{4,I}$	...	...	...	...	...	...	...	$S_{4,Q}$	...	...
5	$S_{5,A}$	...	...	$S_{5,D}$	...	...	...	...	$S_{5,I}$	...	...	...	...	...	...	...	$S_{5,Q}$	...	...
6	$S_{6,A}$	...	...	$S_{6,D}$	...	...	...	...	$S_{6,I}$	...	...	...	...	...	...	...	$S_{6,Q}$	...	...
7	$S_{7,A}$	...	...	$S_{7,D}$	...	...	...	...	$S_{7,I}$	...	...	...	...	...	...	...	$S_{7,Q}$	...	...
8	$S_{8,A}$	...	...	$S_{8,D}$	...	...	...	...	$S_{8,I}$	...	...	...	...	...	...	...	$S_{8,Q}$	...	...
9	$S_{9,A}$	...	...	$S_{9,D}$	...	...	...	...	$S_{9,I}$	...	...	...	...	...	...	...	$S_{9,Q}$	...	...

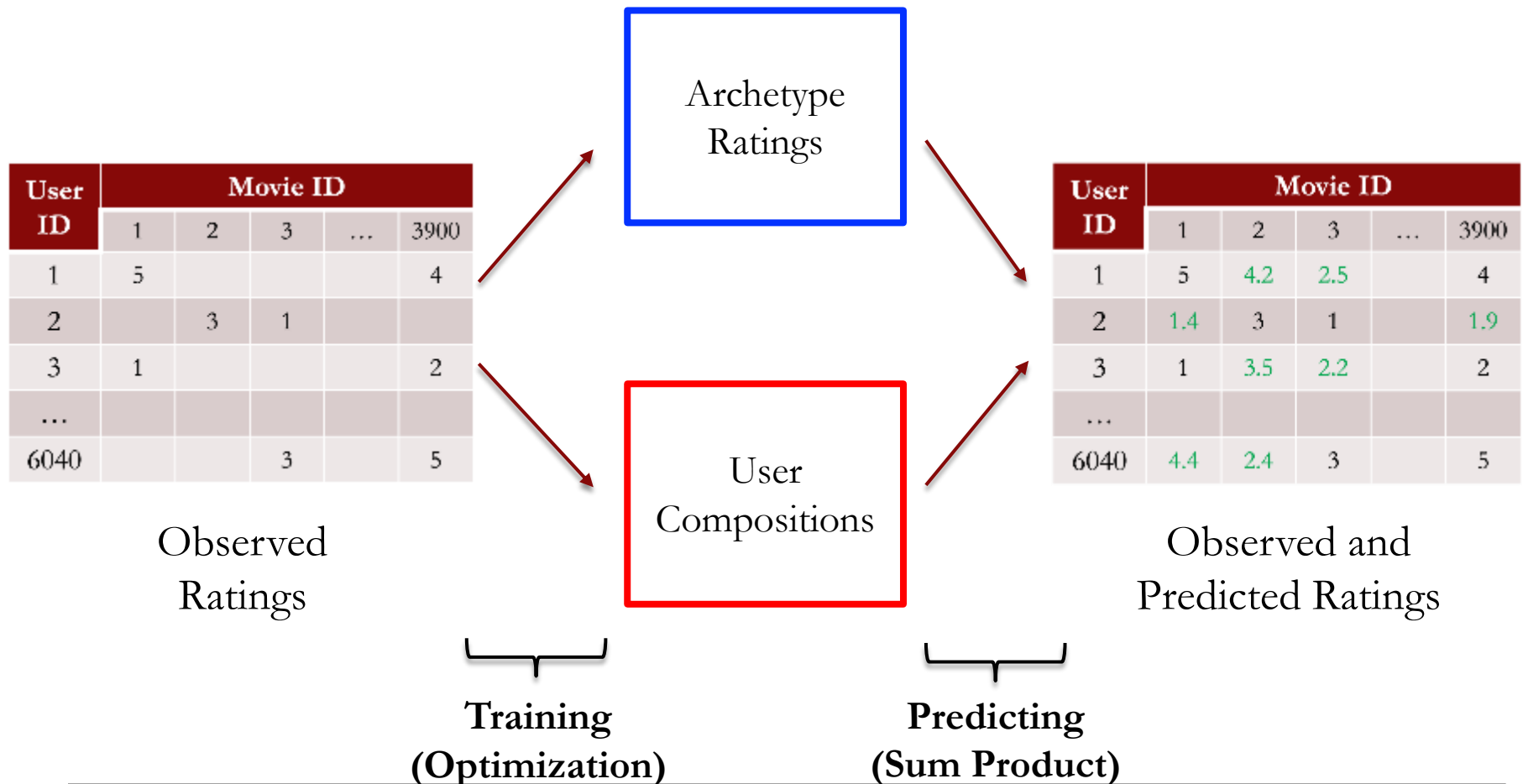
$$R_{User,Movie} = \sum_{a=1}^9 w_{User,a} \cdot s_{a,Movie}$$

For those more mathematically inclined:
R is the matrix product of w and S .

Training the Model

Determining the archetype ratings and user compositions.

Training and Predicting



Notation

User	Movie																											
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	...	
John				5					1				3			4												...
Isabel														4			4							1		2		...
Charles			3				2				2				5													...
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

- Index users by $u = 1, \dots, 6040$ and movies by $m = 1, \dots, 3900$
- *OBS*Pairs: collection of the **observed** user-movie pairs (u, m)
 - $(\text{John}, D), (\text{John}, I), (\text{John}, M), (\text{John}, P), (\text{Isabel}, N), \dots$
- $OBSR_{u,m}$: observed rating of the user-movie pair (u, m)
 - For example $OBSR_{\text{John}, D} = 5$

Least Squares Fitting

- Question: Which objects do we need to determine?
 - $S_{a,m}$: rating of movie m by archetype a
 - $w_{u,a}$: weight of archetype a in the composition of user u
- Let us compute these values to minimize the sum of squared residuals, namely the difference between observed and predicted user-movie ratings for all user-movie pairs in the observed ratings

$$\text{minimize}_{w_{u,a}, S_{a,m}} \sum_{\text{all observed user-movie pairs } (u,m)} (\text{OBSR}_{u,m} - (w_{u,1}S_{1,m} + w_{u,2}S_{2,m} + \cdots + w_{u,9}S_{9,m}))^2$$

OBSPairs

We sum over all observed movie ratings by users

User	Movie																											
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	...	
John				5					1				3			4											...	
Isabel														4				4						1		2		...
Charles			3				2				2				5												...	
Paul					1				1										4		5						...	
Rahul												2												4			...	
Jose		5				3											2			2						1	...	
Caroline								4							1							4					...	
Stephanie	3				2					2	2							3						5			...	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	

Residual for model's predicted value for Jose of movie Q is:

$$\text{Residual}_{\text{Jose},Q} = \text{OBSR}_{\text{Jose},Q} - (w_{\text{Jose},1} \cdot s_{1,Q} + w_{\text{Jose},2} \cdot s_{2,Q} + \cdots + w_{\text{Jose},9} \cdot s_{9,Q})$$

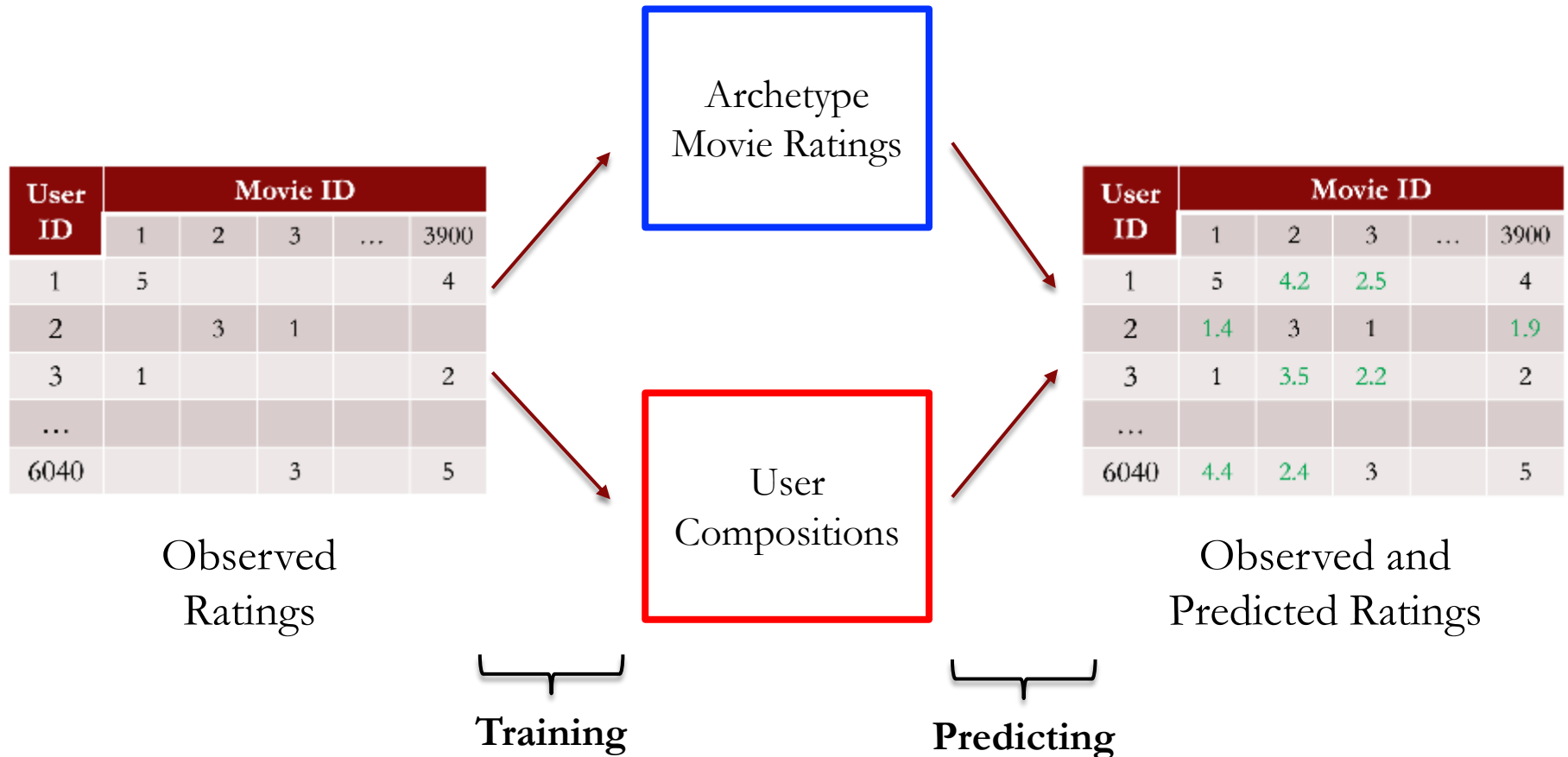
Fitting Procedure, continued

- Question: Which objects do we need to compute from the data?
 - $S_{a,m}$: supposed rating of movie m by archetype a
 - $w_{u,a}$: weight of archetype a in the linear combination of user u

$$\text{minimize} \sum_{\substack{\text{all observed} \\ \text{user-movie} \\ \text{pairs } (u, m)}} (\text{OBSR}_{u,m} - (w_{u,1}S_{1,m} + w_{u,2}S_{2,m} + \cdots + w_{u,9}S_{9,m}))^2$$

- This is a least-square problem—akin to linear regression—but:
 1. The scale is very large, with $9 \times 6,040 + 9 \times 3,900 = 89,460$ variables
 2. The model is no longer linear in the variables (it is “bilinear”)
 3. This is computationally challenging, but we do it anyway...

Model Overview



Model-Based Collaborative Filtering

- The model we just described is an example of “model-based collaborative filtering”, also referred to as matrix factorization.
 - Model user preferences as a linear combination of archetype preferences.
 - Calculate the “best” archetypes and weights to explain the observed data.
- This approach shares similarities with clustering in the sense that it produces archetypes that we hope to be able to “interpret” intuitively.
- Very active area of model research and development—and good R packages
 - **softImpute**, from Prof. Rahul Mazumder (MIT Sloan) and Prof. Trevor Hastie (Stanford Stats)
 - R package much better than Python package...

Performance for MovieLens

Training/Testing Set Split (99/1 Random Split)

Full Dataset ($n = 1,000,209$)

	rating	userID	movieID	date
0	5.0	1	1193	2001-01-01 06:12:40
1	3.0	1	661	2001-01-01 06:35:09
2	3.0	1	914	2001-01-01 06:32:48
3	4.0	1	3408	2001-01-01 06:04:35
4	5.0	1	2355	2001-01-07 07:38:11
...	...	...	...	...
1000204	1.0	6040	1091	2000-04-26 10:35:41
1000205	5.0	6040	1094	2000-04-26 07:21:27
1000206	5.0	6040	562	2000-04-26 07:19:06
1000207	4.0	6040	1096	2000-04-26 10:20:48
1000208	4.0	6040	1097	2000-04-26 10:19:29

99% Training Set ($n = 990,206$)

	rating	userID	movieID	date
0	5.0	1	1193	2001-01-01 06:12:40
1	3.0	1	661	2001-01-01 06:35:09
2	3.0	1	914	2001-01-01 06:32:48
3	4.0	1	3408	2001-01-01 06:04:35
4	5.0	1	2355	2001-01-07 07:38:11
...	...	...	...	...
1000204	1.0	6040	1091	2000-04-26 10:35:41
1000205	5.0	6040	1094	2000-04-26 07:21:27

1% Testing Set ($n = 10,003$)

	rating	userID	movieID	date
417	5.0	5	994	2000-12-31 06:35:40
...	...	...	...	...
999940	4.0	6040	2863	2000-05-07 16:26:35

Training/Testing Split

User	Movie																										
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	...
John				5					1				3			4											...
Isabel														4				4						1		2	...
Charles				3				2				2				5											...
Paul					1				1										4		5						...
Rahul												2													4		...
Jose			5			3											2			2						1	...
Caroline								4								1						4					...
Stephanie	3				2					2	2								3						5		...
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

 **Testing Set.** Hide from the model to evaluate out-of-sample performance.

Model Quality on the Test Set

We use the observed user-movie ratings to assess model performance—leveraging the usual metrics of fit

1. R^2 : how well the model fits the observed user-movie ratings
 - Defined just as in linear regression

2. MAE: Mean Absolute Error

$$\text{MAE} = \frac{1}{N} \left(\sum_{\text{all test pairs } (u,m)} |w_{u,1}S_{1,m} + w_{u,2}S_{2,m} + \cdots + w_{u,9}S_{9,m} - \text{OBSR}_{u,m}| \right)$$

3. RMSE: Root-Mean-Square Error

$$\text{RMSE} = \sqrt{\frac{1}{N} \left(\sum_{\text{all test pairs } (u,m)} (w_{u,1}S_{1,m} + w_{u,2}S_{2,m} + \cdots + w_{u,9}S_{9,m} - \text{OBSR}_{u,m})^2 \right)}$$

Results: MovieLens Ratings Prediction

- Results of our collaborative filtering model on the test set:
 - Fit of 33.5% measured by the OSR^2
 - Average error of 0.71 – 0.92 (on a ratings scale of 1 to 5), as measured by MAE and RMSE

Model	OSR^2	MAE	RMSE
Collaborative Filtering Model	0.333	0.708	0.916

- As a point of reference, Netflix's Cinematch algorithm RMSE was 0.952
 - If we were competing for the Netflix prize, these results would have achieved a 3.78% improvement in RMSE over Netflix's own method!

Results by movie genre

- For interpretability, we show 10 standard genres and here report the average rating per archetype-genre pair

Archetype	arch_1	-4.47	-3.96	-3.93	-2.66	-2.97	-3.77	-4.18	-5.00	-4.23	-4.50
	arch_2	-3.38	-1.54	-0.15	5.00	-0.97	2.69	0.91	0.57	0.97	-2.72
	arch_3	-2.99	-4.41	-1.80	0.42	-5.00	-0.21	-1.91	0.80	-0.49	-1.90
	arch_4	-2.14	-0.58	1.07	1.08	-3.84	1.49	-5.00	-2.89	2.56	-3.92
	arch_5	1.37	-4.41	-0.90	2.85	3.33	-0.21	-1.01	-0.92	-2.95	3.04
	arch_6	1.66	-3.82	-2.70	-0.02	-1.61	1.56	4.02	3.18	-0.48	-2.33
	arch_7	-4.98	-1.82	2.29	-0.47	3.40	0.11	-2.18	-2.67	-0.77	-4.88
	arch_8	-2.20	-2.86	-0.53	-2.06	4.86	0.66	-1.17	-0.33	1.58	-0.82
		Action	Children's	Comedy	Documentary	Horror	Drama	Western	War	Romance	Sci-Fi

- Interpretation:
 - Archetype 2 seems to prefer documentaries and dramas
 - Romance films are preferred by archetype 4 audience and disliked by archetype 1 audience
 - Etc.

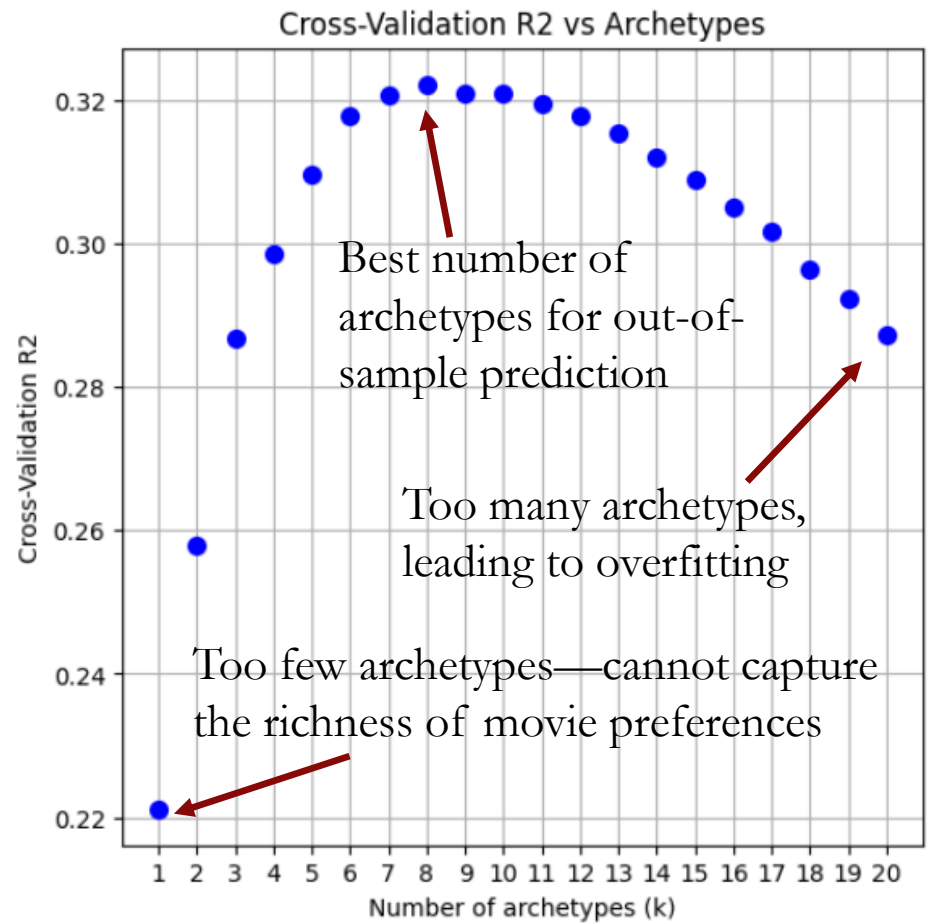
The Number of Archetypal Users

- How many “archetypal users” should we use in the collaborative filtering model?
 - If we have too few archetypal users, the predictions will be too rough—we will not capture important patterns in the data
 - If we have too many archetypal users, the model will overfit to the training set—resulting in poor out-of-sample performance
- We will use cross-validation to select the value of k that yields the best prediction on unseen data
 - For each value of k , perform 20-fold cross-validation*
 - Record the average R^2 using the current value of k . Call it **AvgR2**
 - Choose the value of k that yields the highest value of **AvgR2**

*Please don't confuse the k from k -fold cross-validation with the number of archetypes k

Cross-Validation Results

archetypes	r2	SSE	RMSE	MAE
1	0.221188	962259.383980	0.985787	0.769102
2	0.258028	916742.412182	0.962190	0.749781
3	0.286798	881195.620825	0.943351	0.733145
4	0.298647	866555.281157	0.935482	0.725845
5	0.309700	852899.127599	0.928081	0.718929
6	0.317792	842900.566532	0.922625	0.713809
7	0.320744	839253.395760	0.920627	0.711237
8	0.322188	837469.357909	0.919648	0.709575
9	0.320818	839162.611704	0.920577	0.709882
10	0.320919	839037.467280	0.920509	0.708778
11	0.319478	840817.818669	0.921485	0.709336
12	0.317839	842842.905355	0.922594	0.709407
13	0.315350	845918.011610	0.924275	0.710167
14	0.311990	850070.165638	0.926541	0.711284
15	0.308986	853780.670616	0.928561	0.712326
16	0.305091	858593.276475	0.931174	0.713848
17	0.301598	862909.166328	0.933512	0.715027
18	0.296486	869225.184118	0.936922	0.717081
19	0.292268	874437.678836	0.939727	0.718868
20	0.287245	880643.517395	0.943056	0.720992



Adding Content Features

Additional Data

- Our Collaborative Filtering model has leveraged data on user-movie ratings to “reconstruct” missing data
- But we have access to additional metadata on the movies
 - Genre (out of 18 genres): Action, Adventure, ..., Western
- And we also have access to additional data on the users
 - Gender
 - Age
 - Occupation (out of 21 categories): Administrator, Artist, ..., Writer
 - Zip code — hence estimates of income, rural/urban living, etc.
- Last, we also know the date and time of each movie rating

Ensemble Model

- Combine the collaborative filtering (CF) model into another method (linear regression, CART, random forests, etc.)
 - Use the CF output as an additional variable!
 - For instance, let $CF_{u,m}$ be the predicted rating of user-movie pair (u,m) using our CF model. Then consider the following regression model:
- This **ensemble model** combines different predictive methods

$$R_{u,m} = \beta_0 + \beta_1 \cdot CF_{u,m} + \text{linear model of movie data} + \text{linear model of user data}$$

Extended Training Data

k : # of independent variables ($k = 51$); n : # of observations ($n = 990,206$)

	rating	weekday	mon	year	hour	AgeRange	Male	MedianIncome	Urban	Job_academic	...	Film-Noir	Horror	Musical	Mystery	Romance	Sci-Fi	Thriller	War	Western	CF
0	5.0	1	1	2001	6	1	0	63015	1	0	...	0	0	0	0	0	0	0	0	0	4.520302
1	3.0	1	1	2001	6	1	0	63015	1	0	...	0	0	1	0	0	0	0	0	0	3.677090
2	3.0	1	1	2001	6	1	0	63015	1	0	...	0	0	1	0	1	0	0	0	0	4.498897
3	4.0	1	1	2001	6	1	0	63015	1	0	...	0	0	0	0	0	0	0	0	0	4.650397
4	5.0	0	1	2001	7	1	0	63015	1	0	...	0	0	0	0	0	0	0	0	0	4.165368
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
990201	1.0	3	4	2000	10	25	1	44031	1	0	...	0	0	0	0	0	0	0	0	0	1.000000
990202	5.0	3	4	2000	7	25	1	44031	1	0	...	0	0	0	0	1	0	0	1	0	3.960501
990203	5.0	3	4	2000	7	25	1	44031	1	0	...	0	0	0	0	0	0	0	0	0	3.440465
990204	4.0	3	4	2000	10	25	1	44031	1	0	...	0	0	0	0	0	0	0	0	0	3.981944
990205	4.0	3	4	2000	10	25	1	44031	1	0	...	0	0	0	0	0	1	0	0	0	3.406581

Dependent
variable

Date/time
data

User data

Movie data

CF prediction

Linear Regression: Results

Regression Results

R-squared: 0.4268

Adjusted R-squared: 0.4268

No. Observations: 990206

Coefficients:

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
const	0.59	0.03	17.56	0.00	0.52	0.65	***
AgeRange	0.00	0.00	16.08	0.00	0.00	0.00	***
Male	-0.02	0.00	-7.32	0.00	-0.02	-0.01	***
MedianIncome	0.00	0.00	0.03	0.98	-0.00	0.00	
Urban	0.01	0.00	3.61	0.00	0.01	0.02	***
Job_academic	-0.08	0.01	-11.09	0.00	-0.09	-0.06	***
Job_artist	-0.07	0.01	-9.95	0.00	-0.09	-0.06	***
Job_clerical/admin	-0.06	0.01	-7.96	0.00	-0.08	-0.05	***
Job_college/grad student	-0.06	0.01	-8.85	0.00	-0.07	-0.04	***
Job_customer service	-0.07	0.01	-8.19	0.00	-0.09	-0.05	***
Job_doctor/health care	-0.07	0.01	-8.49	0.00	-0.08	-0.05	***
Job_executive/managerial	-0.06	0.01	-9.39	0.00	-0.08	-0.05	***
Job_farmer	-0.10	0.02	-5.53	0.00	-0.13	-0.06	***
Job_homemaker	-0.06	0.01	-5.60	0.00	-0.08	-0.04	***
Job_lawyer	-0.06	0.01	-7.47	0.00	-0.08	-0.05	***
Job_other or not specified	-0.07	0.01	-10.85	0.00	-0.08	-0.06	***
Job_programmer	-0.05	0.01	-6.52	0.00	-0.06	-0.03	***
Job_retired	-0.02	0.01	-2.34	0.02	-0.04	-0.00	*
Job_sales/marketing	-0.06	0.01	-8.28	0.00	-0.07	-0.05	***
Job_scientist	-0.05	0.01	-5.80	0.00	-0.06	-0.03	***
Job_self-employed	-0.07	0.01	-9.83	0.00	-0.09	-0.06	***
Job_technician/engineer	-0.05	0.01	-6.85	0.00	-0.06	-0.03	***

- We knew we would do at least as well as CF on the training set
- What about out-of-sample performance?

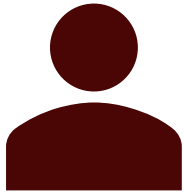
Updated Prediction Results

- The ensemble model enhances predictive performance, as compared to “just” the collaborative filtering model

Model	OSR ²	MAE	RMSE
Collaborative Filtering Model	0.335	0.708	0.916
Ensemble Model	0.359	0.705	0.900

- Recall that Netflix’s Cinematch reported an RMSE of 0.952
→ These results would have achieved a 5.46% RMSE improvement
- Ensemble models are often winners in machine learning contests
 - For instance, the winners of the Netflix Prize merged their teams and used their different models to create an ensemble model

An Important Practical Consideration – The Cold Start Problem



For a new user, what do we recommend?

If you have user metadata (e.g., demographics, device, location) data, you can make recommendations that are popular for the average user in that ‘metadata bucket’

You can recommend items popular across the entire user base and rapidly adapt based on how the user reacts to those recommendations



For a new item, how to proceed?

If you have item metadata (e.g, genre, director, lead actors for movies), you can assign the average rating of the ‘metadata bucket’

Identify ‘sibling items’ and ‘borrow’ the sibling’s ratings as a starting point

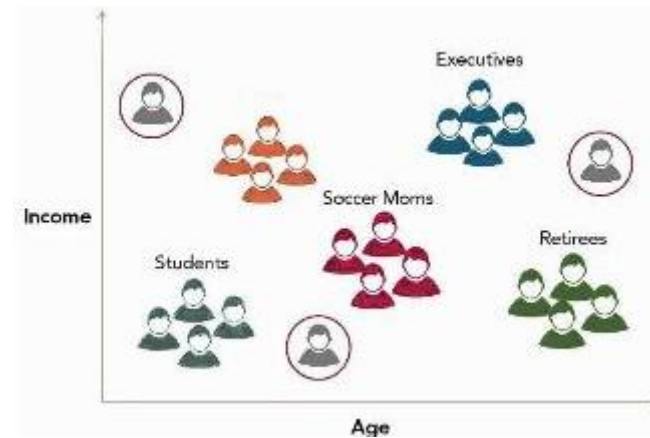
Recommender Systems

- Recommender systems are critical to many businesses: e-retail, brick-and-mortar retail, music listening services, mobile apps, etc.
- The challenge of building recommender systems is to predict user preferences based on very “sparse” data
- Recommender systems have enabled businesses to achieve critical progress in customer engagement and customer retention



Today's Modeling Techniques

- **Model-Based Collaborative Filtering:** reconstruction of unobserved user-item ratings from observed ratings, based on inter-user similarity
 - Concept of a small number of archetype users
 - Concept of each user as a linear combination of archetype users
 - Prediction of user ratings: weighted average of archetype users' ratings
- **Ensemble Learning:** combination of several prediction models
 - Previously: Random Forest is an ensemble of CART models
 - In this lecture: use of collaborative filtering (CF) predictions as an independent variable in a larger linear regression model
 - Of course, other predictive approaches can be combined into other types of ensemble models



CLUSTERING

COLLABORATIVE FILTERING

Customer Segmentation; Recommendation Engine

Humans are Good at Grouping

- Grouping similar “things” into buckets allows us derive insights about sets of diverse items.
 - Fuqua Courses → Area Subject
 - Movies/Music/Books → Genres
 - Food → Type of cuisine, meal (breakfast), or course (appetizer)
 - Living Things → Kingdom, Phylum..., Species
 - History → Eras (Renaissance, Industrial Revolution)
- But what if we don't know what the “types” should be?
- How can analytics help us define these groups?

Agenda

- Customer Segmentation in the Airline Industry
- Introduction to Clustering Methods
 - K-Means
 - Hierarchical clustering
 - **Optional:** Gaussian Mixture Models
- Unsupervised learning/descriptive analytics

Question

- Who knows what “**scree**” is?



Customer Segmentation

Customer Segmentation: Airline Industry

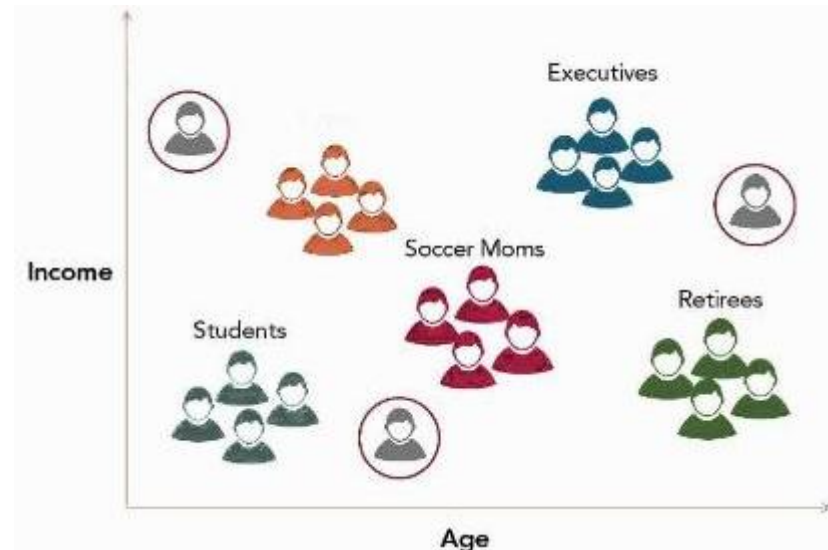
- Suppose you are an airline, and you want to better understand your passengers.
 - You can leverage the (massive) data from your frequent flyer program.
 - What are the different “types” of customers?
- 
- A photograph showing the interior of an airplane cabin. Passengers are seated in rows, looking forward. The cabin is filled with people, illustrating the large volume of data available for segmentation.
- Why might it be useful for an airline to know the types of customers?
 - Targeted promotions and marketing
 - Pricing
 - Customer experience and retention

Why Segmentation Matters?

- **Customer/User Segmentation:** subdivision of customers into a relatively small number of groups that share similar characteristics.
 - Moving away from “one size fits all” offerings for competitive edge.
- Divide the market into meaningful and measurable segments according to customers’ needs, past behaviors, and demographic profiles
- **Targeted Marketing:** design of tailored products/promotions/services to meet customer needs within each segment
- Moving from “one size fits all” to personalized marketing

Clustering

- Divide a dataset into groups (clusters) based on shared similarities.
- Part of exploratory data analysis (descriptive analytics).
- Derive insights that can lead to better business decisions.



Airline Customer Data

- Data on 3,999 customers obtained from the loyalty program of a former airline
- Six numerical values describing customers:
 - Number of miles eligible for award travel from historical activity
 - Number of miles earned from non-flight bonus transactions in the past 12 months
 - Number of non-flight bonus transactions in the past 12 months
 - Number of flight miles in the past 12 months
 - Number of flight transactions in the past 12 months
 - Tenure in the loyalty program (days)

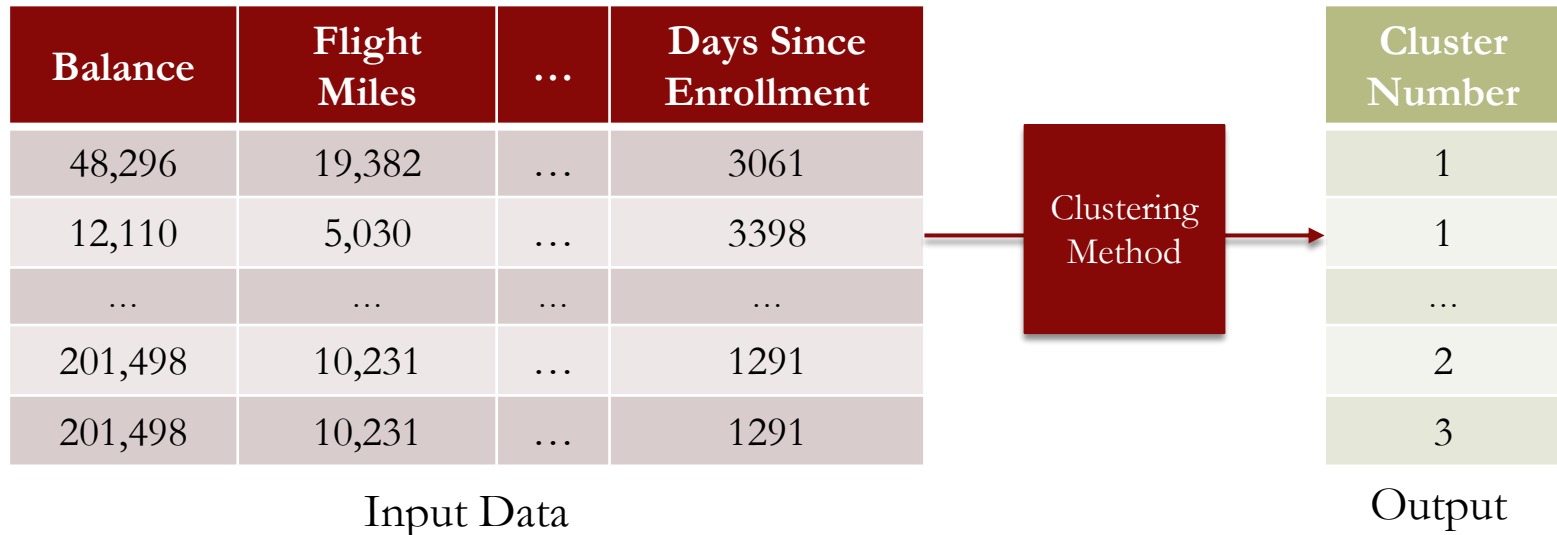
Airline Customer Data

n : # of observations ($n = 3,999$)

$n = 3,999$

	Balance	BonusMiles	BonusTrans	FlightMiles	FlightTrans	DaysSinceEnroll
0	48296	31329	9	500	1	3061
1	10021	0	0	0	0	7879
2	49280	22370	16	0	0	3312
3	213539	2750	15	0	0	4751
4	125465	14750	9	0	0	7206
5	7698	0	0	0	0	1734
6	201259	40755	34	0	0	3398
7	350608	50988	26	2643	5	3630
8	146232	83783	19	375	1	3566
9	2080	0	0	0	0	4635
...	...	...	...	...	...	...
3994	3016	0	0	0	0	1398
3995	28577	48564	14	0	0	3586
3996	276571	42044	23	0	0	7872
3997	28848	0	0	0	0	3069
3998	96522	61105	19	0	0	6924

Clustering Method (Algorithm)



- Clustering methods typically takes as input the number of clusters (often called k) and assigns each data point a number between 1 and k .
- **Goal:** High similarity (low dissimilarity) within each cluster

Unsupervised Learning

- Previously, we have seen datasets with a *dependent variable* that we are trying to predict as a function of some *independent variables*
 - These are called **supervised learning** problems
- In clustering, there is no dependent variable to predict; instead, we aim to create groups of “similar” samples/observations
 - These are called **unsupervised learning** problems
- No training/testing split for unsupervised learning
 - Without a dependent variable, out-of-sample performance is irrelevant
 - Unsupervised learning is used for summary / exploratory analysis
 - Unsupervised models are more subjective to interpret
 - Unsupervised models are also more difficult to validate

Distances and Standardization

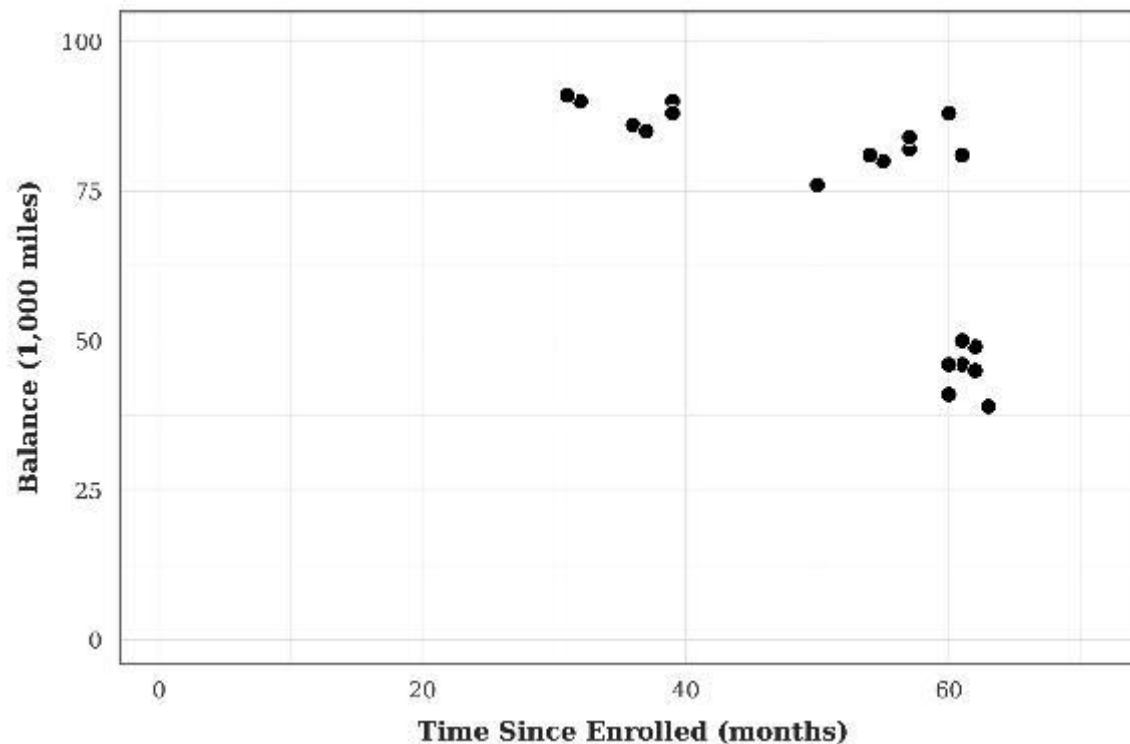
What does “similar” mean?

Mini-dataset for Airline Loyalty Program Customers

Member Number	Time Since Enrolled (Months)	Current Mileage Balance (1,000 miles)
1	60	41
2	61	46
3	62	49
4	63	39
5	62	45
6	61	50
7	60	46
8	50	76
9	57	82
10	60	88
11	55	80
12	54	81
13	57	84
14	61	81
15	32	90
16	36	86
17	39	90
18	31	91
19	37	85
20	39	88

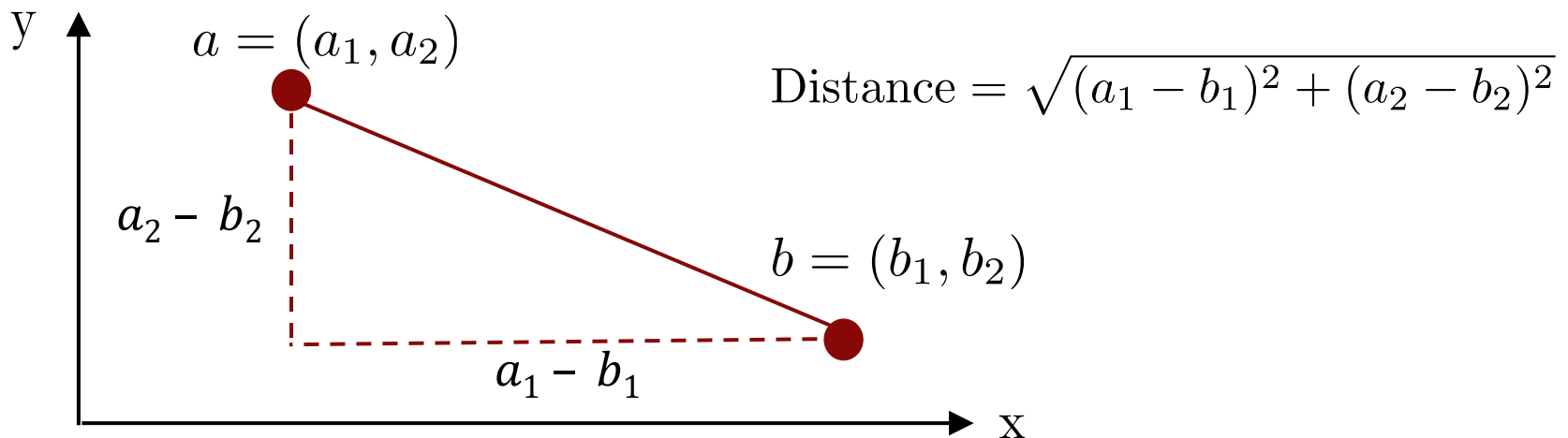
Mini-dataset

- For visualization purposes, let us consider a simple dataset with 20 customers and 2 characteristics



Measuring the distance between 2 observations

- Clustering relies on the *distance* between pairs of observations
 - Typically, Euclidean distance (recall the Pythagorean theorem!)

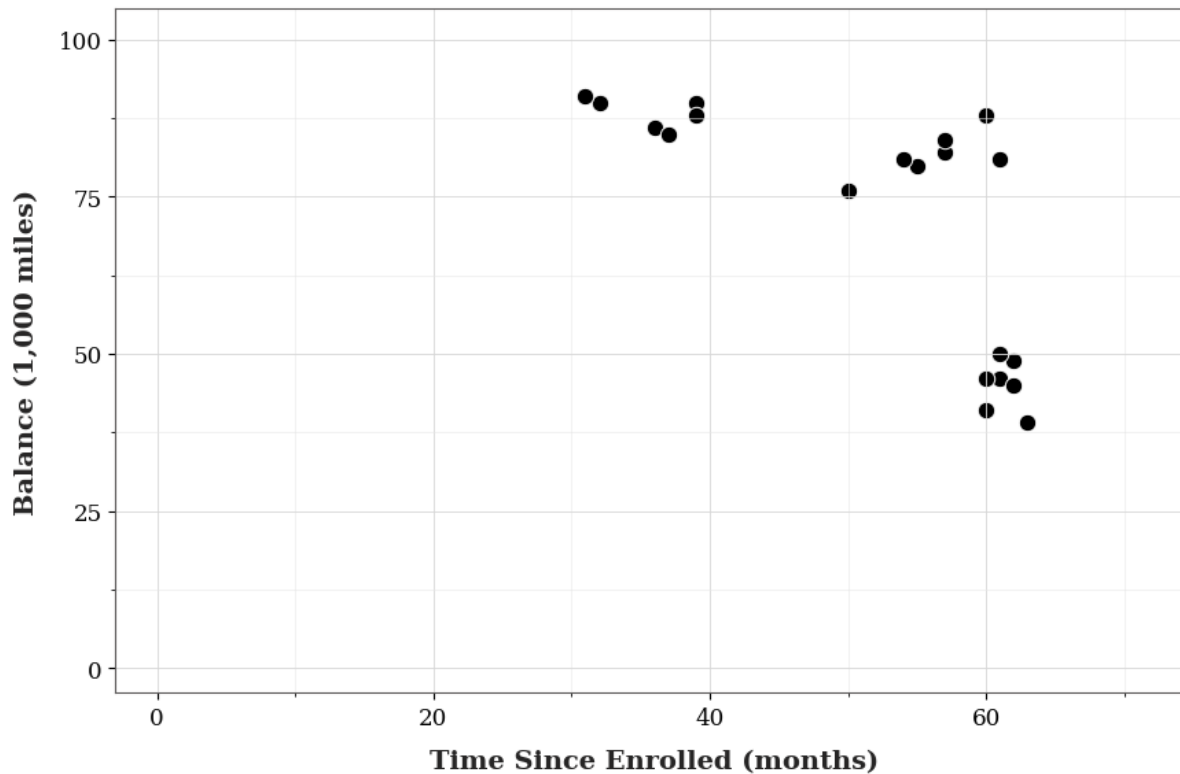


- Can be directly generalized to instances with more variables

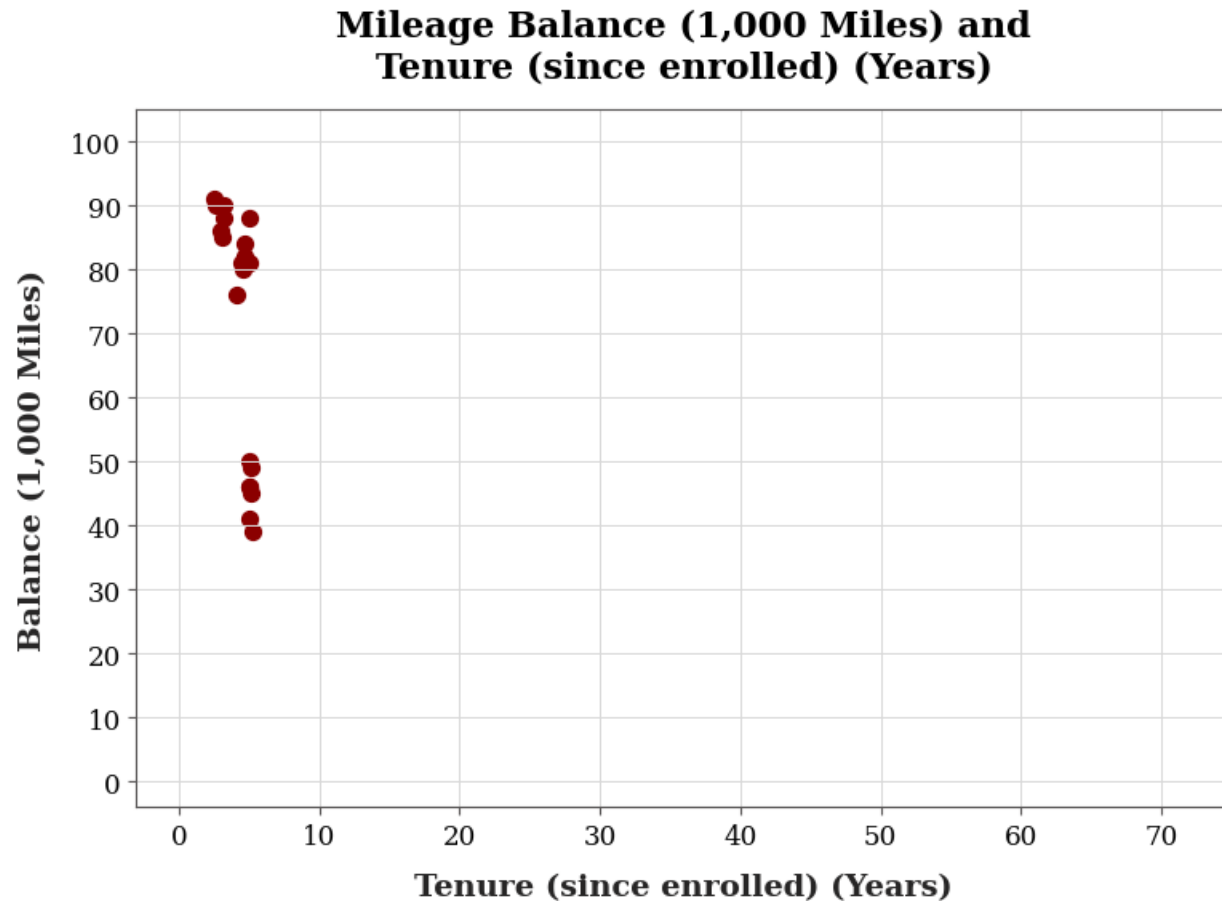
$$\text{Distance} = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \cdots + (a_6 - b_6)^2}$$

Mini-dataset

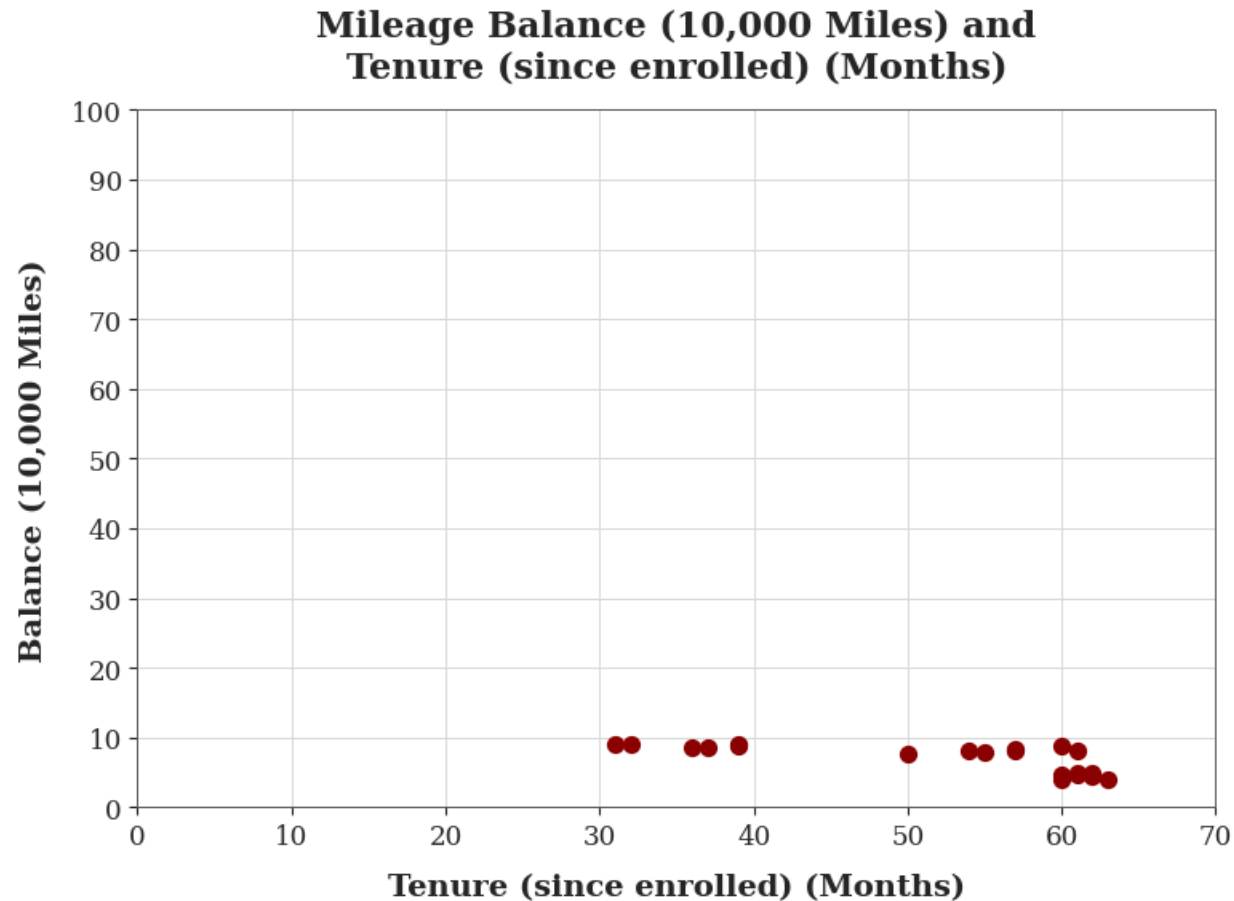
- How many clusters would you say there are?



Plot with Changed Units from Months to Years



Plot with Changed Units from 1,000 to 10,000 Miles

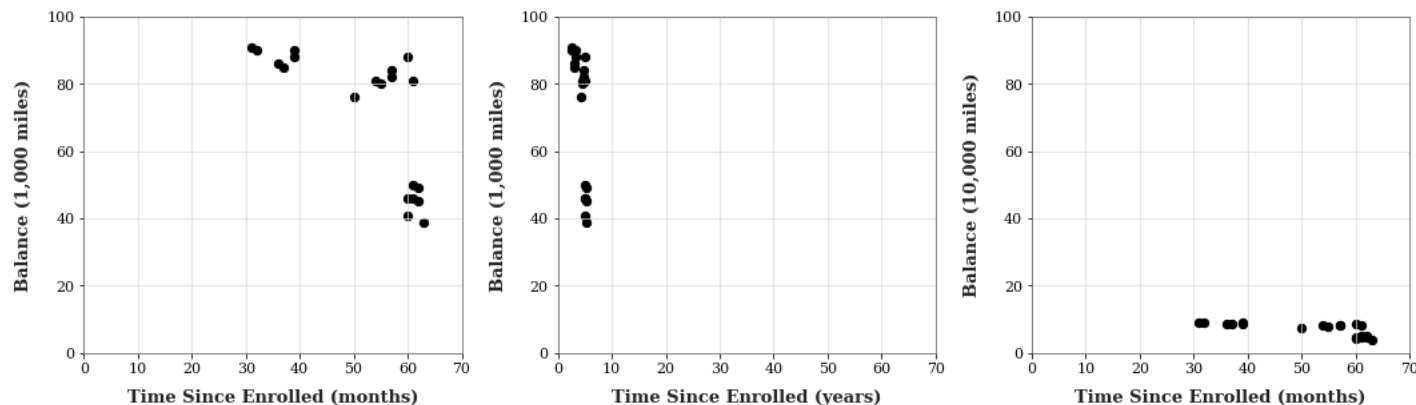


Scaling Issue

- Hard to compare different units (e.g., days vs. miles)
- Distance is primarily driven by the variables of larger scale

Variable	Balance	BonusMiles	BonusTrans	FlightMiles	FlightTrans	Tenure
Mean	73,601	17,145	12	460	1	4,118
Std. dev.	100,776	24,151	10	1,400	4	2,065

- The distance metric is sensitive to the choice of units



Standardizing the Data

Standardized variables are expressed in terms of “number of standard deviations from the mean”

$$\text{Standardized data} = \frac{\text{Data} - \text{Sample mean}}{\text{Sample standard deviation}}$$

Original Data

Member Number	Tenure (Months)	Balance (1,000 miles)
1	60	41
2	61	46
3	62	49
4	63	39
5	62	45
...	...	...
19	37	85
20	39	88
Sample mean	51.85	70.90
Sample S.D.	11.43	19.86

Standardized Data

Member Number	Tenure (standardized)	Balance (standardized)
1	0.71	-1.51
2	0.80	-1.26
3	0.89	-1.10
4	0.98	-1.61
5	0.89	-1.30
...	...	...
19	-1.30	0.71
20	-1.12	0.86

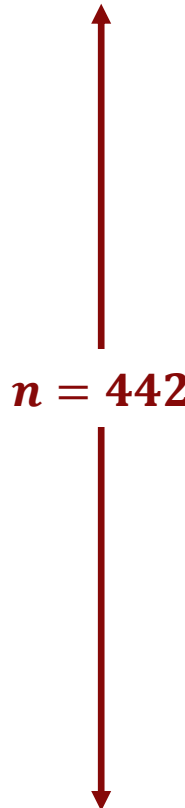
[For example, $0.71 = (60 - 51.85) / 11.43$]

Diabetes Data from Last Class

- The (regularization) methods we learned last time work best when the regression coefficients are comparable in terms of scale. This is accomplished by “standardizing” the variables.

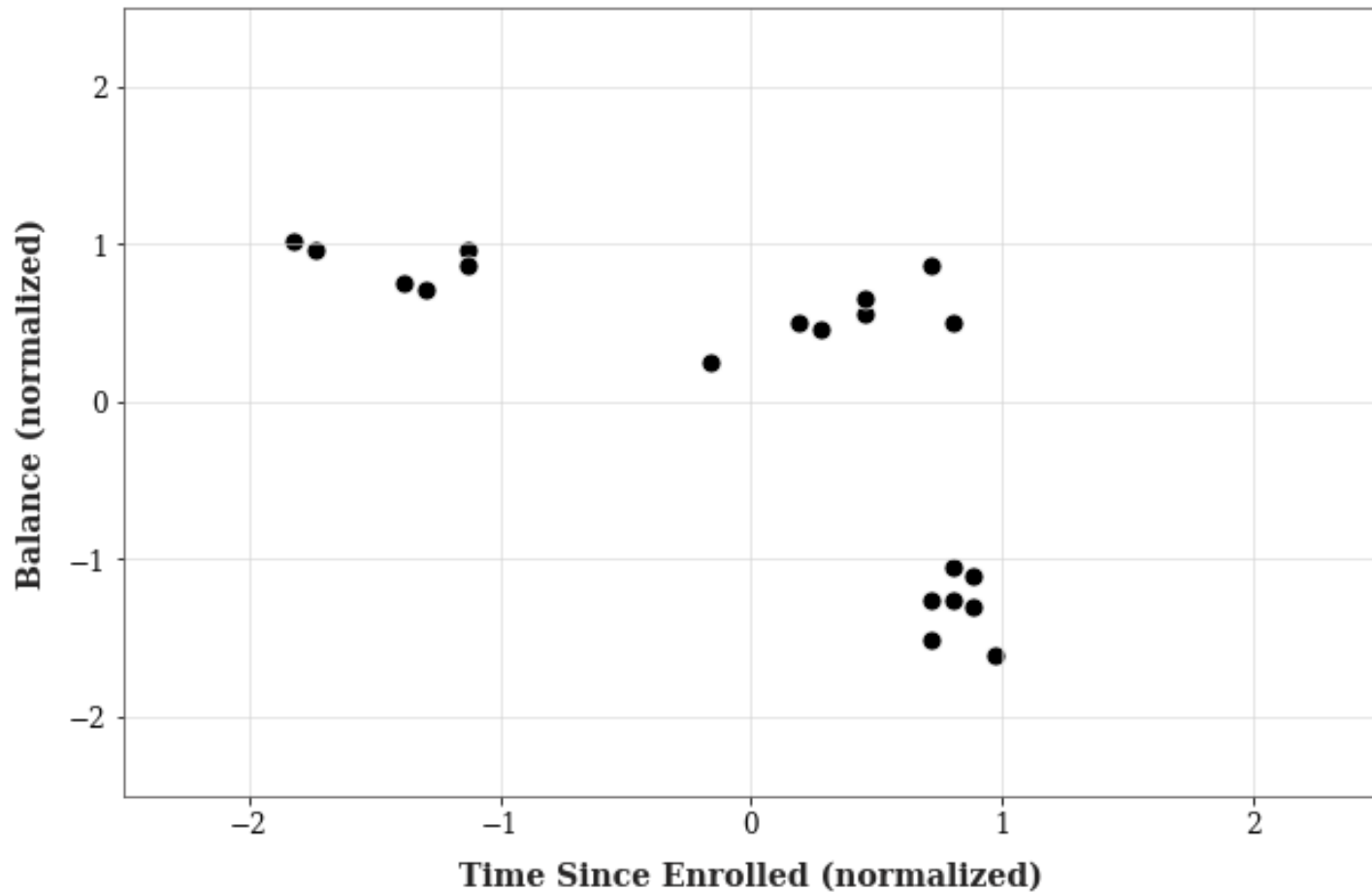
Standardized Expanded Diabetes Dataset

k : # of independent variables ($k = 64$); n : # of observations ($n = 442$)



	y	age		glu	age^2		glu^2		age:glu		ltg:glu
0	151	0.0381	...	-0.0176	-0.0149	...	-0.0275	...	-0.0303	...	-0.0278
1	75	-0.0019	...	-0.0922	-0.0413	...	0.0880	...	-0.0114	...	0.1040
2	141	0.0853	...	-0.0259	0.0916	...	-0.0224	...	-0.0647	...	-0.0223
3	206	-0.0891	...	-0.0094	0.1036	...	-0.0307	...	0.0034	...	-0.0250
4	135	0.0054	...	-0.0466	-0.0408	...	-0.0012	...	-0.0209	...	0.0087
5	97	-0.0927	...	-0.0963	0.1157	...	0.0990	...	0.1844	...	0.0578
6	138	-0.0455	...	-0.0384	-0.0036	...	-0.0112	...	0.0237	...	0.0270
7	63	0.0635	...	0.0031	0.0324	...	-0.0318	...	-0.0109	...	-0.0230
8	110	0.0417	...	0.0113	-0.0096	...	-0.0301	...	-0.0047	...	-0.0242
9	310	-0.0709	...	-0.0135	0.0505	...	-0.0293	...	0.0061	...	-0.0390
10	101	-0.0963	...	-0.0342	0.1283	...	-0.0154	...	0.0584	...	0.0218
11	69	0.0272	...	-0.0591	-0.0279	...	0.0173	...	-0.0511	...	0.0920
12	179	0.0163	...	-0.0425	-0.0365	...	-0.0064	...	-0.0307	...	0.0051
13	185	0.0054	...	-0.0135	-0.0408	...	-0.0293	...	-0.0169	...	-0.0311
14	118	0.0453	...	-0.0756	-0.0038	...	0.0488	...	-0.0919	...	0.0271
...	...	...	...	...	...	...	...	...	...	...	...
437	178	0.0417	...	0.0072	-0.0096	...	-0.0312	...	-0.0085	...	-0.0164
438	104	-0.0055	...	0.0445	-0.0408	...	-0.0040	...	-0.0207	...	-0.0368
439	132	0.0417	...	0.0155	-0.0096	...	-0.0285	...	-0.0008	...	-0.0352
440	220	-0.0455	...	-0.0259	-0.0036	...	-0.0224	...	0.0111	...	-0.0437
441	57	-0.0455	...	0.0031	-0.0036	...	-0.0318	...	-0.0184	...	-0.0211

Plot of Standardized Mini-dataset



Standardized Airline Customer Data

n : # of observations ($n = 3,999$)

$n = 3,999$

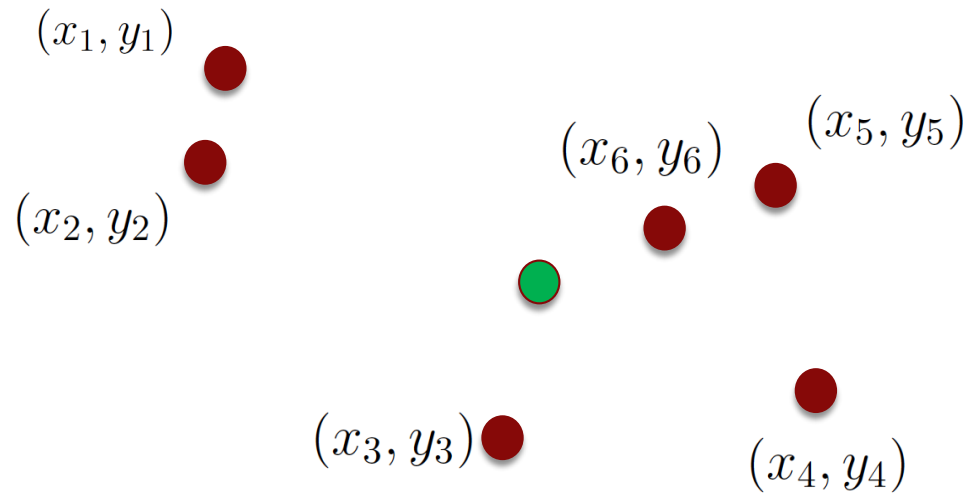
	Balance	BonusMiles	BonusTrans	FlightMiles	FlightTrans	DaysSinceEnroll
0	-0.25	0.59	-0.27	0.03	-0.1	-0.51
1	-0.63	-0.71	-1.21	-0.33	-0.36	1.82
2	-0.24	0.22	0.46	-0.33	-0.36	-0.39
3	1.39	-0.6	0.35	-0.33	-0.36	0.31
4	0.51	-0.1	-0.27	-0.33	-0.36	1.5
5	-0.65	-0.71	-1.21	-0.33	-0.36	-1.15
6	1.27	0.98	2.33	-0.33	-0.36	-0.35
7	2.75	1.4	1.5	1.56	0.96	-0.24
8	0.72	2.76	0.77	-0.06	-0.1	-0.27
9	-0.71	-0.71	-1.21	-0.33	-0.36	0.25
...	...	...	...	...	...	...
3994	-0.7	-0.71	-1.21	-0.33	-0.36	-1.32
3995	-0.45	1.3	0.25	-0.33	-0.36	-0.26
3996	2.01	1.03	1.19	-0.33	-0.36	1.82
3997	-0.44	-0.71	-1.21	-0.33	-0.36	-0.51
3998	0.23	1.82	0.77	-0.33	-0.36	1.36

Clustering Methods

1. k -Means Clustering
2. (Agglomerative) Hierarchical Clustering
3. **Optional:** Gaussian Mixture Models GMMs

Method 1: k -Means Clustering

Centroid/Center of Some Points



$$\text{Centroid} = \left(\frac{x_1 + \cdots + x_6}{6}, \frac{y_1 + \cdots + y_6}{6} \right)$$

k -Means Clustering

Set number of clusters

- Select the number of clusters, denoted by k

Initialize

- Randomly select/create k centroid locations
 - A “centroid” can be thought of as a point that is “representative” of each cluster
 - A centroid need not correspond to a particular sample observation

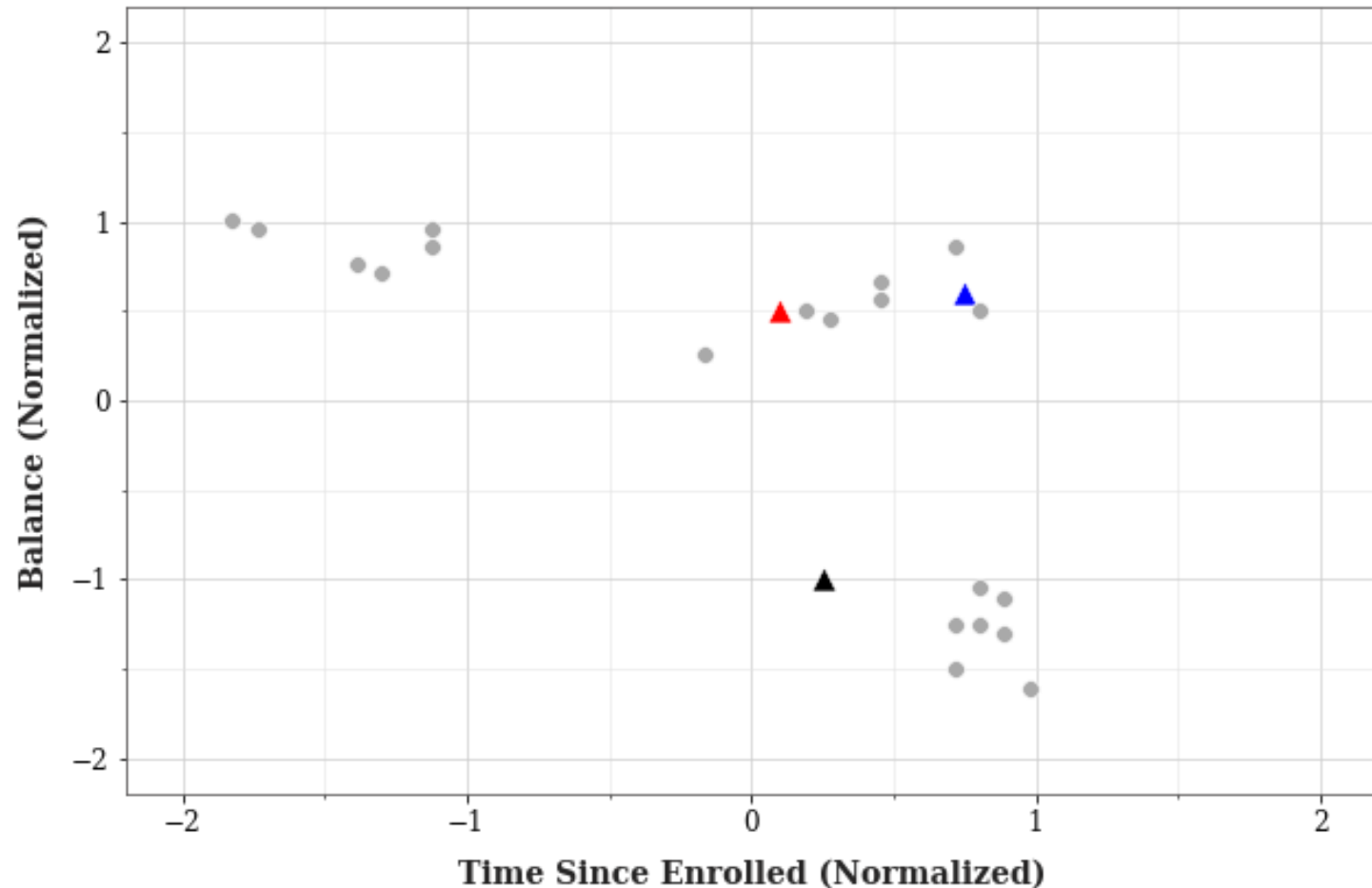
Iterate

- Repeat the following two steps until convergence:
 - Assign each observation to the nearest centroid
 - Recalculate the k centroids as the average of their assigned observations

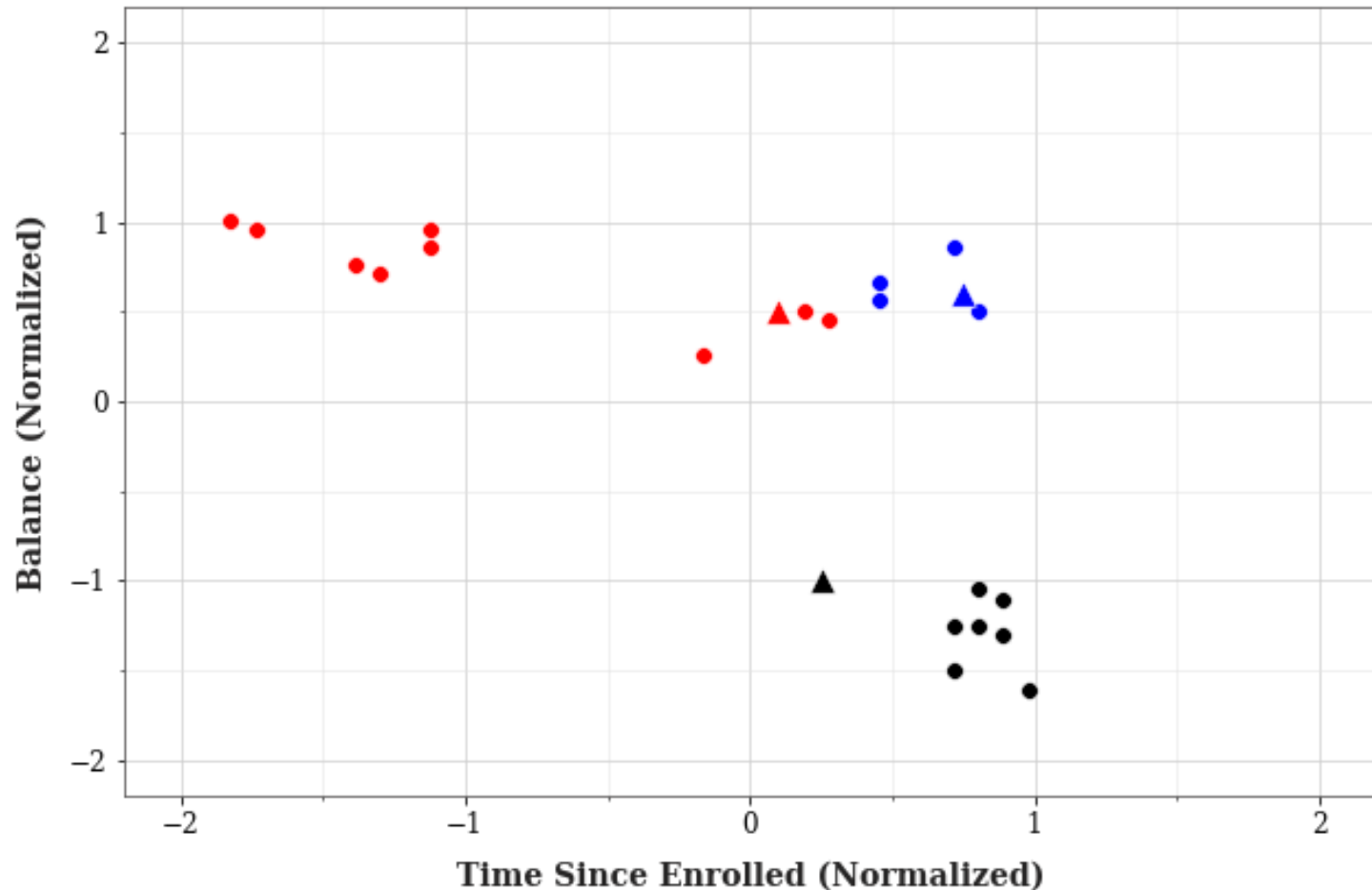
Terminate

- Terminate when centroids no longer move

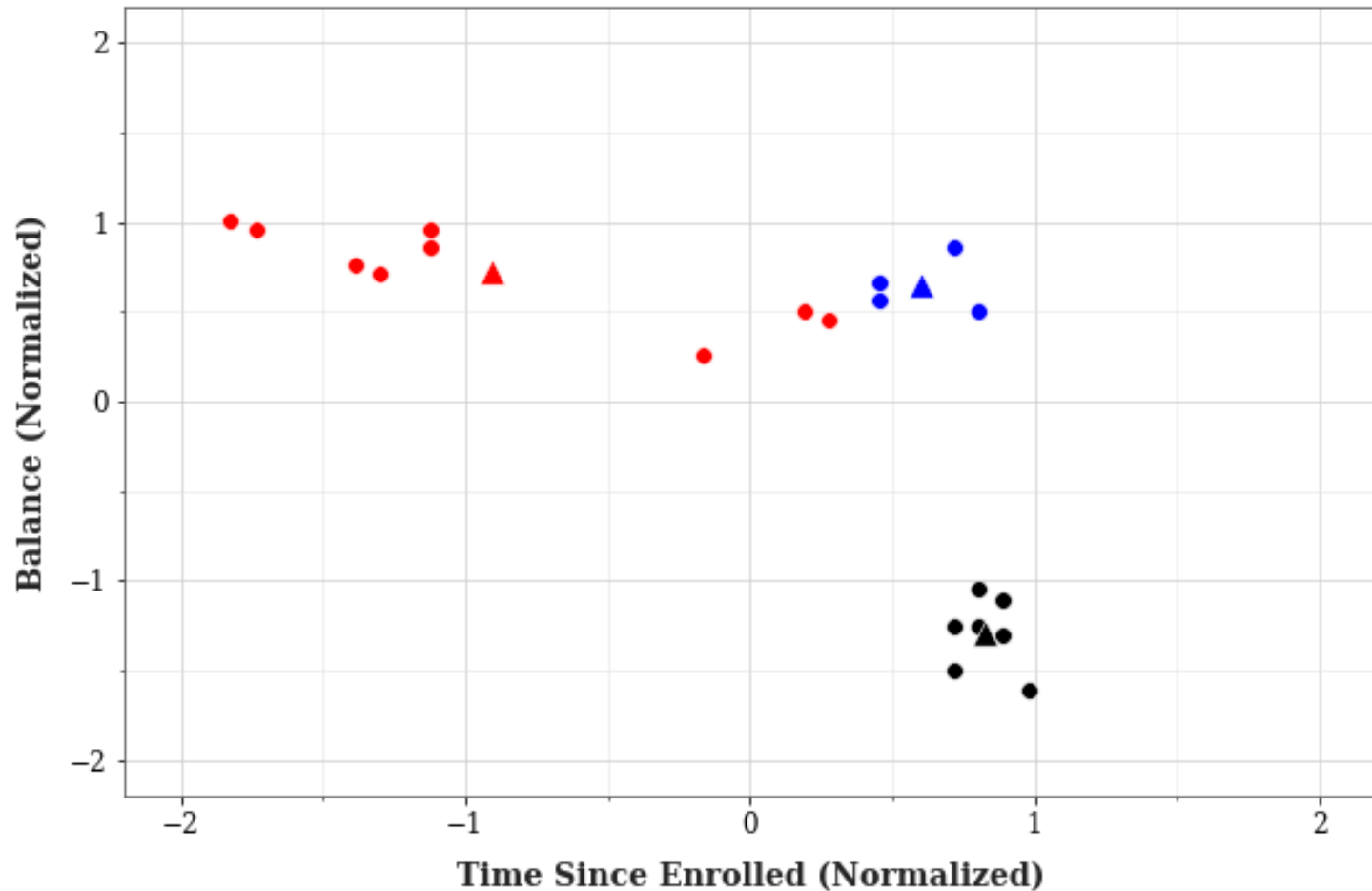
1. Select k Random Centroid Locations



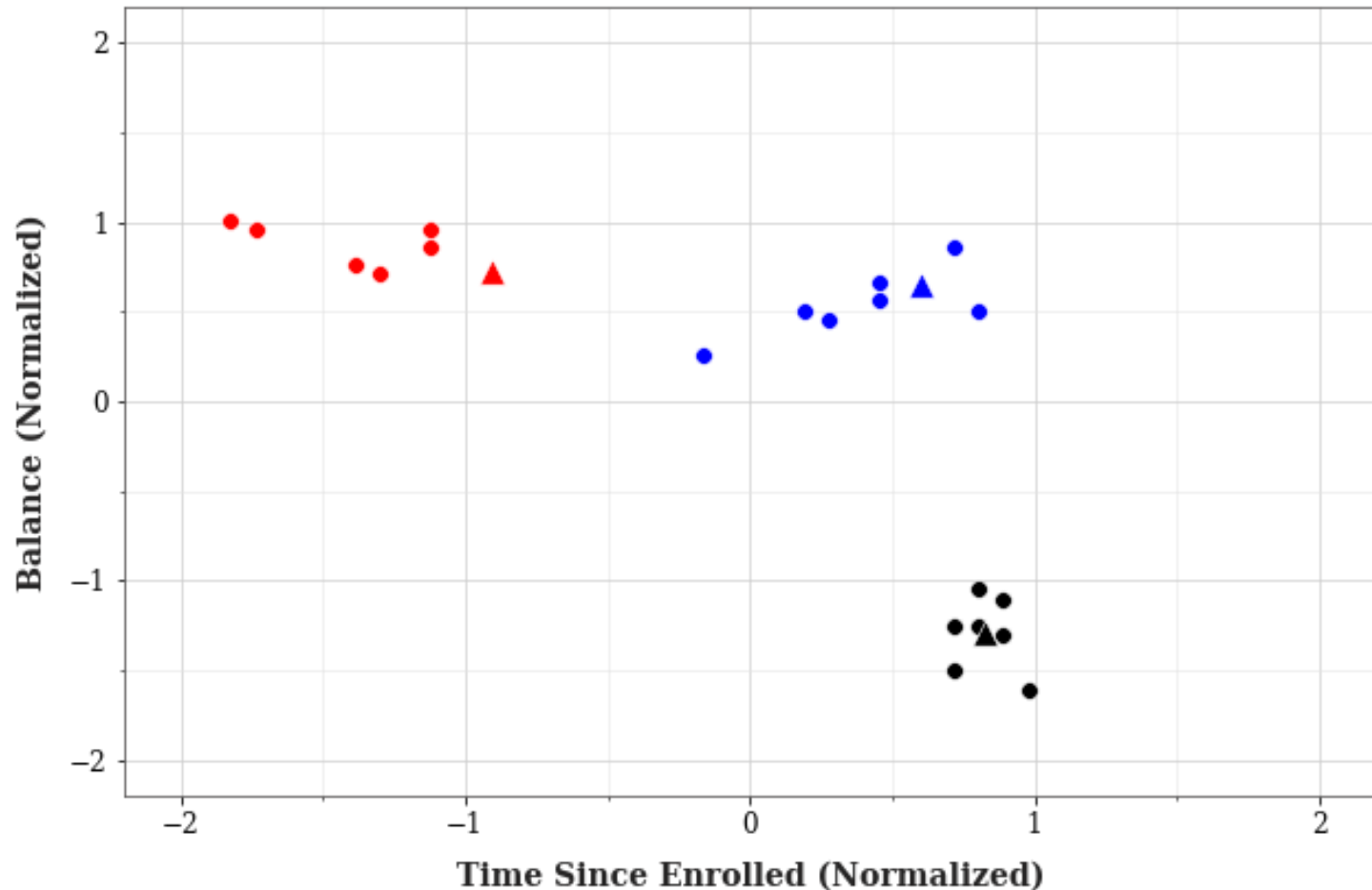
2a. Assign every observation to its nearest centroid



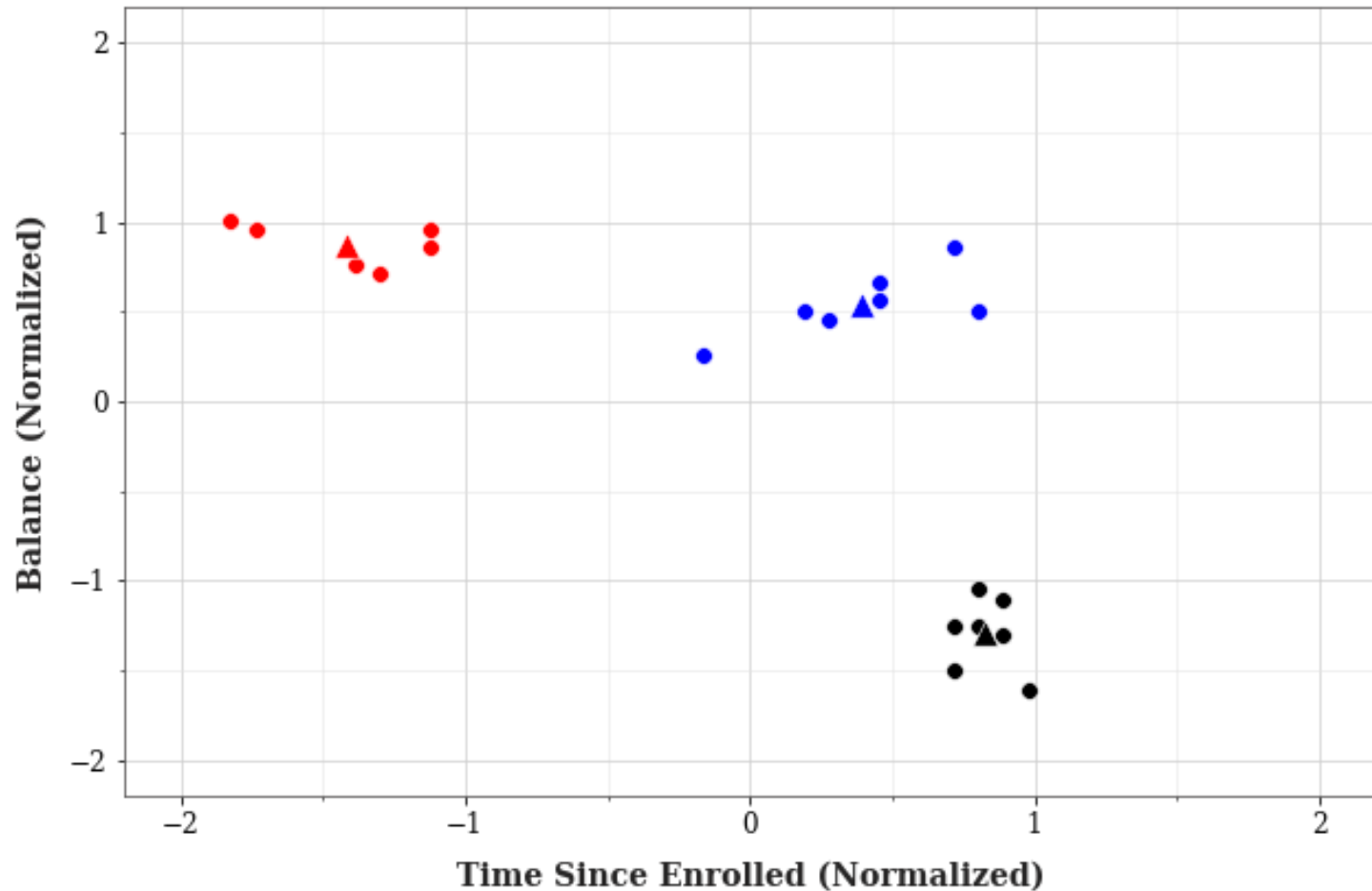
2b. Re-calculate the centroids



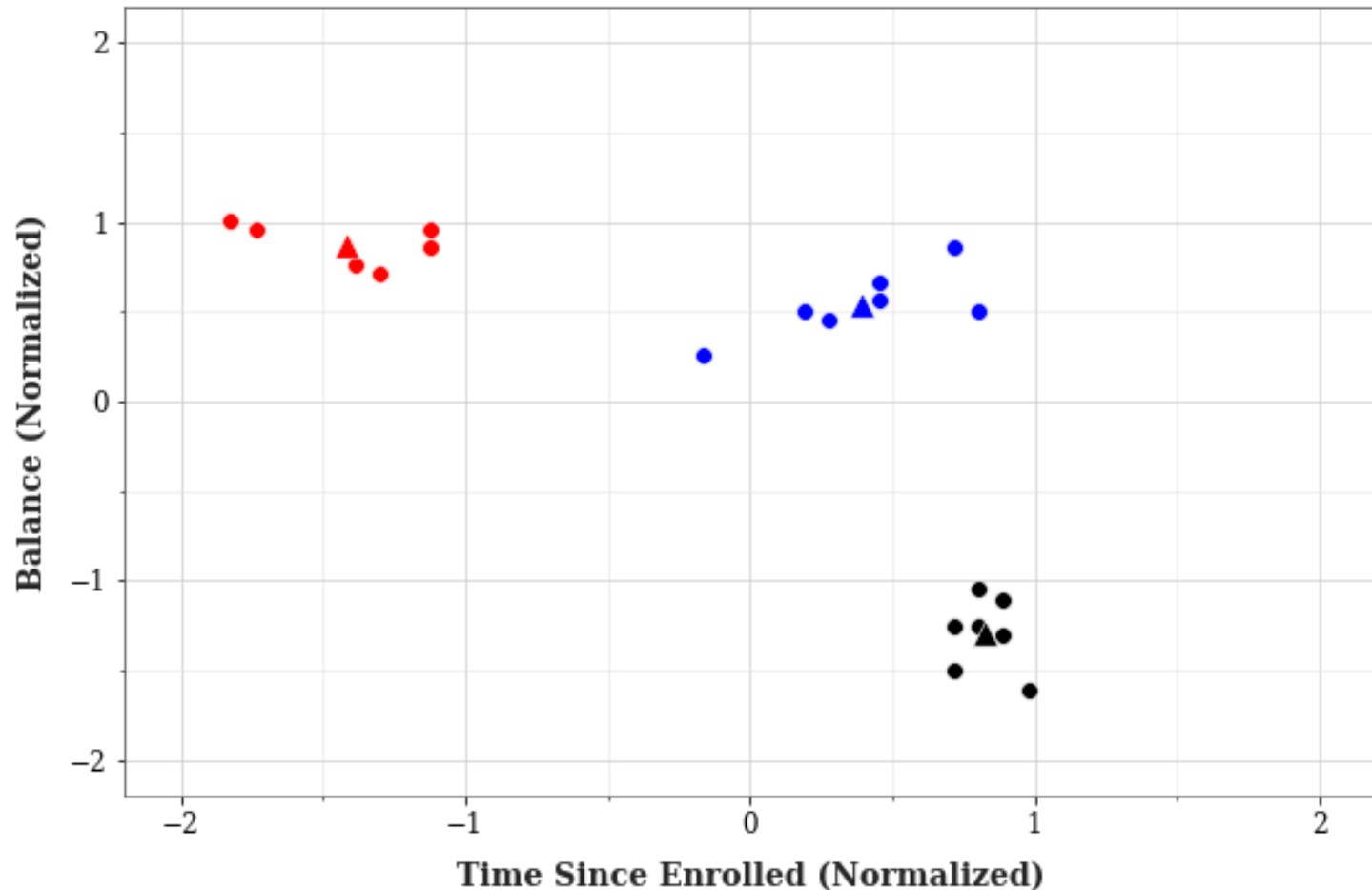
2a. Assign every observation to its nearest centroid



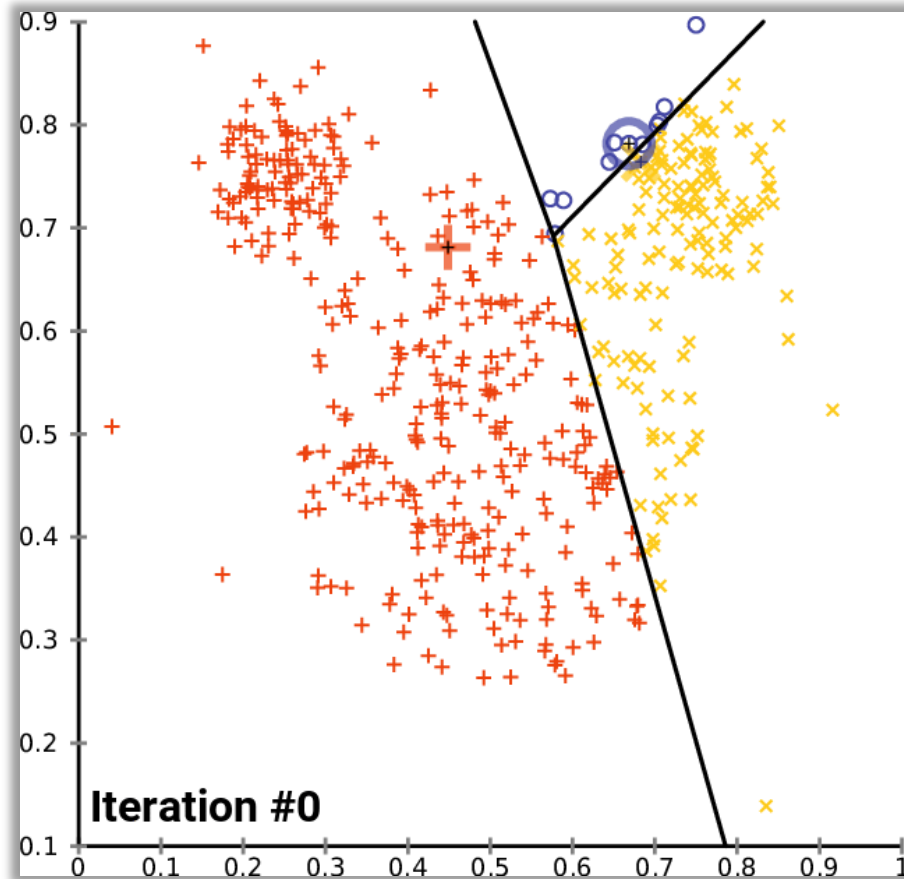
2b. Re-calculate the centroids



2a. Assign every observation to its nearest centroid



Another example: k-means



k -Means Clustering applied to the Airline Customers Data

Standardized Cluster Centroids

Standardized Variable	Cluster							
	1	2	3	4	5	6	7	8
Balance	-0.16	-0.35	0.76	0.41	0.77	-0.42	5.99	0.30
BonusMiles	0.09	-0.58	2.26	0.16	0.86	-0.61	2.09	0.21
BonusTrans	0.55	-0.75	1.13	0.83	2.49	-0.86	1.11	0.46
FlightMiles	-0.24	-0.22	-0.08	1.88	5.84	-0.24	0.80	-0.18
FlightTrans	-0.26	-0.23	-0.05	1.92	6.07	-0.24	1.11	-0.19
DaysSinceEnroll	-0.65	0.70	0.50	0.03	0.15	-1.02	1.17	1.02
Cluster Size	845	888	313	220	54	963	46	670

- How would you describe these groups of customers?
- How might we market to these clusters?

Clusters 1, 3 and 8

Standardized Variable	Cluster		
	1	3	8
Balance	-0.16	0.76	0.30
BonusMiles	0.09	2.26	0.21
BonusTrans	0.55	1.13	0.46
FlightMiles	-0.24	-0.08	-0.18
FlightTrans	-0.26	-0.05	-0.19
DaysSinceEnroll	-0.65	0.50	1.02
Cluster Size	845	313	670

Original Variable	Cluster		
	1	3	8
Balance	57,477	150,191	103,834
BonusMiles	19,318	71,726	22,217
BonusTrans	17	22	16
FlightMiles	124	348	208
FlightTrans	0	1	1
DaysSinceEnroll	2,227	5,151	6,225
Cluster Size	845	313	670

- The point addicts: focus on bonus transactions
- Target bonuses for flying? Special offers for bonus transactions?

Clusters 2 and 6

Standardized Variable	Cluster		Original Variable	Cluster	
	2	6		2	6
Balance	-0.35	-0.42	Balance	38,330	31,276
BonusMiles	-0.58	-0.61	BonusMiles	3,137	2,413
BonusTrans	-0.75	-0.86	BonusTrans	4	3
FlightMiles	-0.22	-0.24	FlightMiles	152	124
FlightTrans	-0.23	-0.24	FlightTrans	1	0
DaysSinceEnroll	0.70	-1.02	DaysSinceEnroll	5,564	2,012
Cluster Size	888	963	Cluster Size	888	963

- Low-activity customers, but for different reasons?
→ Promotional one-time events to incentivize new purchases?

Clusters 4 and 5

Standardized Variable	Cluster		Original Variable	Cluster	
	4	5		4	5
Balance	0.41	0.77	Balance	114,919	151,199
BonusMiles	0.16	0.86	BonusMiles	21,010	37,915
BonusTrans	0.83	2.49	BonusTrans	20	36
FlightMiles	1.88	5.84	FlightMiles	3,092	8,637
FlightTrans	1.92	6.07	FlightTrans	9	24
DaysSinceEnroll	0.03	0.15	DaysSinceEnroll	4,181	4,428
Cluster Size	220	54	Cluster Size	220	54

- Frequent Travelers: high flight miles and transactions
→ Thank for loyalty? Special offers? Refer a friend? Tier upgrades?

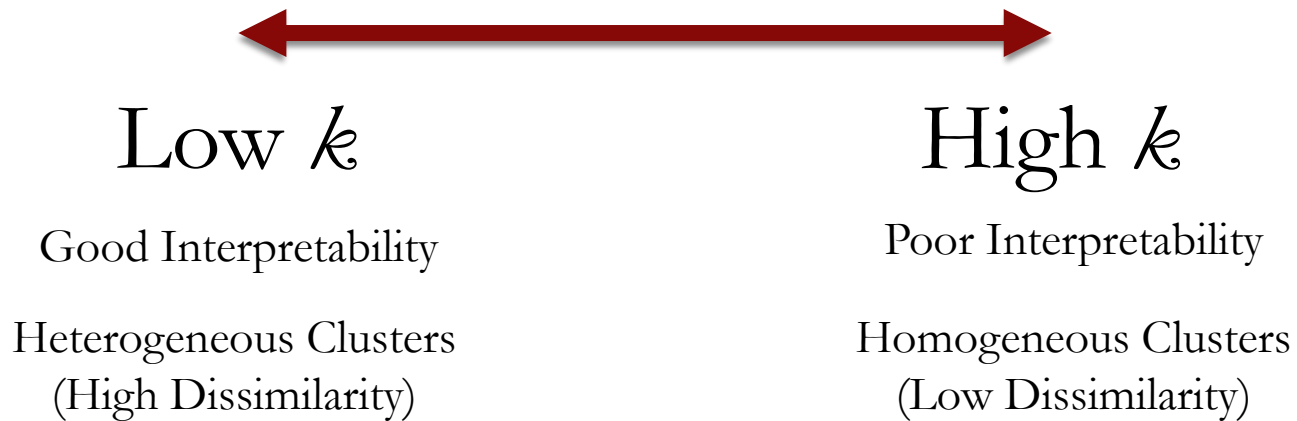
Clusters 7

Standardized Variable	Cluster	Original Variable	Cluster
	7		7
Balance	5.99	Balance	677,248
BonusMiles	2.09	BonusMiles	67,620
BonusTrans	1.11	BonusTrans	22
FlightMiles	0.80	FlightMiles	1,580
FlightTrans	1.11	FlightTrans	6
DaysSinceEnroll	1.17	DaysSinceEnroll	6,535
Cluster Size	46	Cluster Size	46

- Loyal Elites: long-tenured, high-value members with exceptional balances
→ Personalize experiences, offer exclusivity and premium recognition

Selecting the number of clusters k

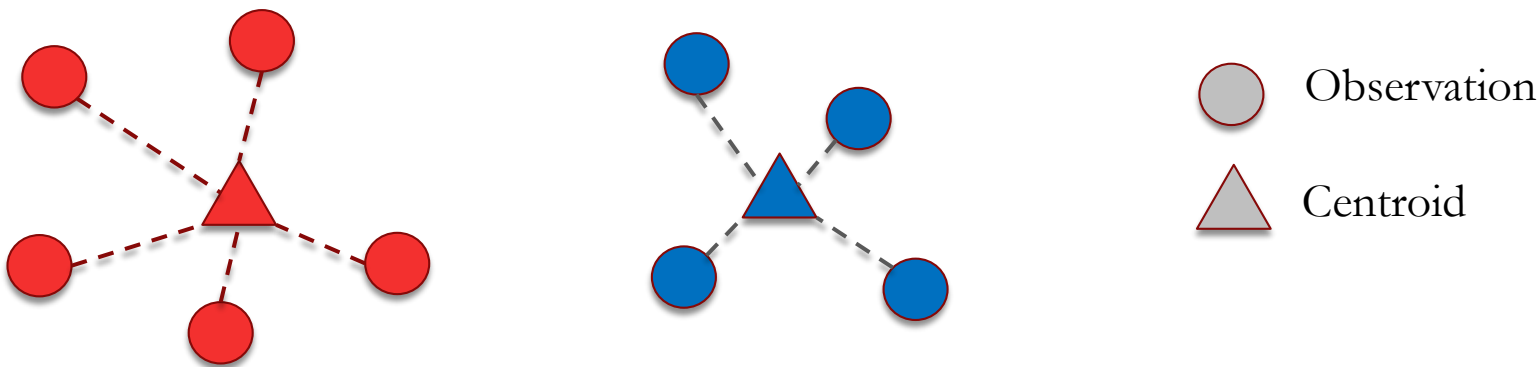
Trade-Off



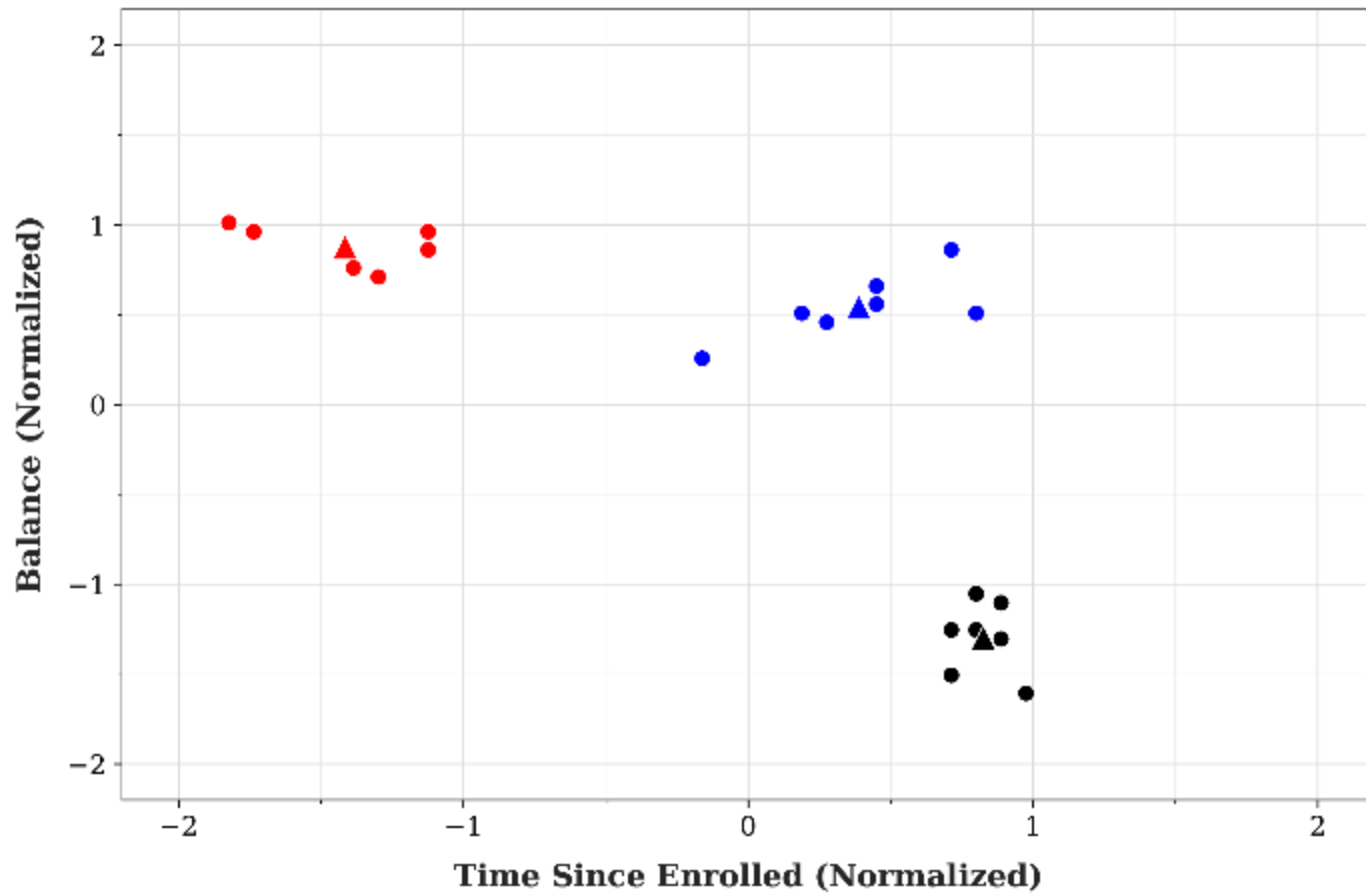
- Trade-off in selecting the number of clusters k
 - **Low k** : dissimilar customers will be combined into the same cluster, leading to heterogeneous clusters
 - **High k** : clusters will be too specific, leading to unactionable clusters — and likely inefficiencies in targeting each cluster differently

Quantifying this Trade-Off

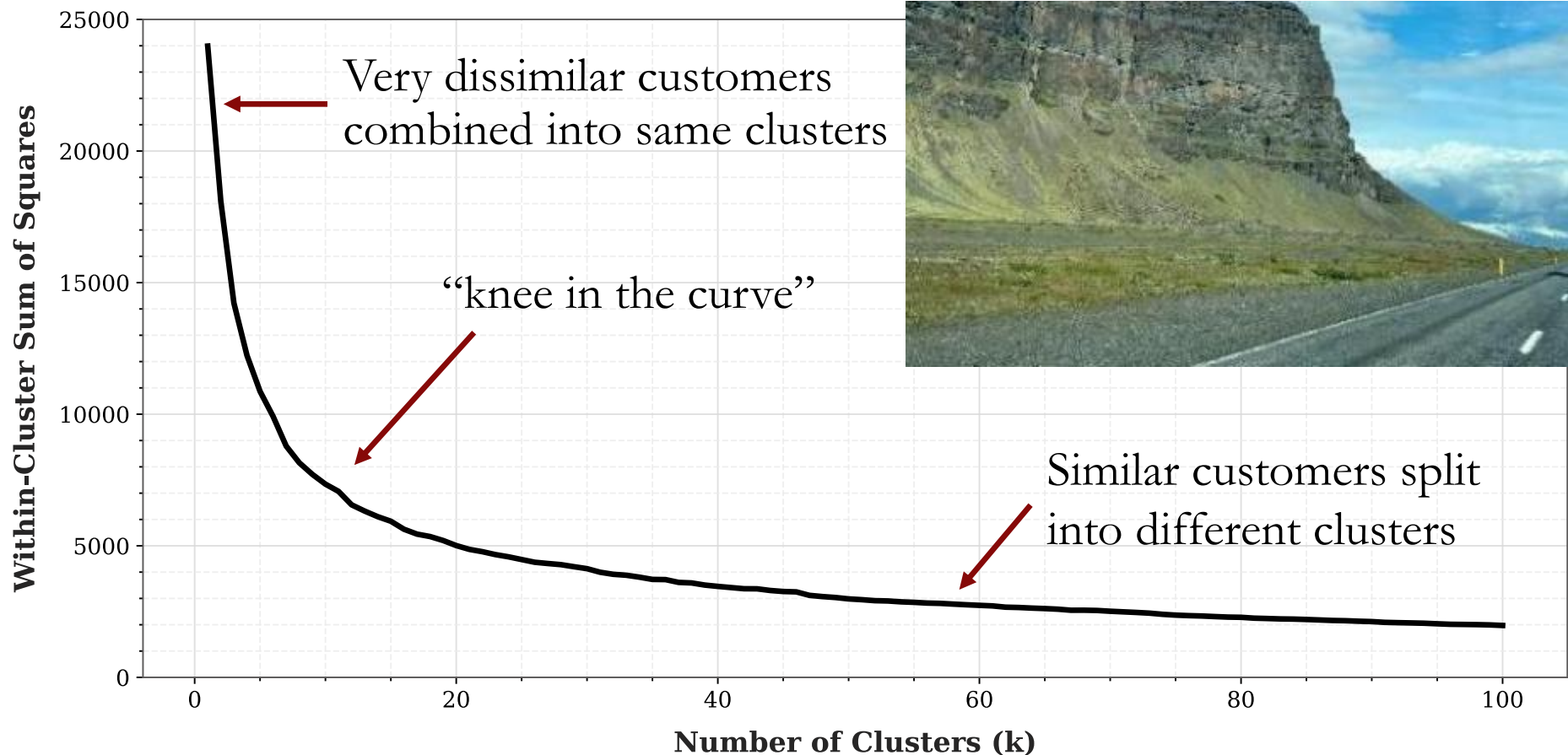
- For each value of k from 2 to 100, run a K-Means clustering algorithm.
- For each value of k , use the clusters to compute the **cluster dissimilarity**.
 - **Cluster Dissimilarity**: Measures the total amount of heterogeneity in the clusters, commonly the **sum of squared distances** from each observation to the centroid of the cluster it is assigned to.
- A *scree plot* displays the tradeoff between number of clusters k and the cluster dissimilarity — to find a “sweet spot”.



3 Clusters



Scree Plot for Airline Clusters



Method 2. Hierarchical Clustering

Hierarchical Clustering

Initialize

- Start with each observation in its own cluster
 - Initially, as many clusters as data points (3,999 here)

Iterate

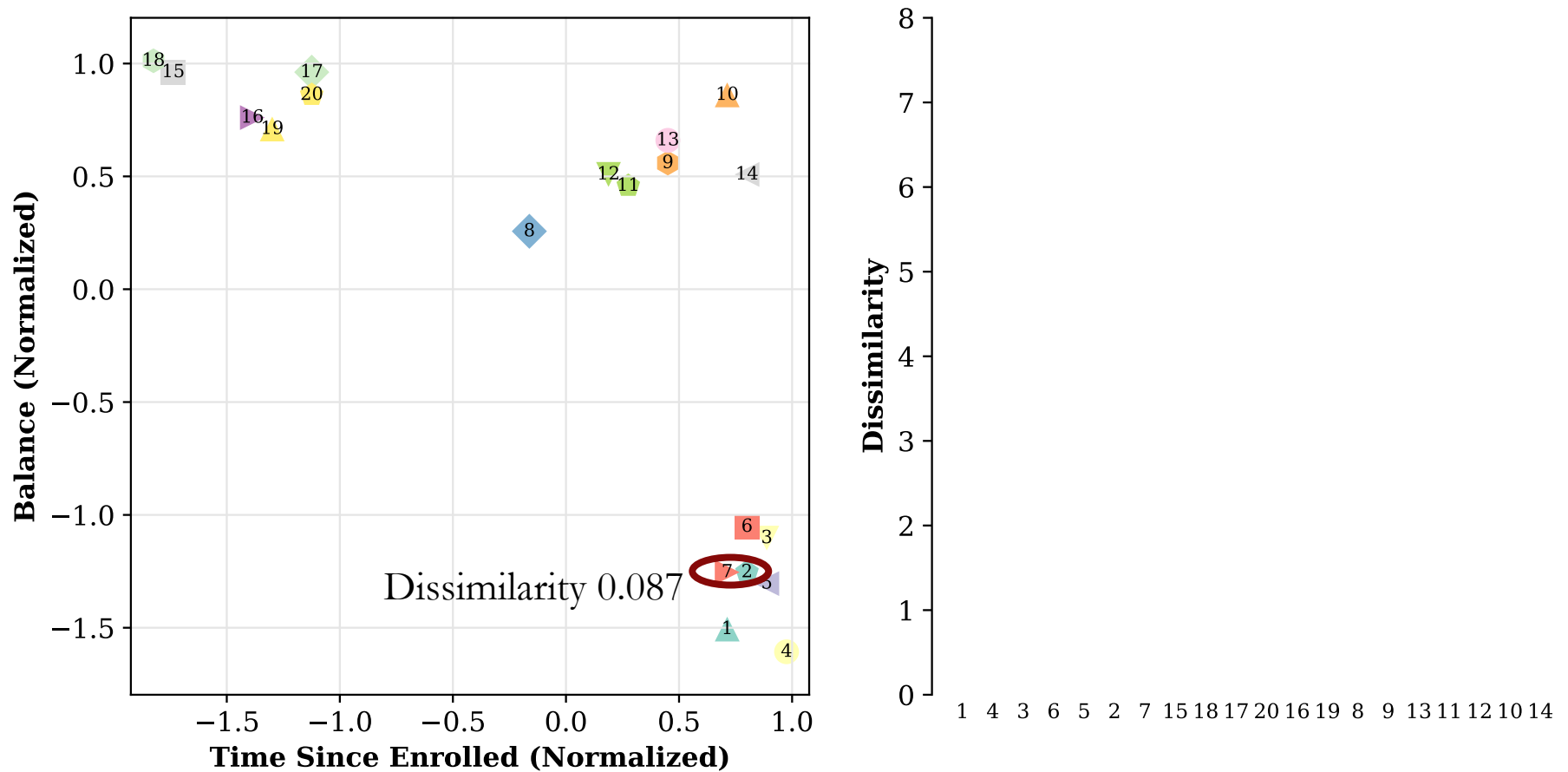
- Iteratively combine the pair of clusters that have the smallest cluster dissimilarity
 - Prefer combining clusters that are “close” to each other
 - Prefer combining small clusters instead of large clusters
- Number of clusters goes down by 1 at each iteration

Terminate

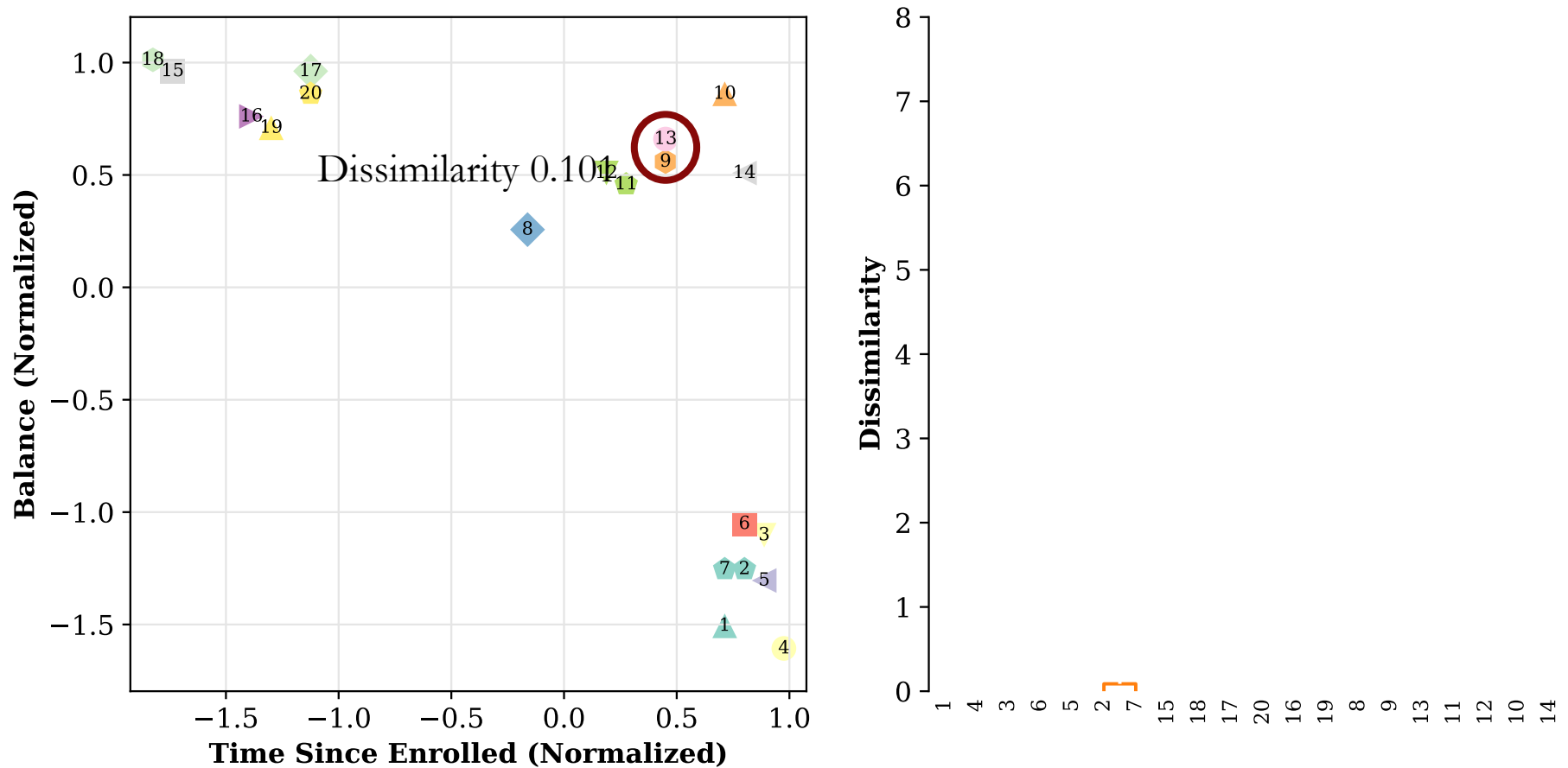
- Ultimately, there is just 1 cluster with all data points
- Enables you to select the appropriate number of clusters



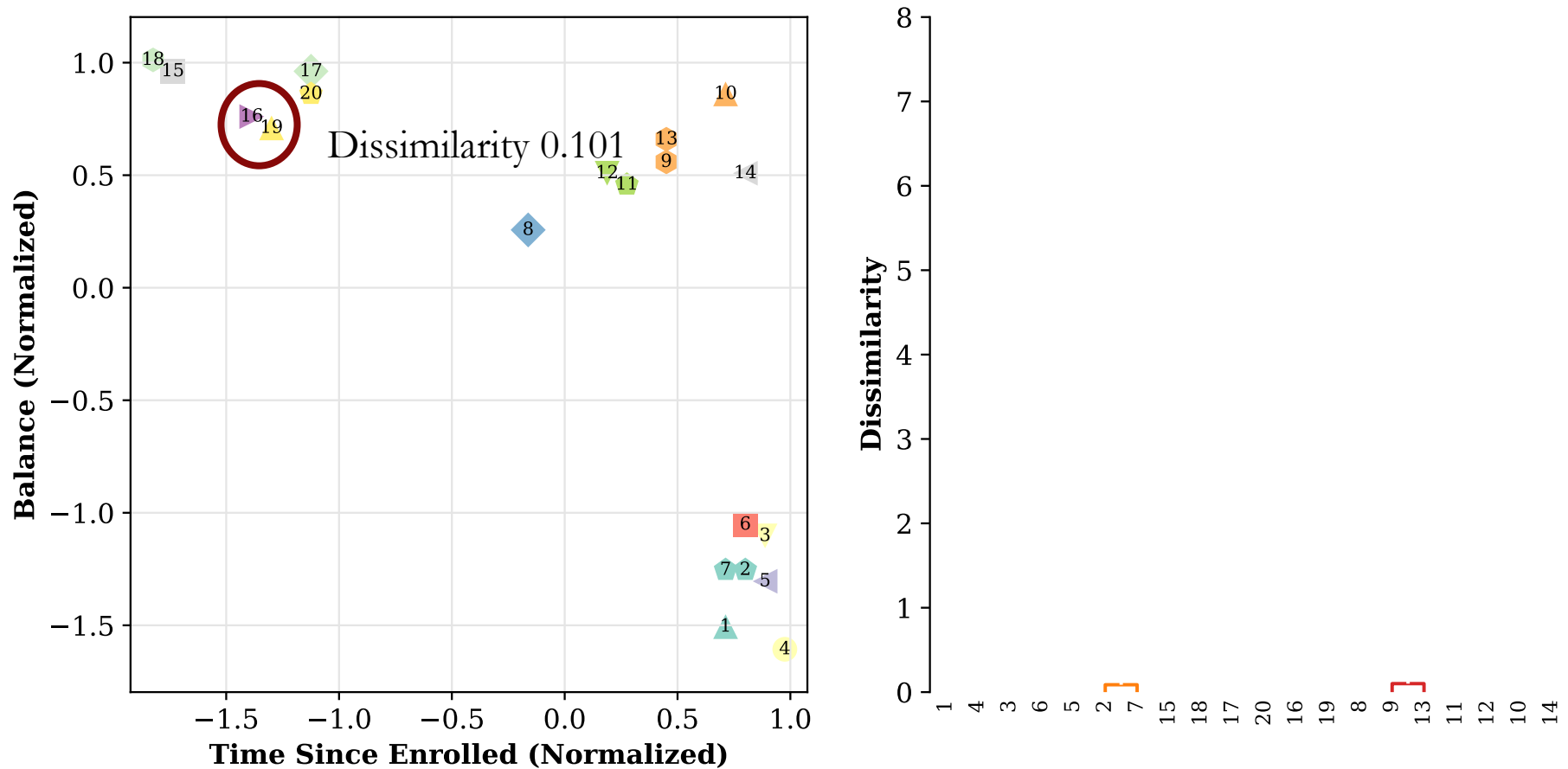
Hierarchical Clustering on Mini-dataset



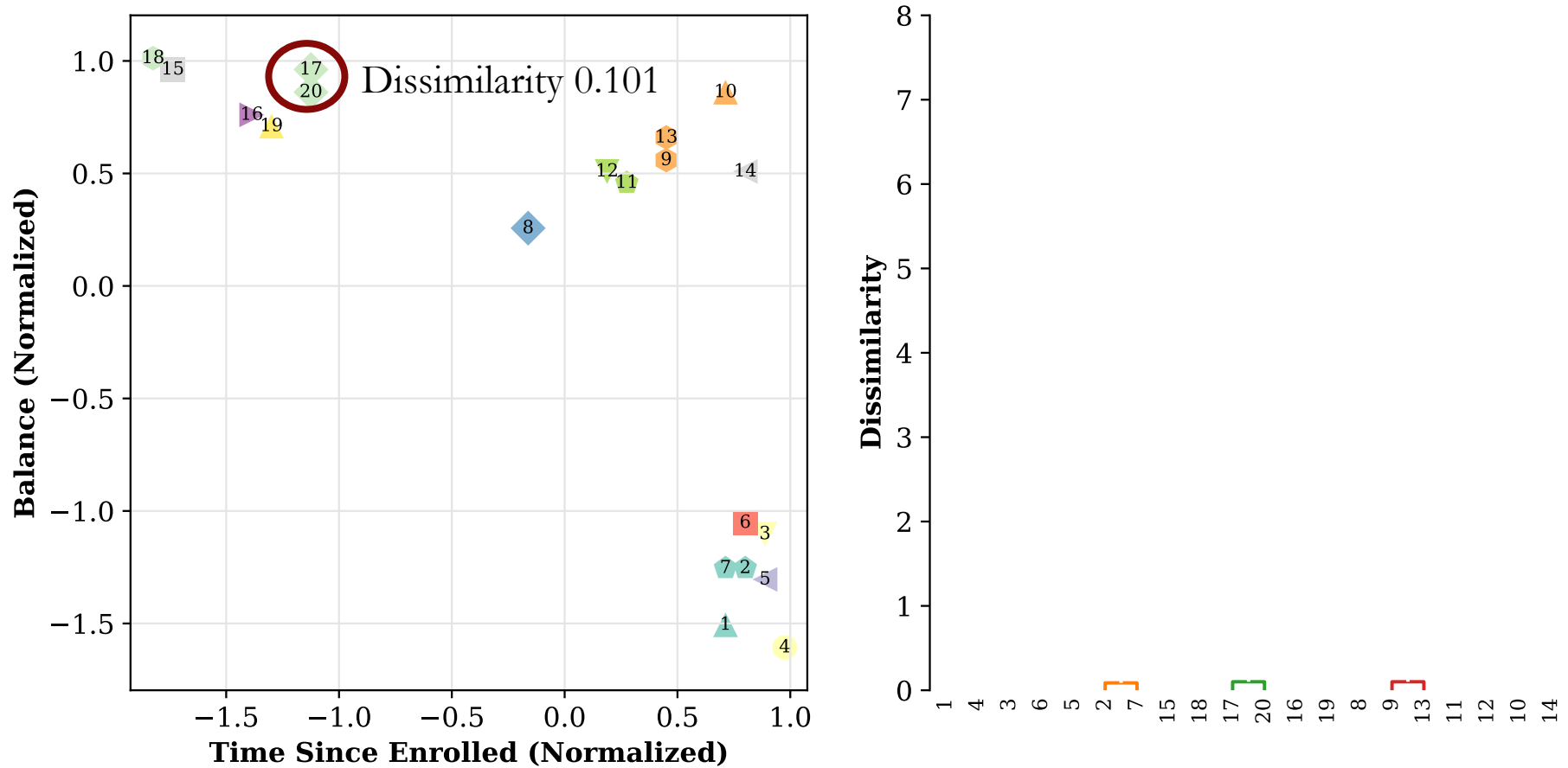
Hierarchical Clustering on Mini-dataset



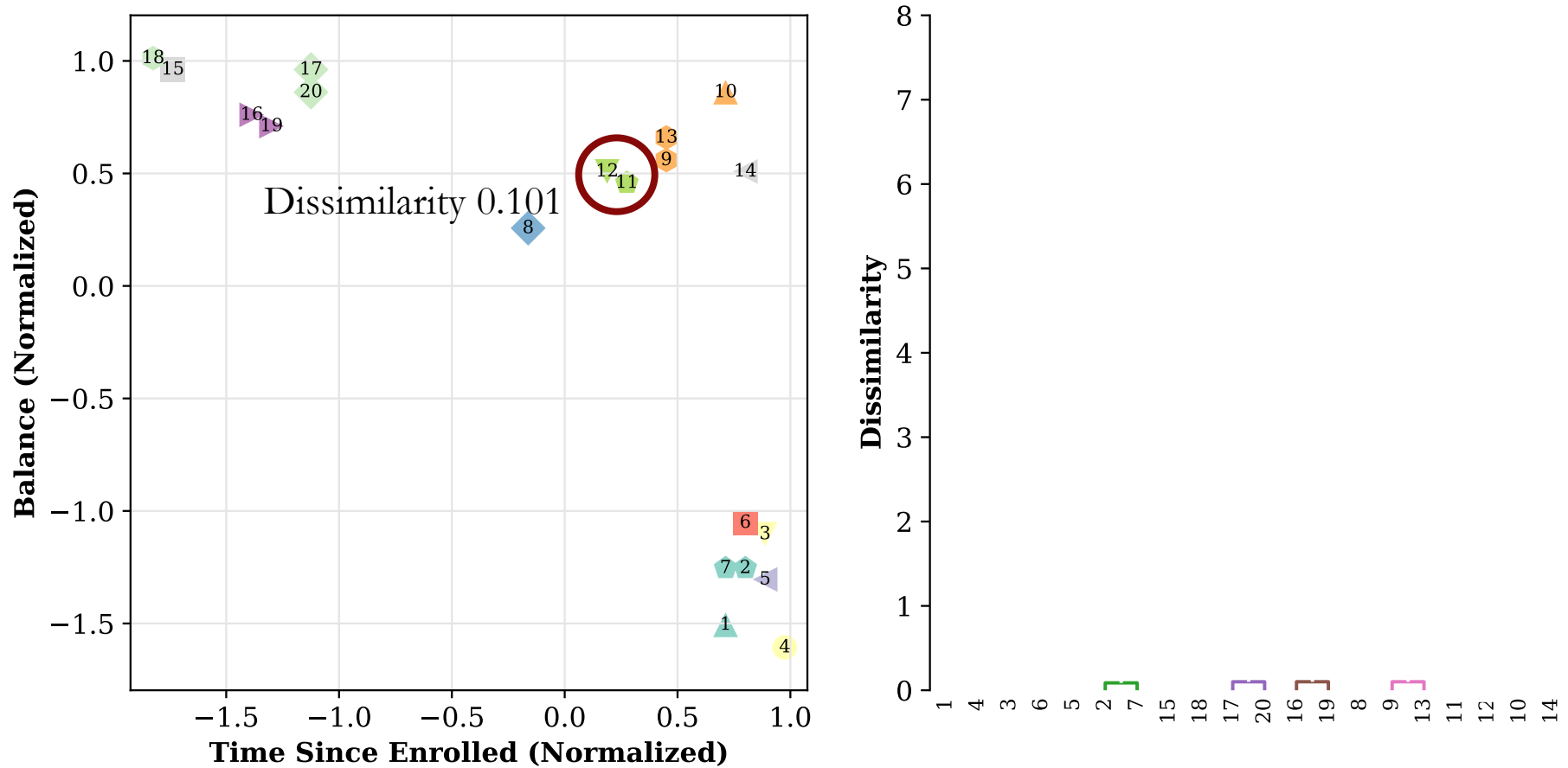
Hierarchical Clustering on Mini-dataset



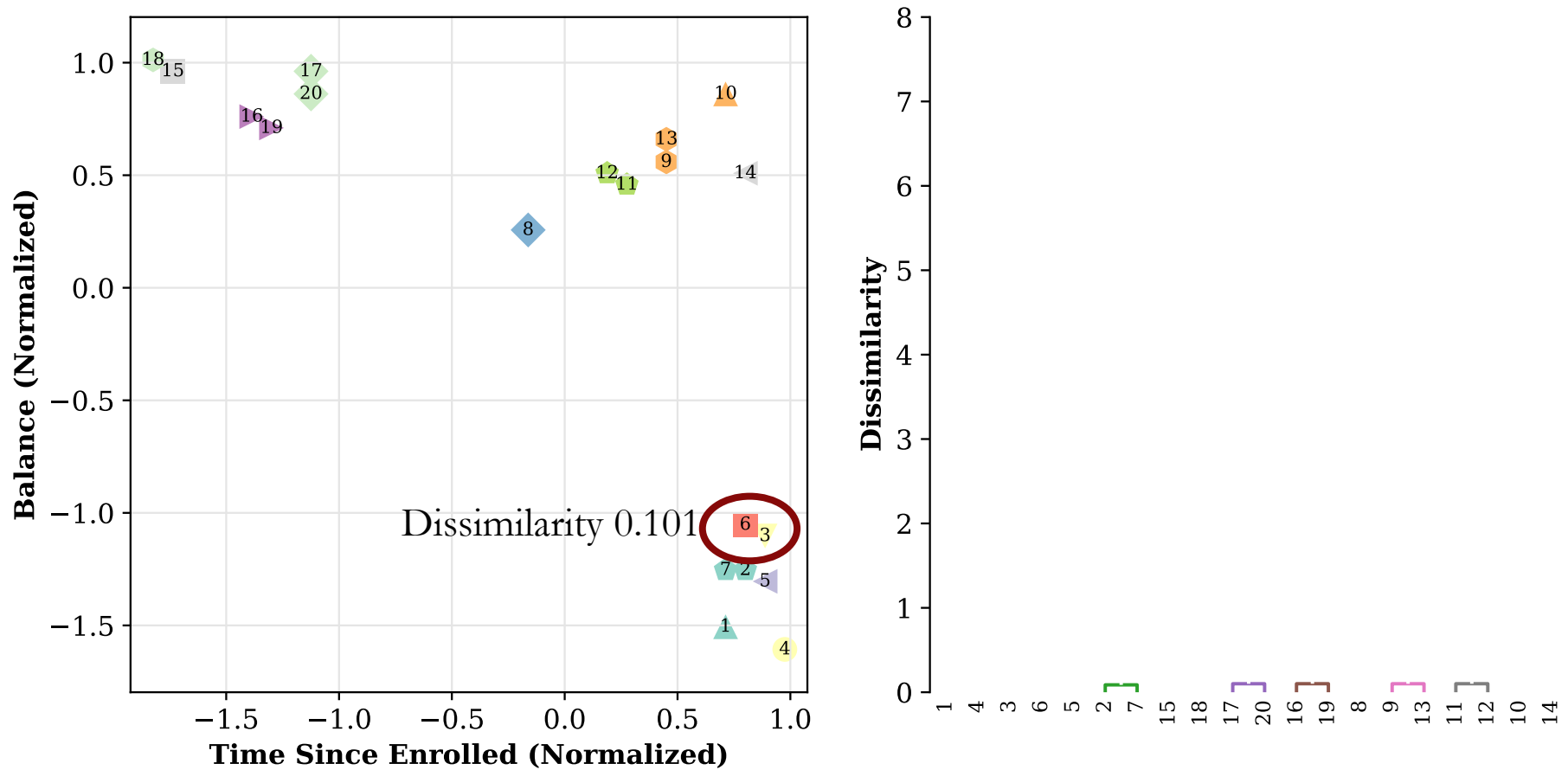
Hierarchical Clustering on Mini-dataset



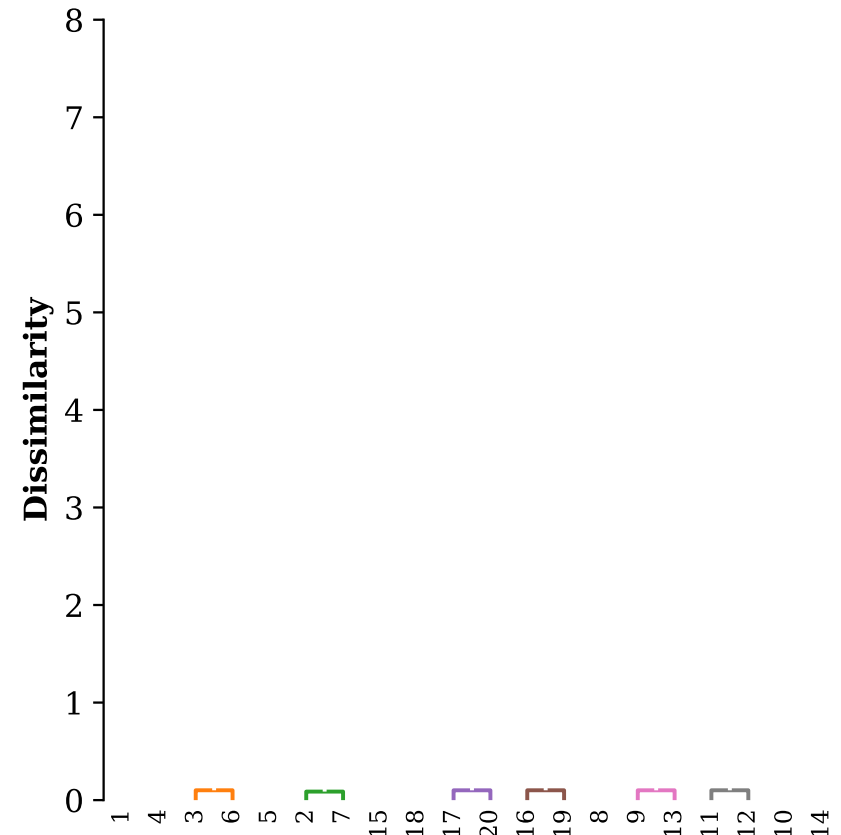
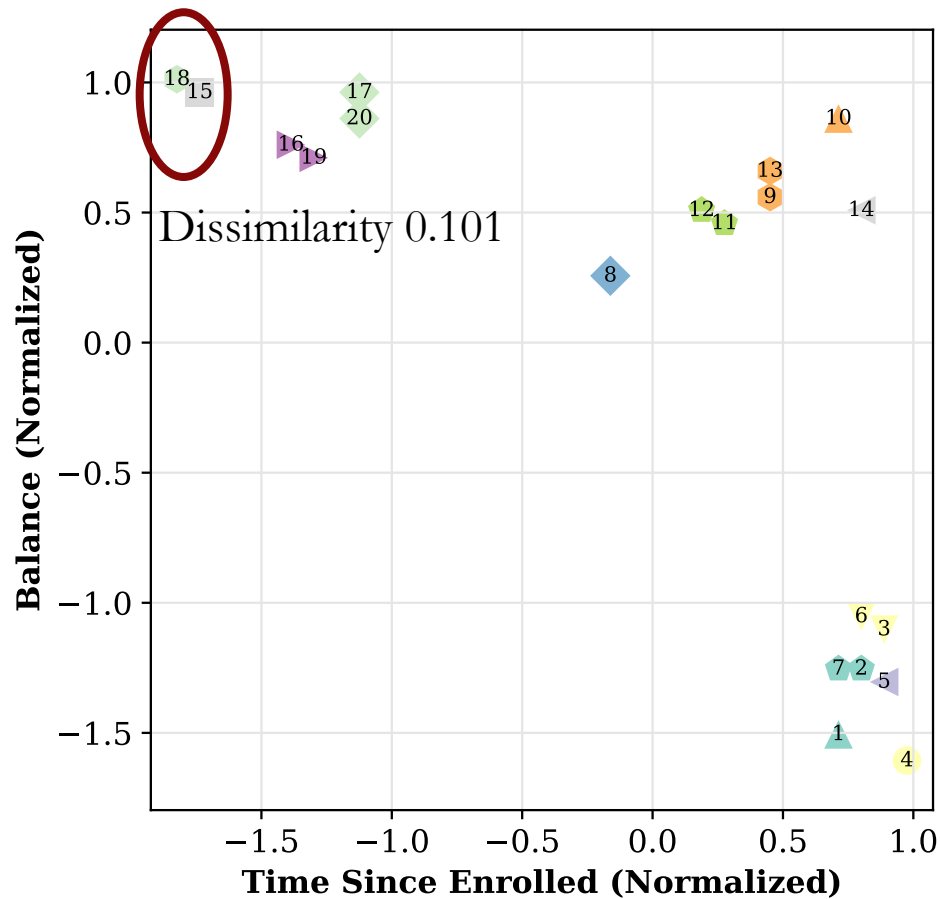
Hierarchical Clustering on Mini-dataset



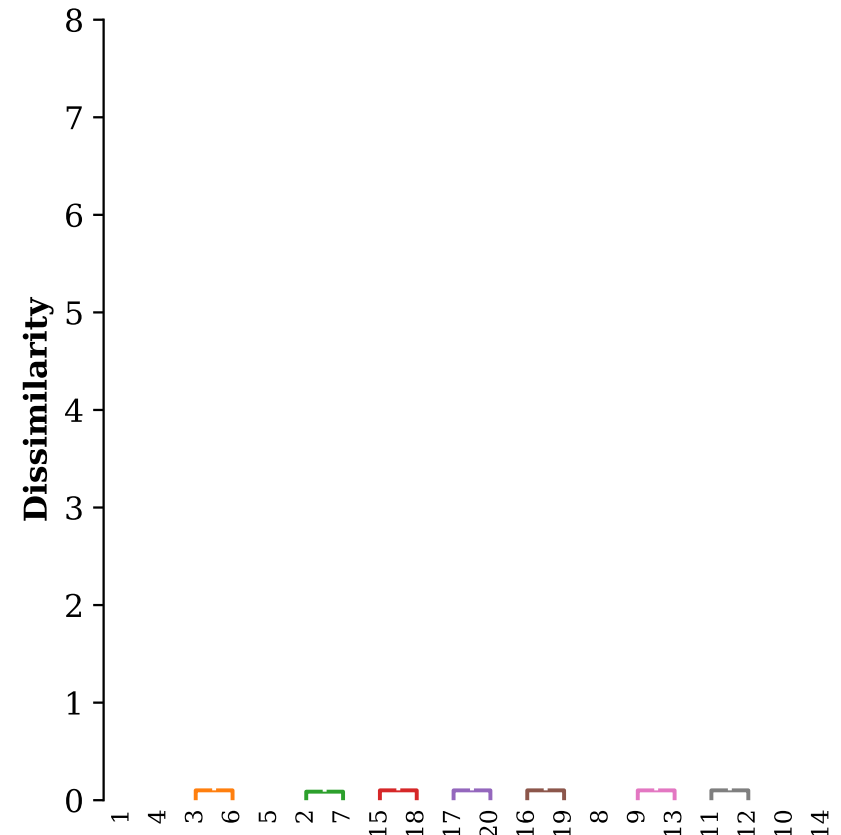
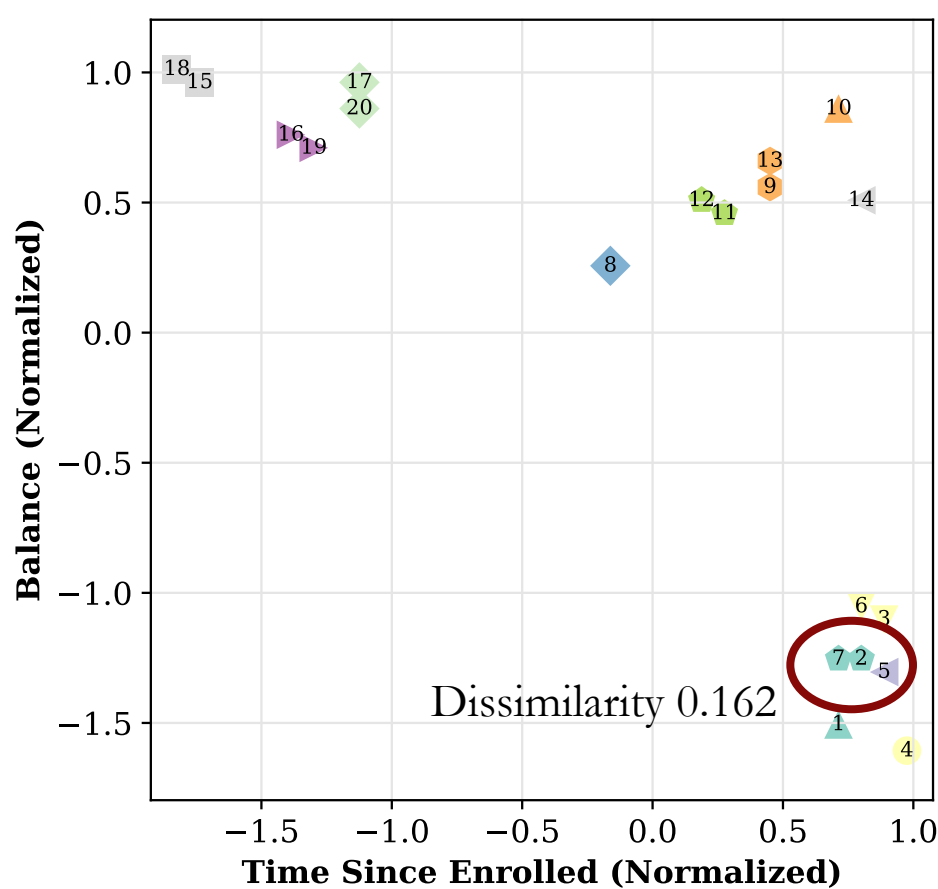
Hierarchical Clustering on Mini-dataset



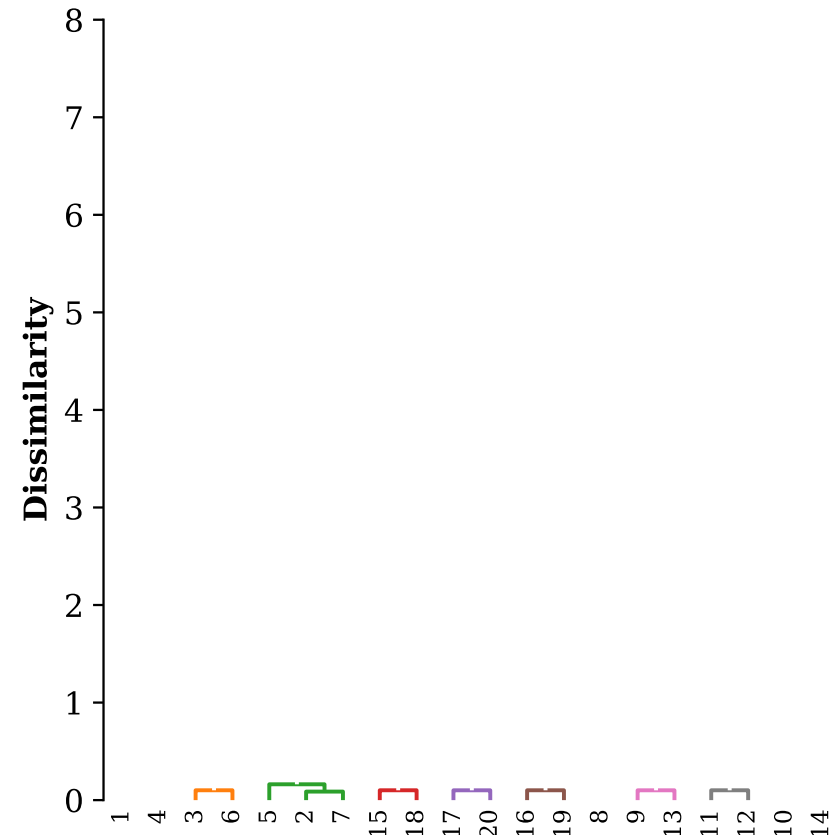
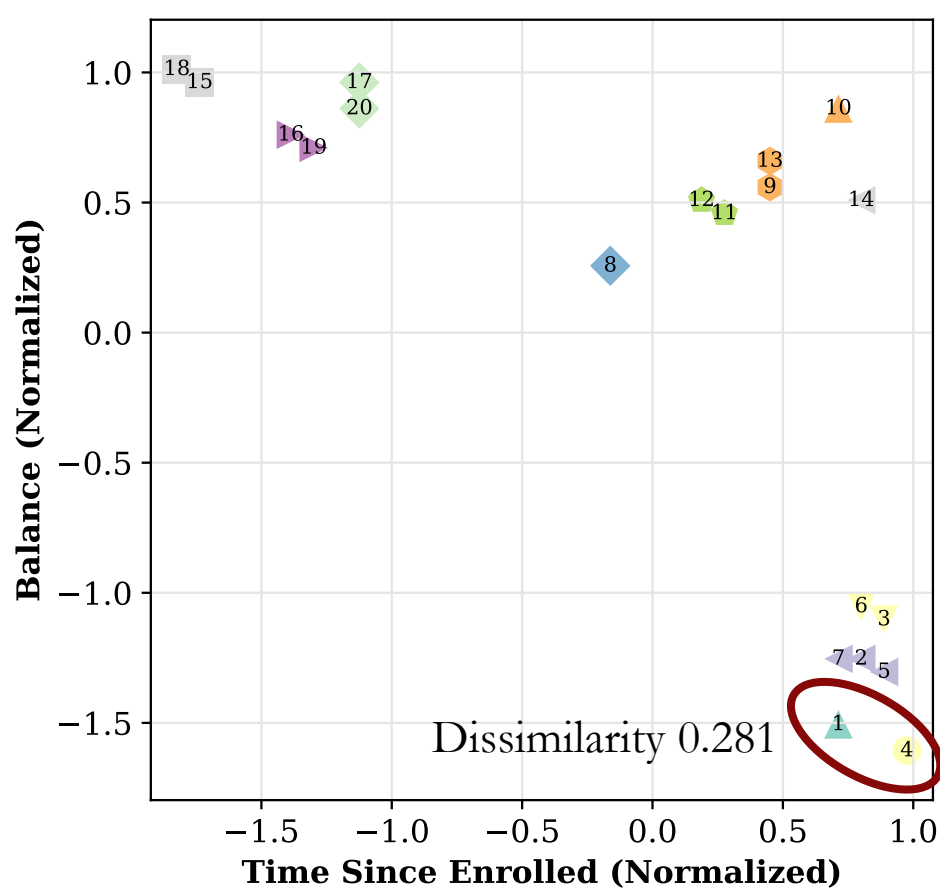
Hierarchical Clustering on Mini-dataset



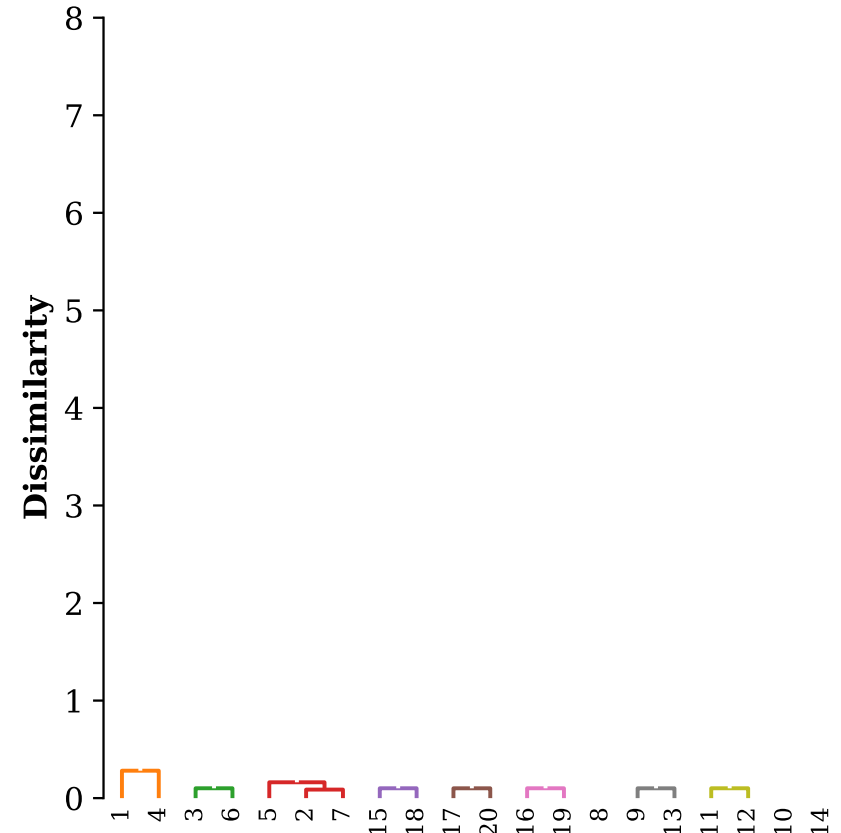
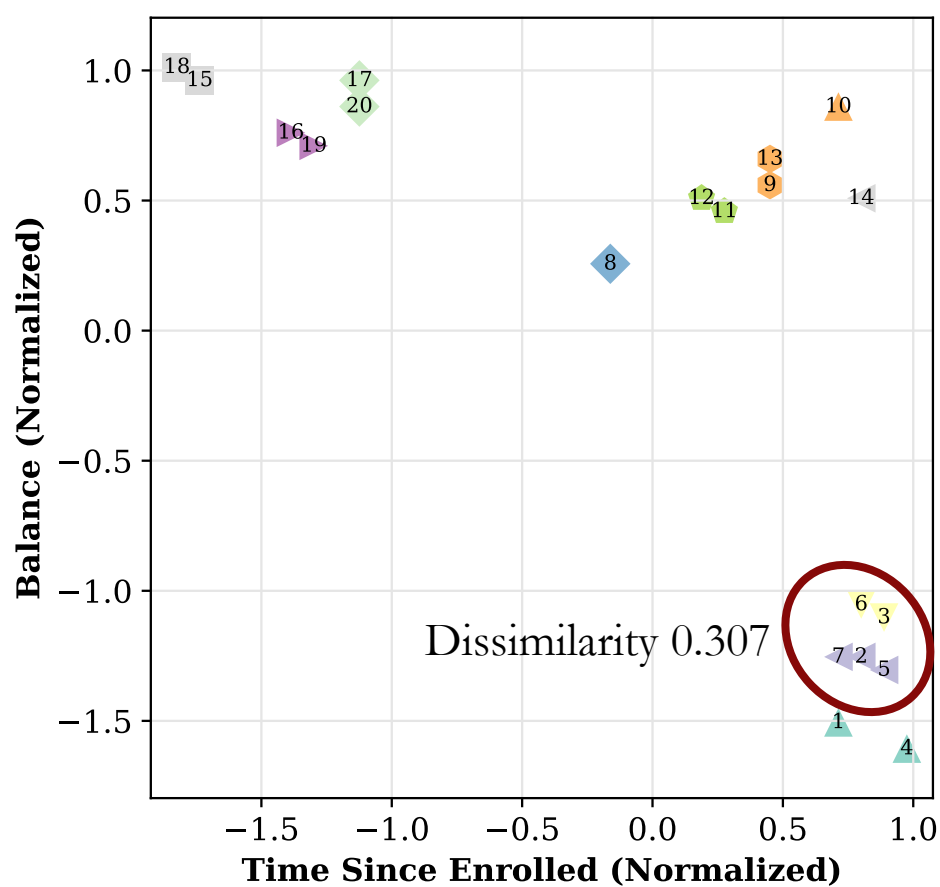
Hierarchical Clustering on Mini-dataset



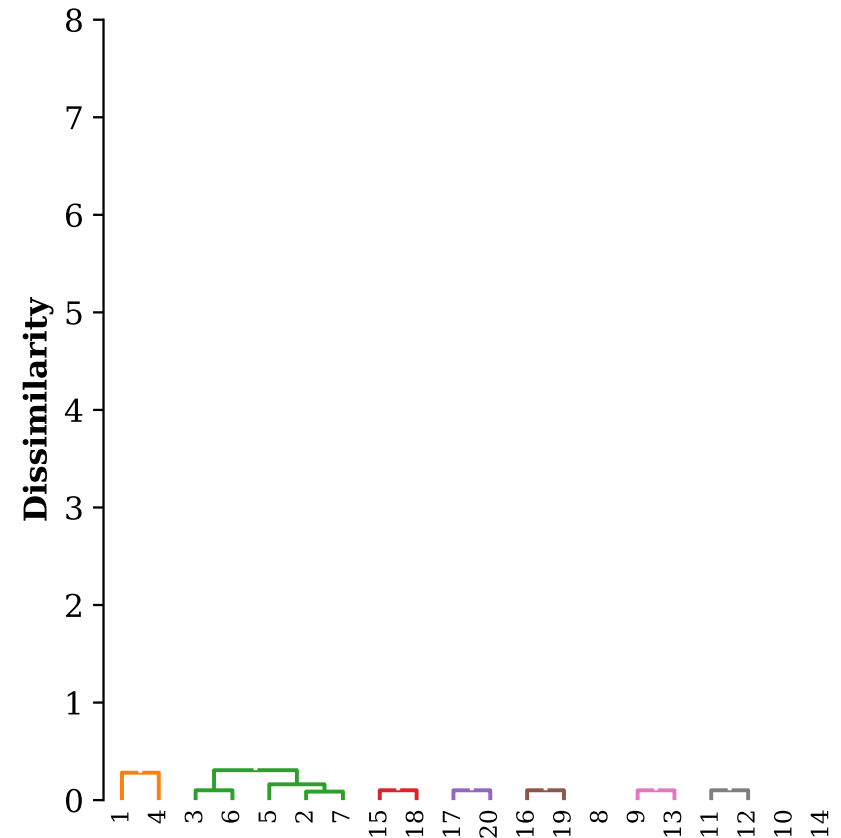
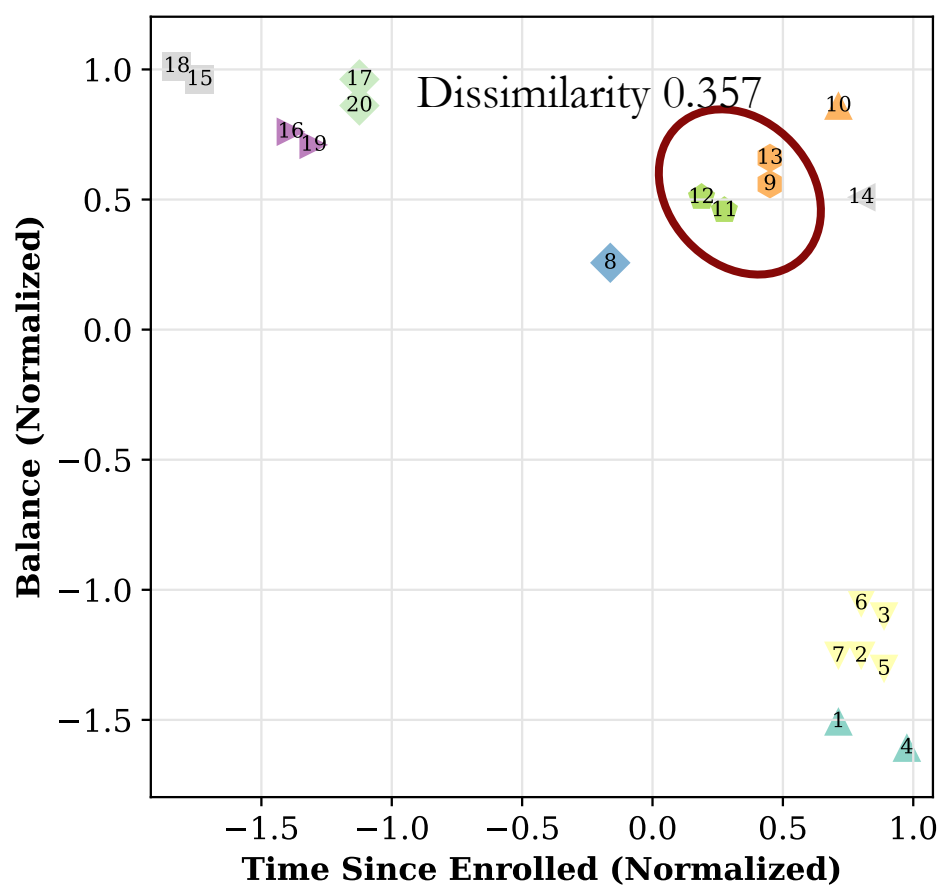
Hierarchical Clustering on Mini-dataset



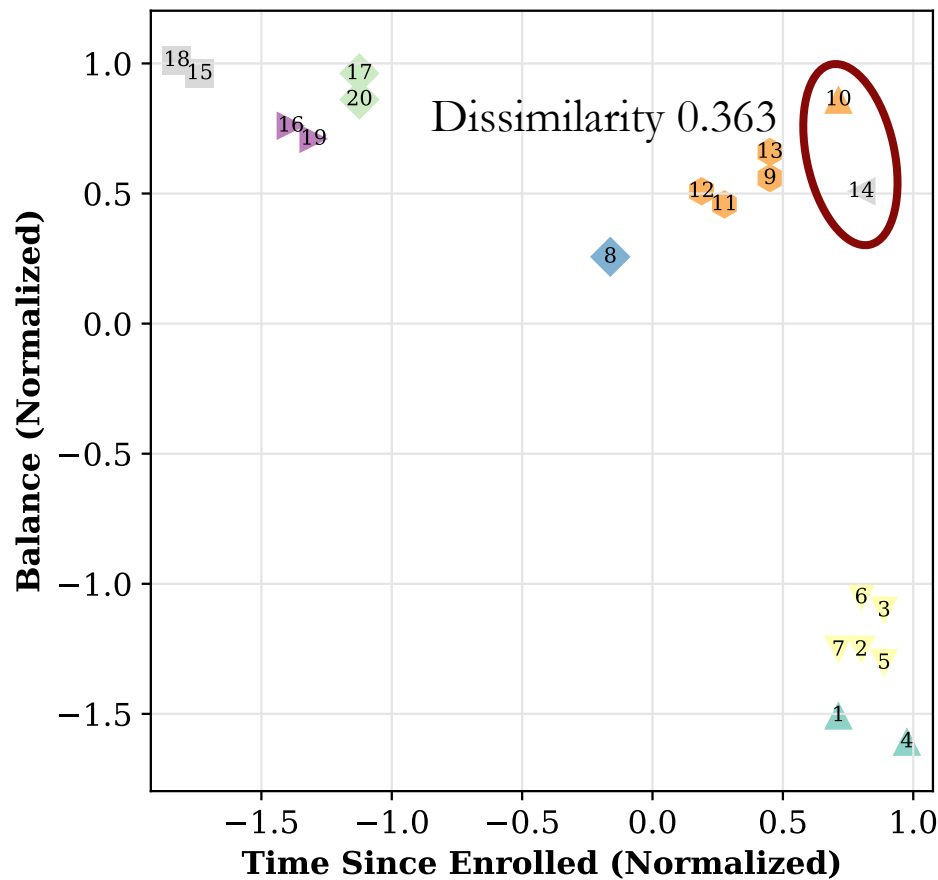
Hierarchical Clustering on Mini-dataset



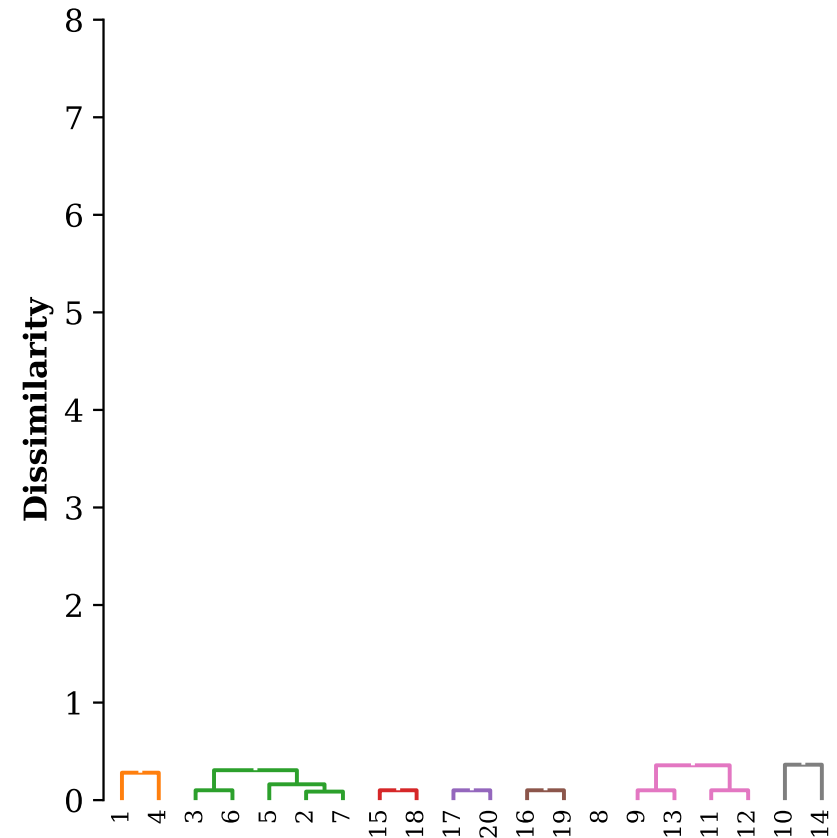
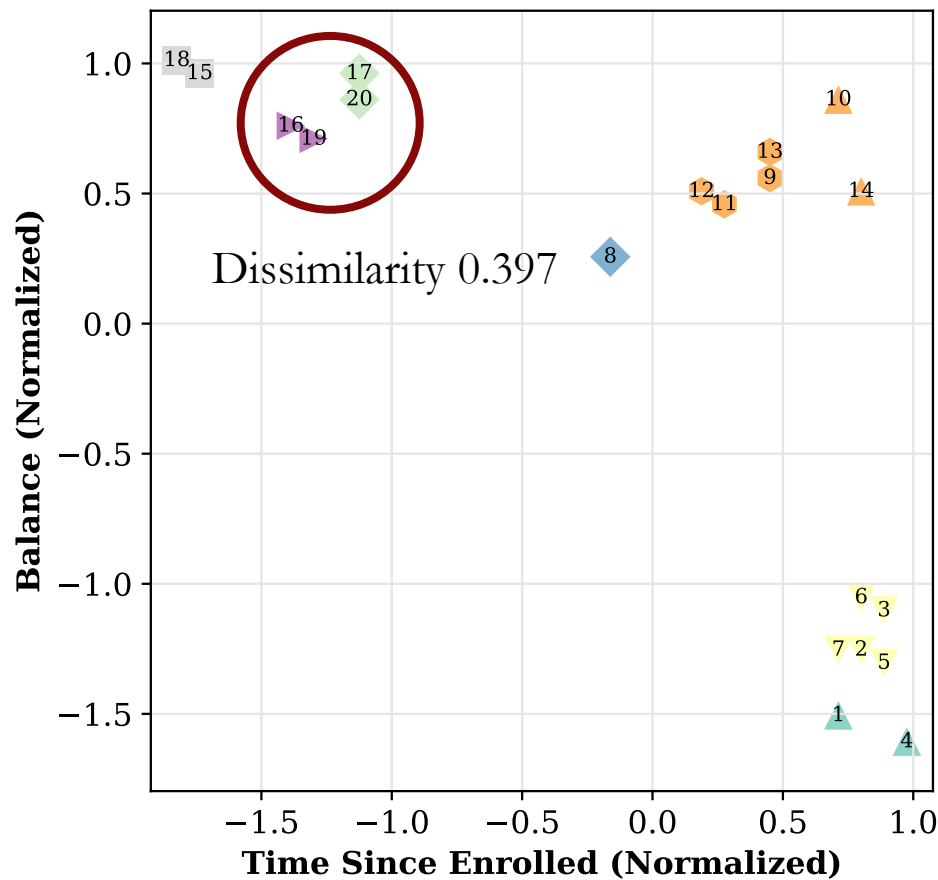
Hierarchical Clustering on Mini-dataset



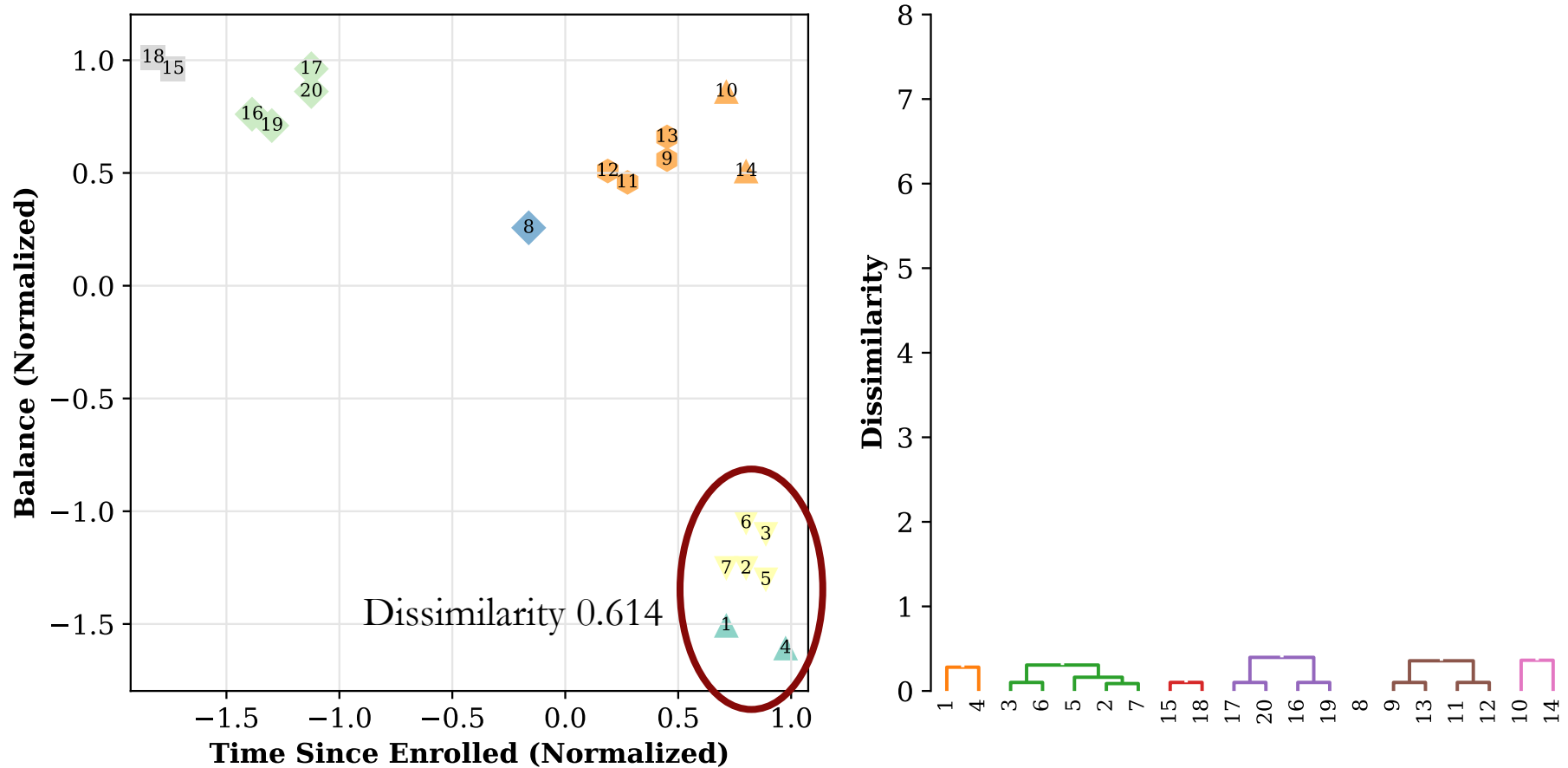
Hierarchical Clustering on Mini-dataset



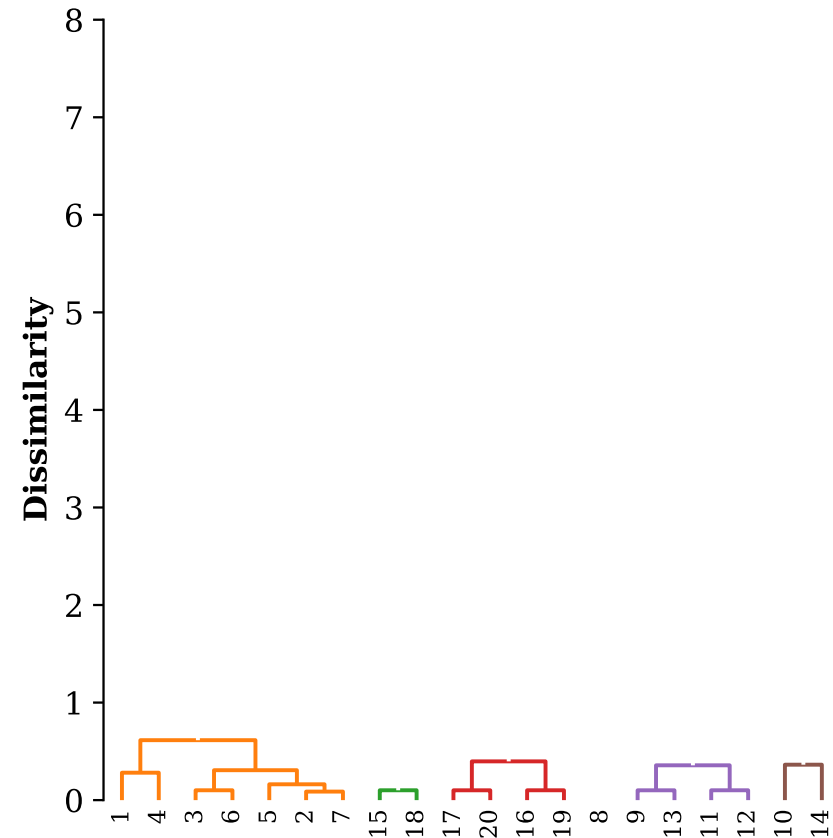
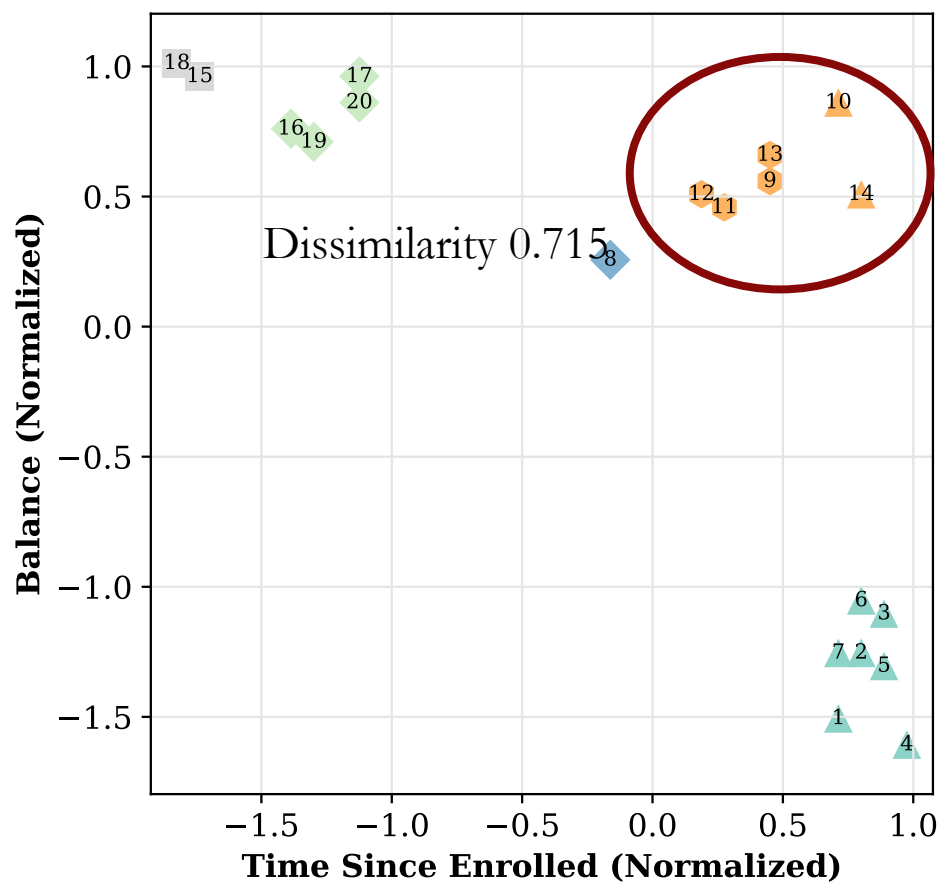
Hierarchical Clustering on Mini-dataset



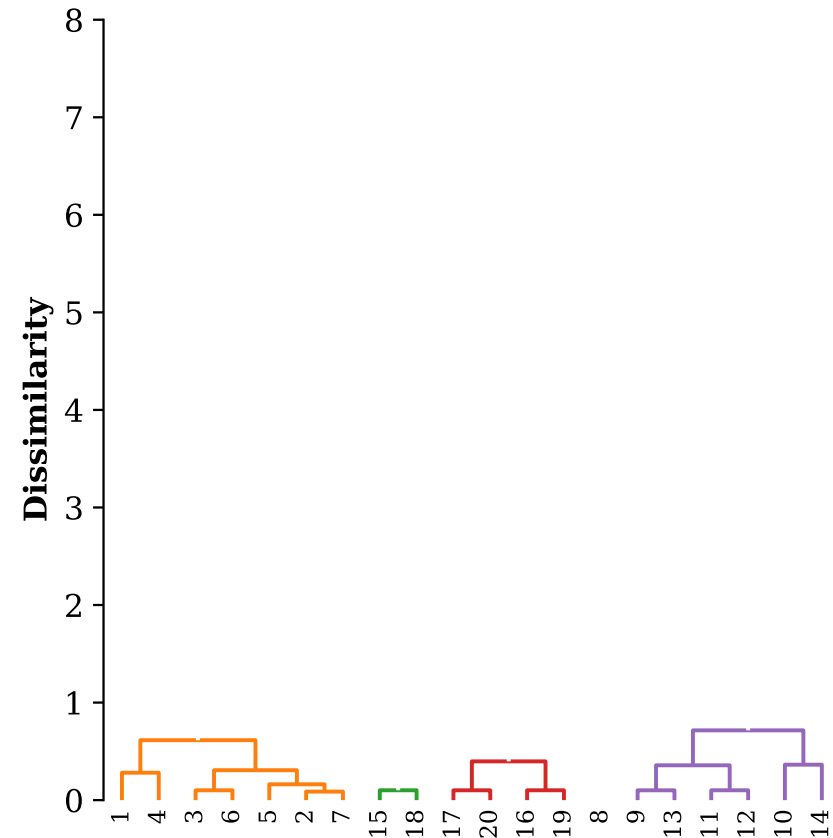
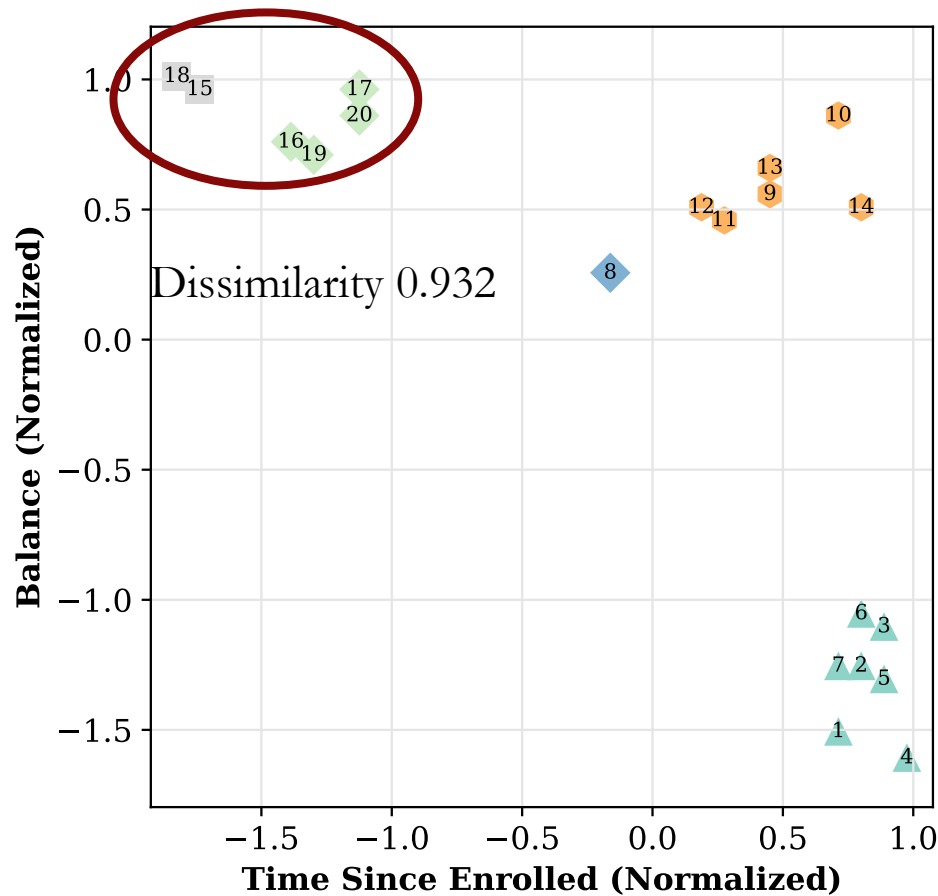
Hierarchical Clustering on Mini-dataset



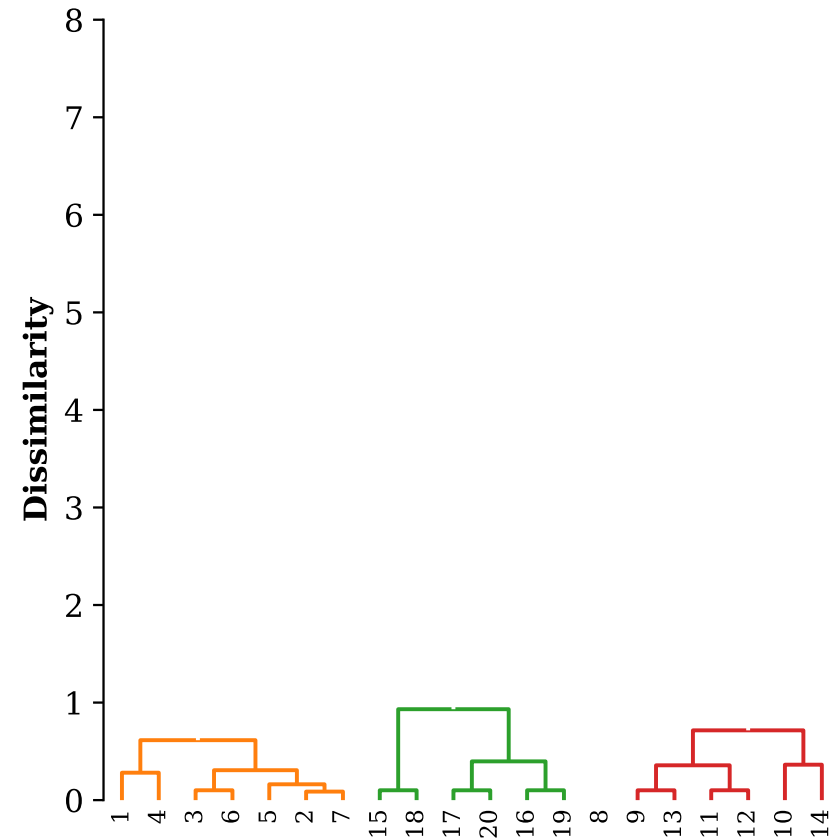
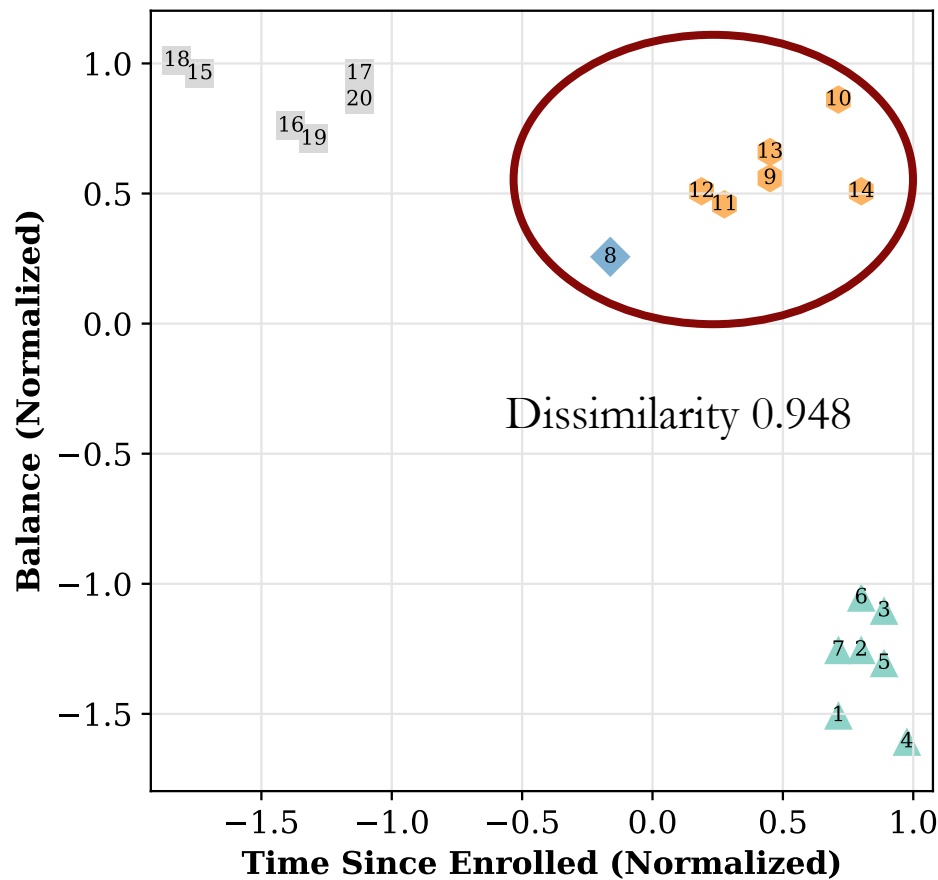
Hierarchical Clustering on Mini-dataset



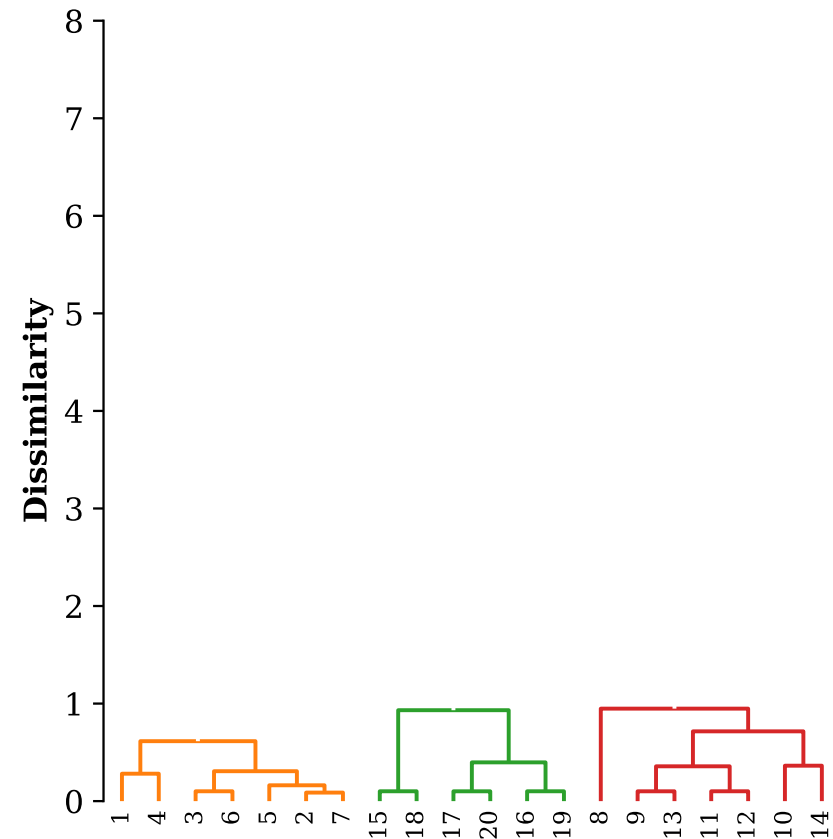
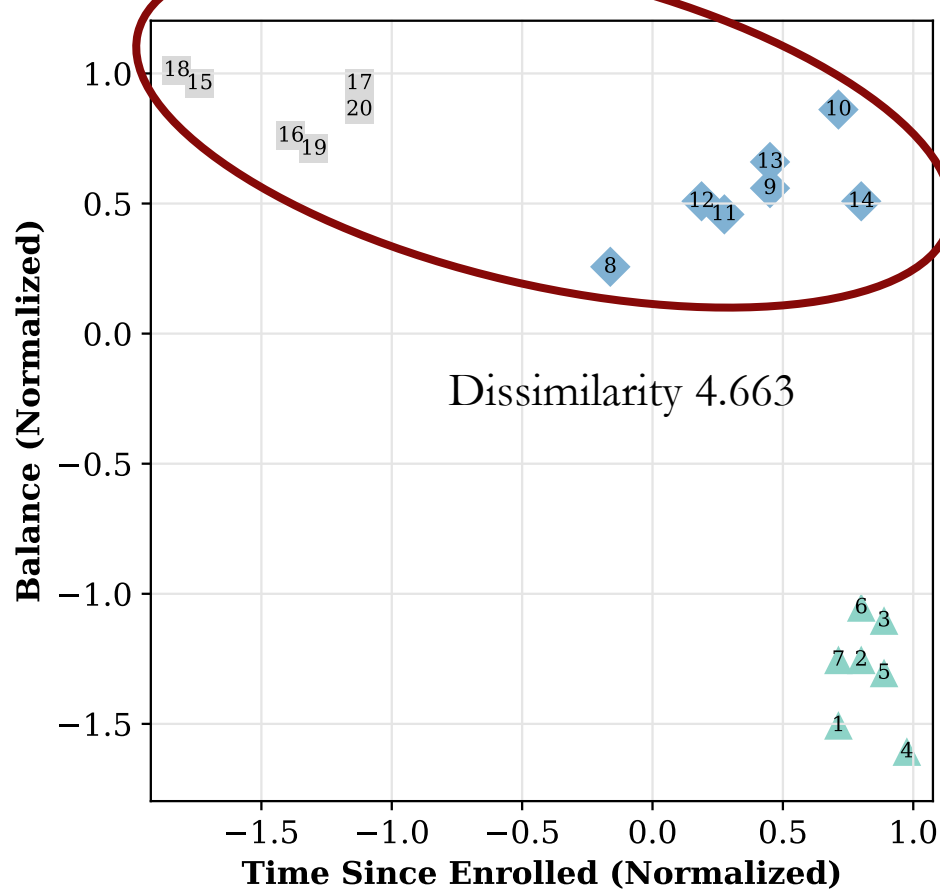
Hierarchical Clustering on Mini-dataset



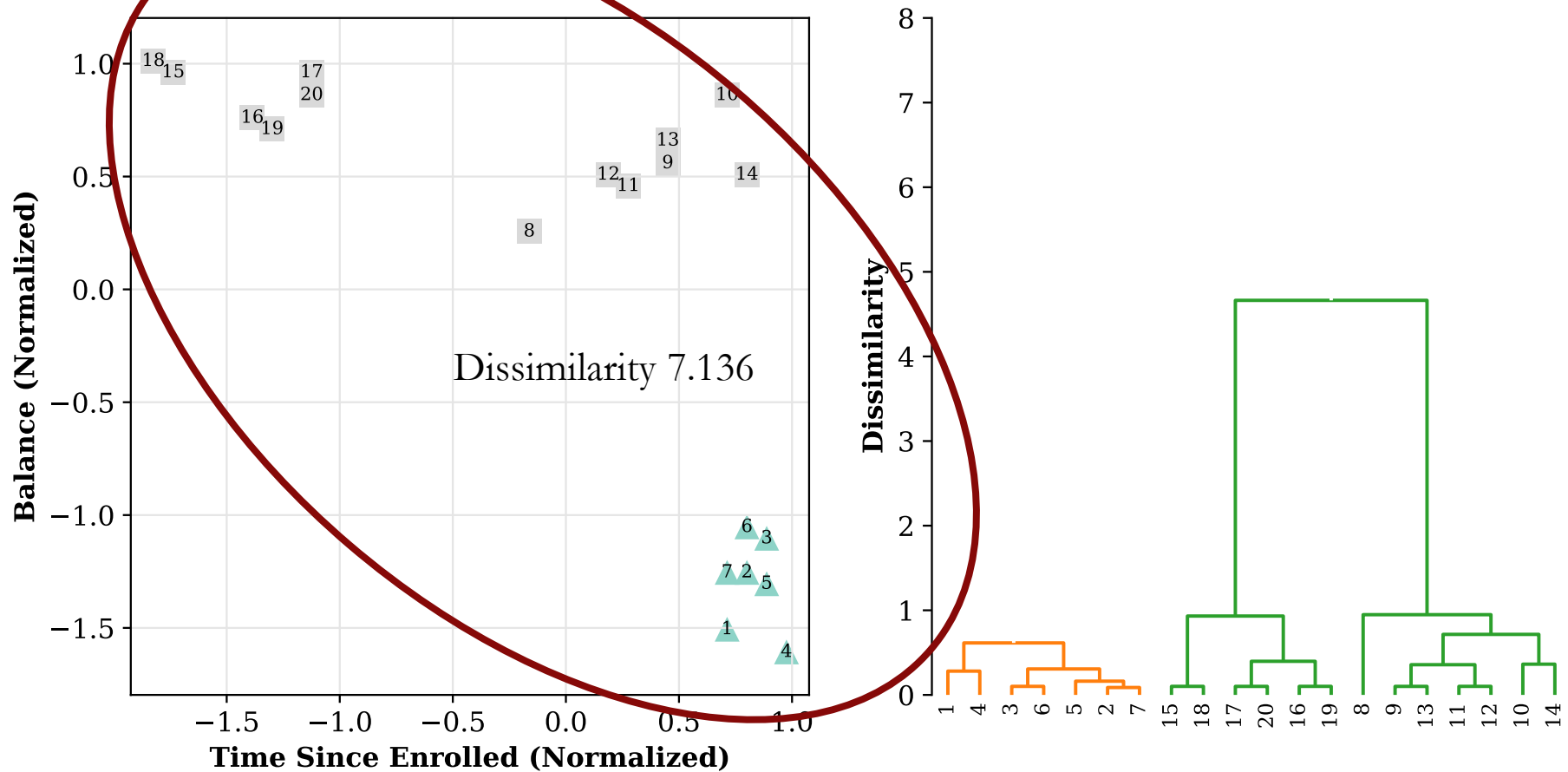
Hierarchical Clustering on Mini-dataset



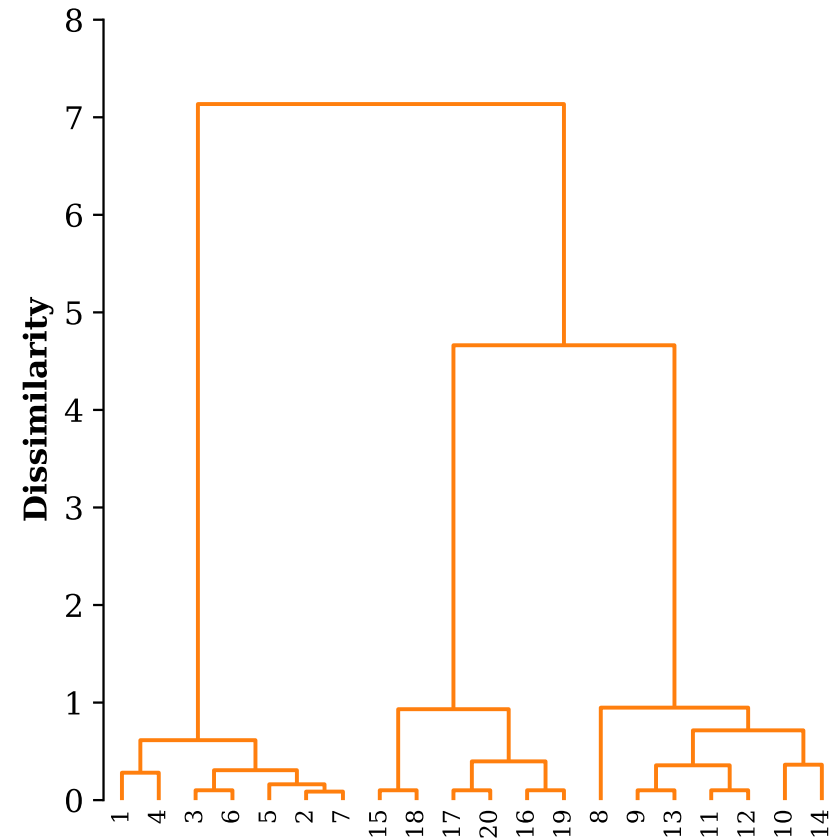
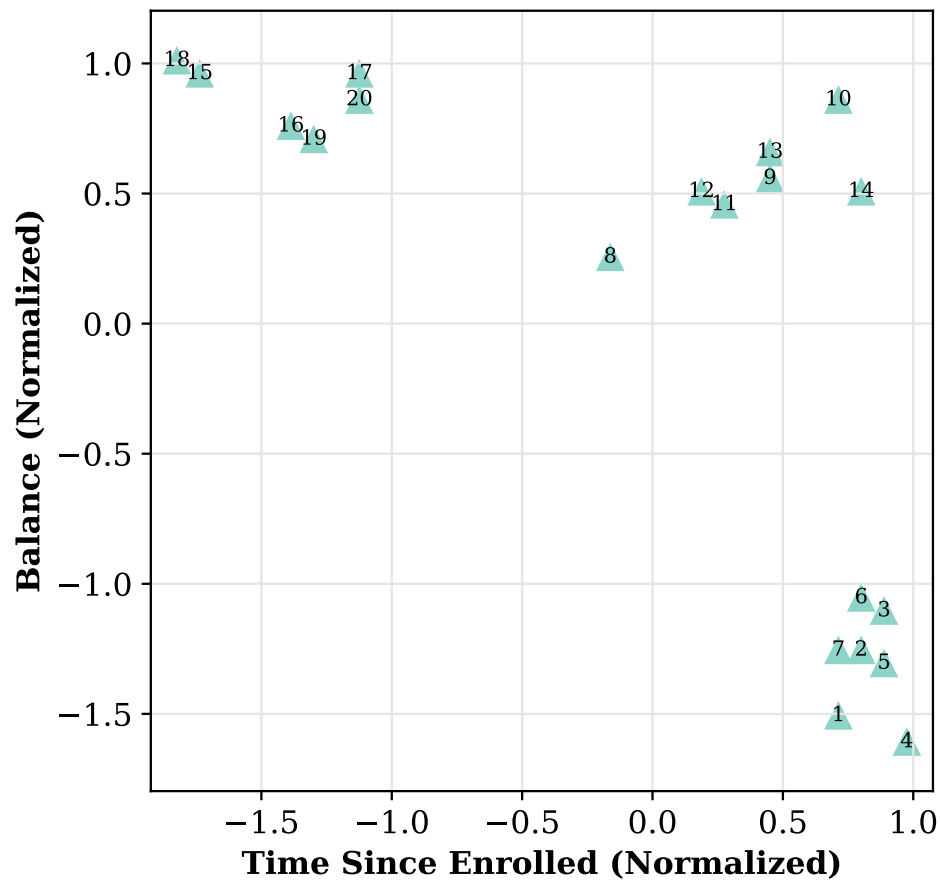
Hierarchical Clustering on Mini-dataset



Hierarchical Clustering on Mini-dataset

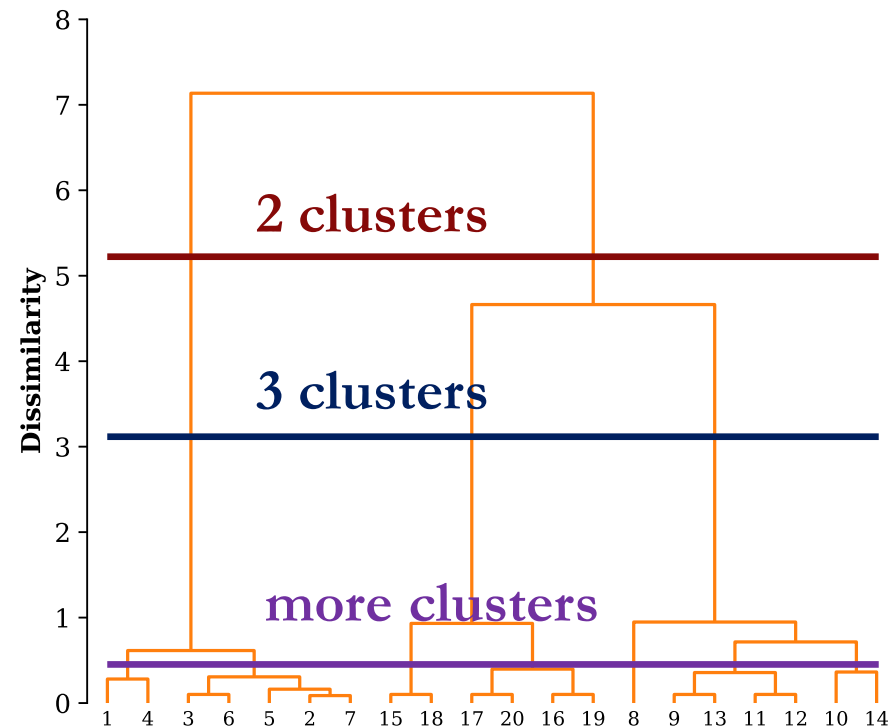


Hierarchical Clustering on Mini-dataset



Use of Dendrogram

- Full hierarchy of clusters, all the way to single observations
- Height indicates dissimilarity
 - E.g., 3 and 6 are close to each other, as are 2 and 7
- One can “reconstruct” a cluster set for each value of k
- Common mistake: the x-axis does not indicate similarity
 - Observation 4 is far from 3



Popular Hierarchical Clustering Dissimilarity Measures Between Clusters A and B , cont.

- `scipy.cluster.hierarchy.linkage`
- `method = single`

$$\min_{a \in A, b \in B} \|a - b\|$$

- `method = complete`

$$\max_{a \in A, b \in B} \|a - b\|$$

Popular Hierarchical Clustering Dissimilarity Measures Between Clusters A and B

- `method = ward`

$$\frac{|A| \cdot |B|}{|A| + |B|} \left(\frac{2}{|A| \cdot |B|} \sum_{a \in A} \sum_{b \in B} \|a - b\|^2 - \frac{1}{|A|^2} \sum_{a \in A} \sum_{z \in A} \|a - z\|^2 - \frac{1}{|B|^2} \sum_{b \in B} \sum_{z \in B} \|b - z\|^2 \right)$$

- `method = average`

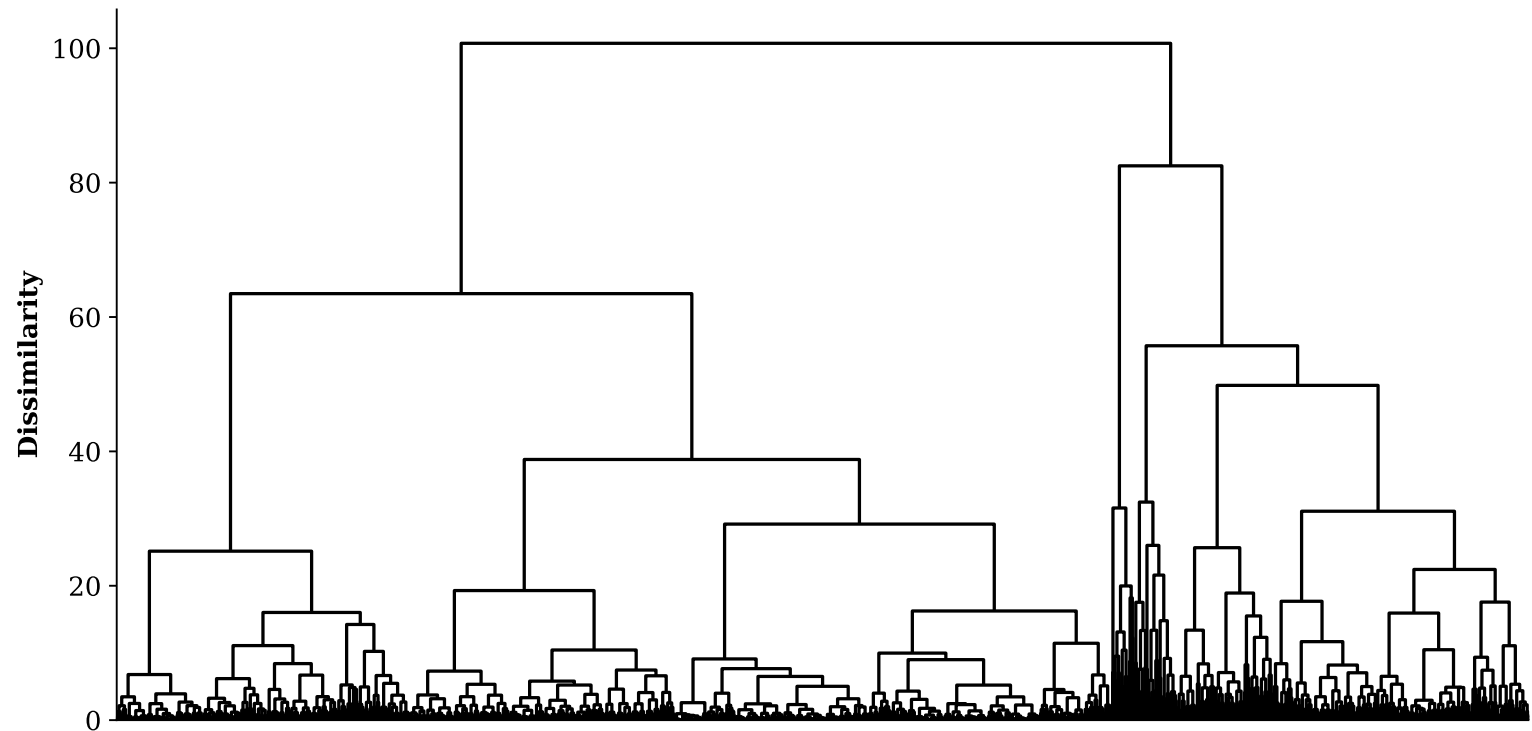
$$\frac{1}{|A| \cdot |B|} \sum_{a \in A} \sum_{b \in B} \|a - b\|$$

Hierarchical Clustering

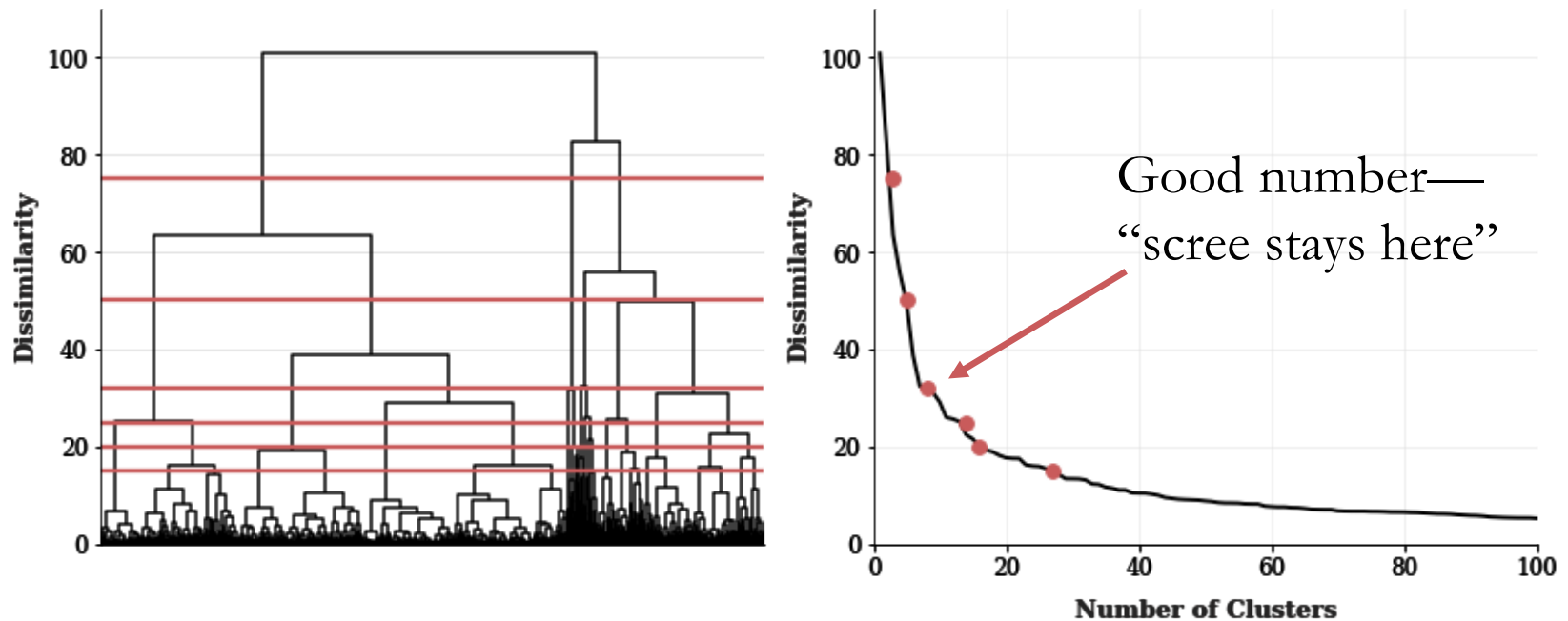
Applied to the Airline Customers Data

Hierarchical Clusters for Airline Data

- Output of the hierarchical clustering approach applied to the airline customer dataset:

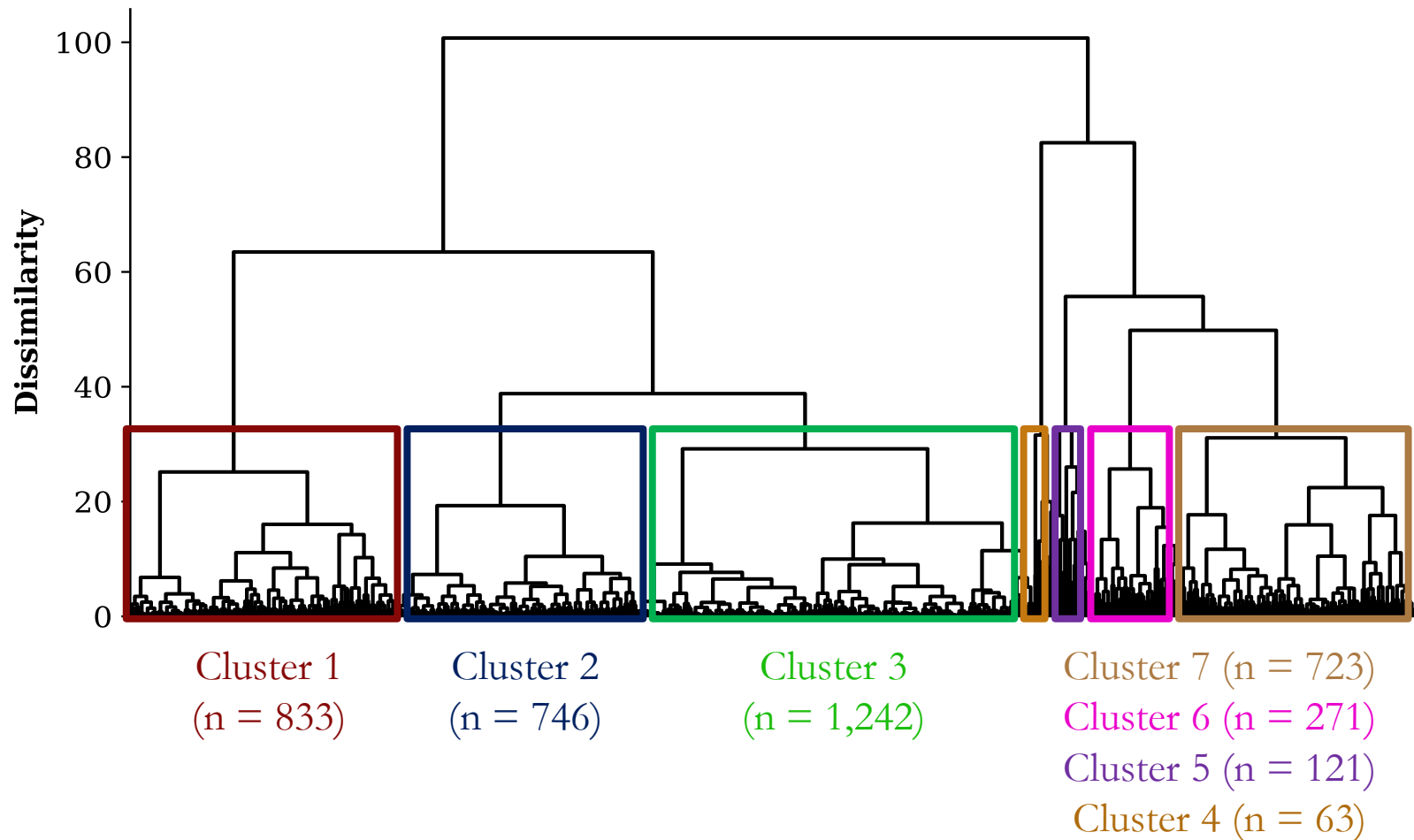


Scree Plot for Airline Clusters



- We will agglomerate any clusters that have dissimilarity 32 or smaller, yielding 7 final clusters

Dendrogram for Airline Clusters

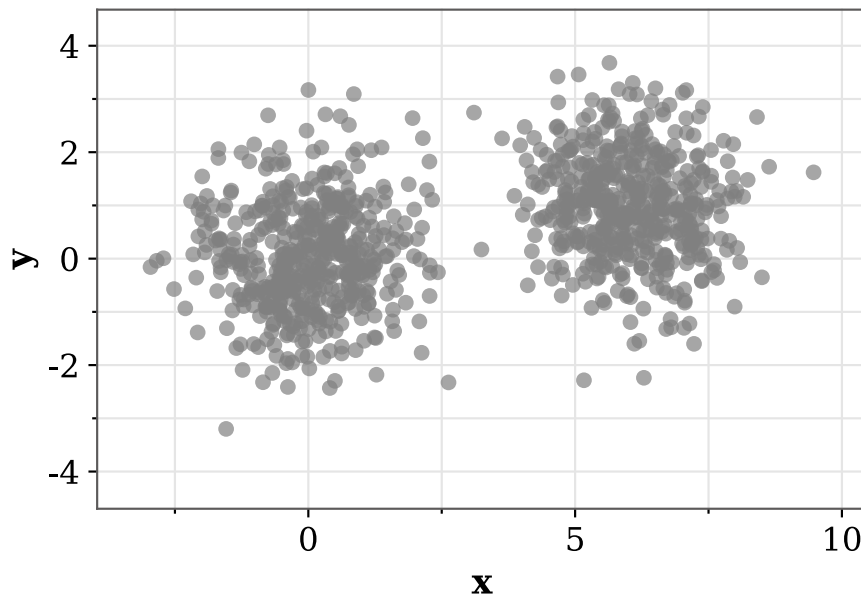


Method 3. Gaussian Mixture Models (Optional)

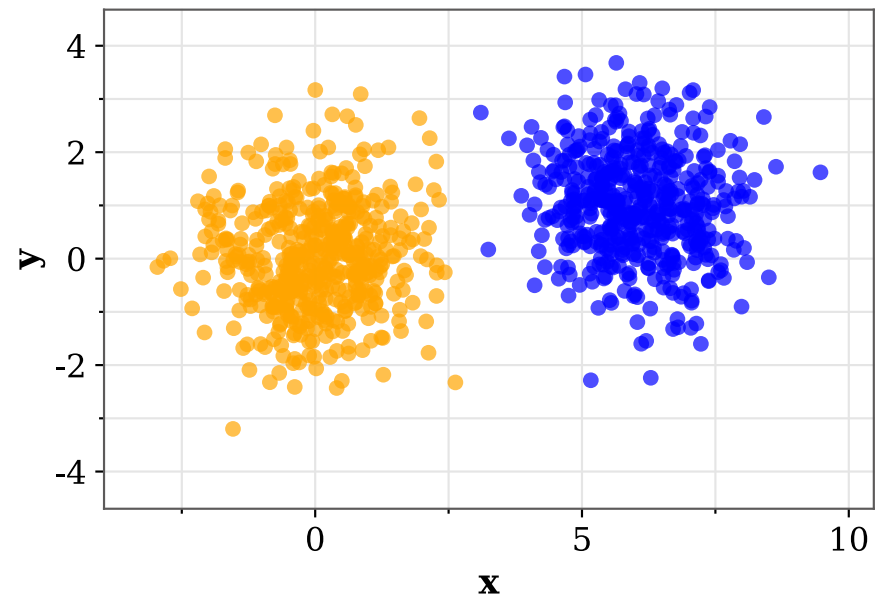
Issue with k -means and/or AHC: different types of “correlation” inside of clusters

- Here is a scatterplot that is particularly well-suited to k -means and/or AHC clustering:

Before

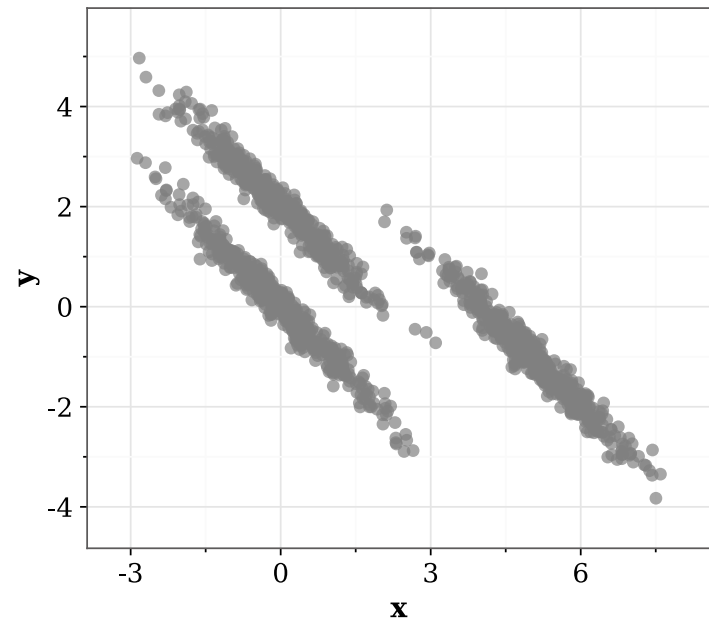


After: 2 clusters



k -means/AHC and “correlation”

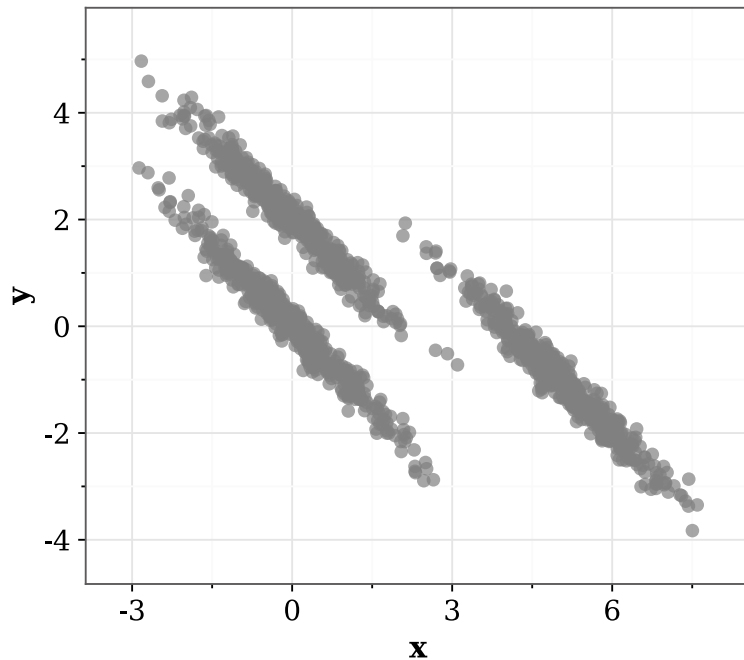
- How many clusters do you think there are?
- What shapes are the clusters?
- Is there strong or weak correlation between the variables within each cluster?



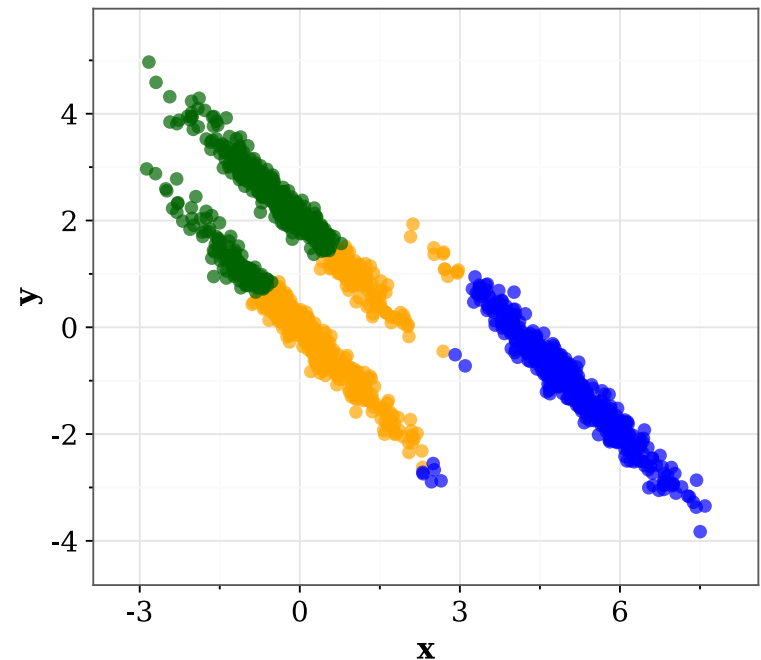
k -means/AHC and correlation, cont.

- Here are the results from clustering with $k = 3$

Before

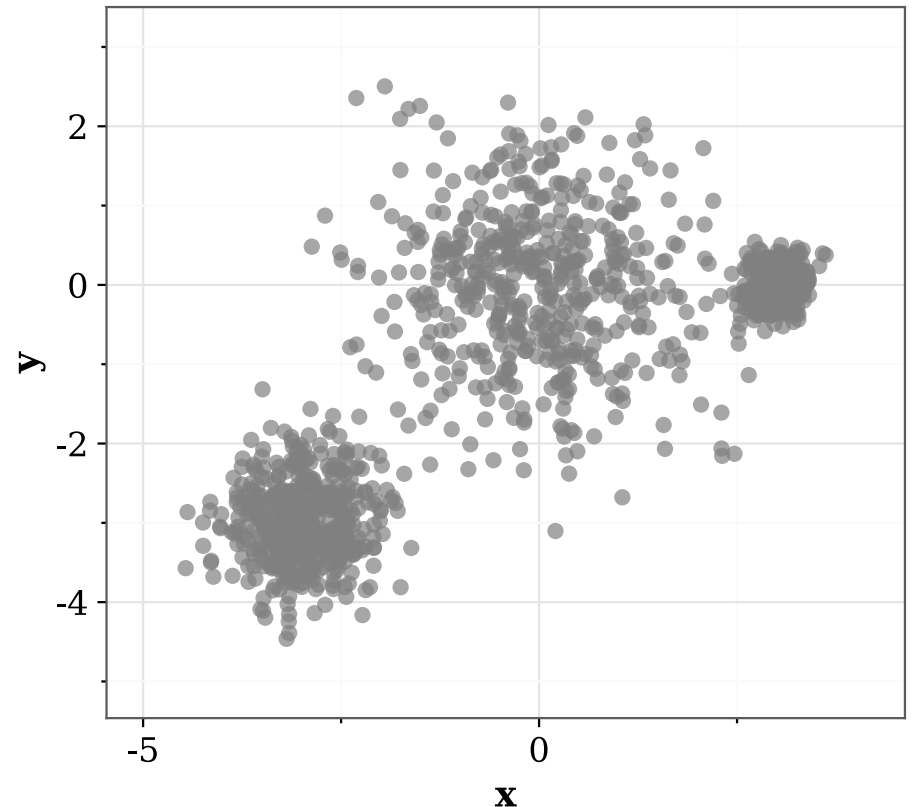


After



Another Example

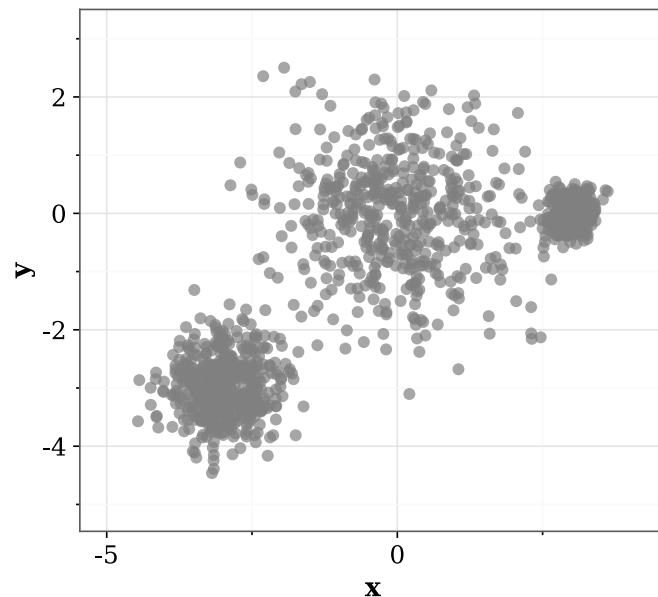
- How many clusters do you think there are?
- What shapes are the clusters?
- How do the distributions within each cluster differ?



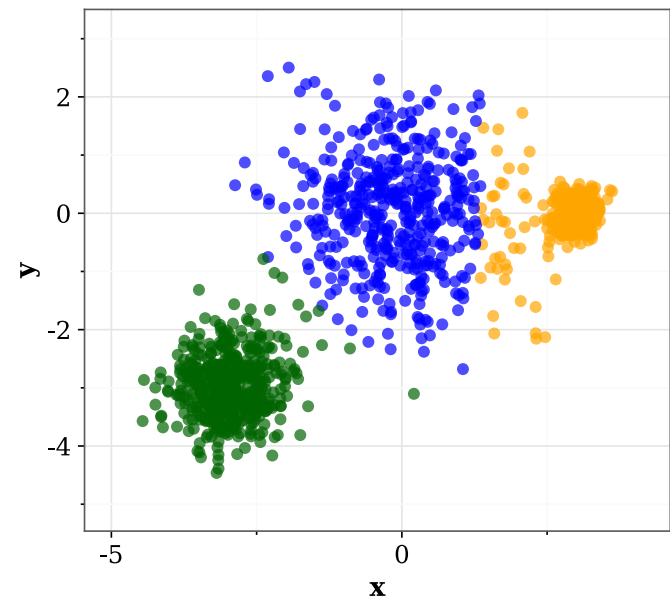
k -means/AHC and unequal variances

- Here are the results from k -means/AHC clustering

Before



After



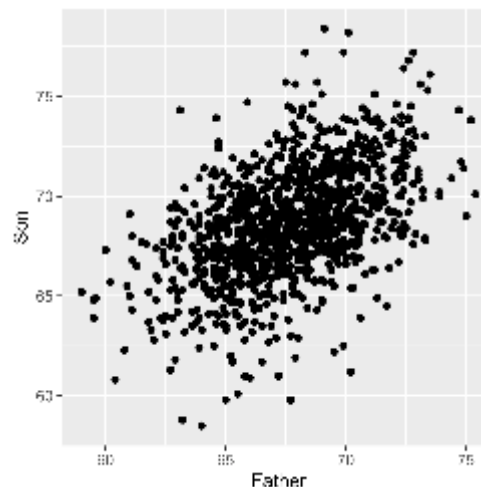
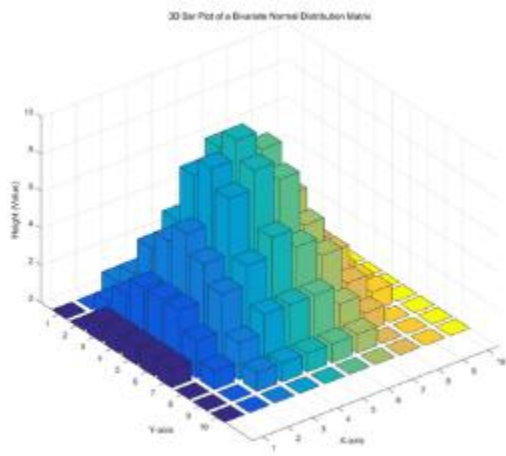
- OK for the most part, but we can see a few problems here...

Problems with k -means/AHC

- k -means/AHC do best when the clusters exhibit:
 1. Circular/spherical shaped (uncorrelated variables); and
 2. Similar variances (“dispersion patterns”) across clusters
 - Unfortunately, oftentimes clusters are:
 1. Best described by elliptical shapes (correlated variables); and/or
 2. Have different types of variances (different types of ellipses)
- In these situations, k -means/AHC will often not perform well

Gaussian Mixture Models

- **Gaussian Mixture Models (GMMs)** provide an improved method for clustering when clusters are elliptical and/or may have different variances among the clusters
- Clustering is based upon the multivariate analogue of the Normal (**Gaussian**) distribution

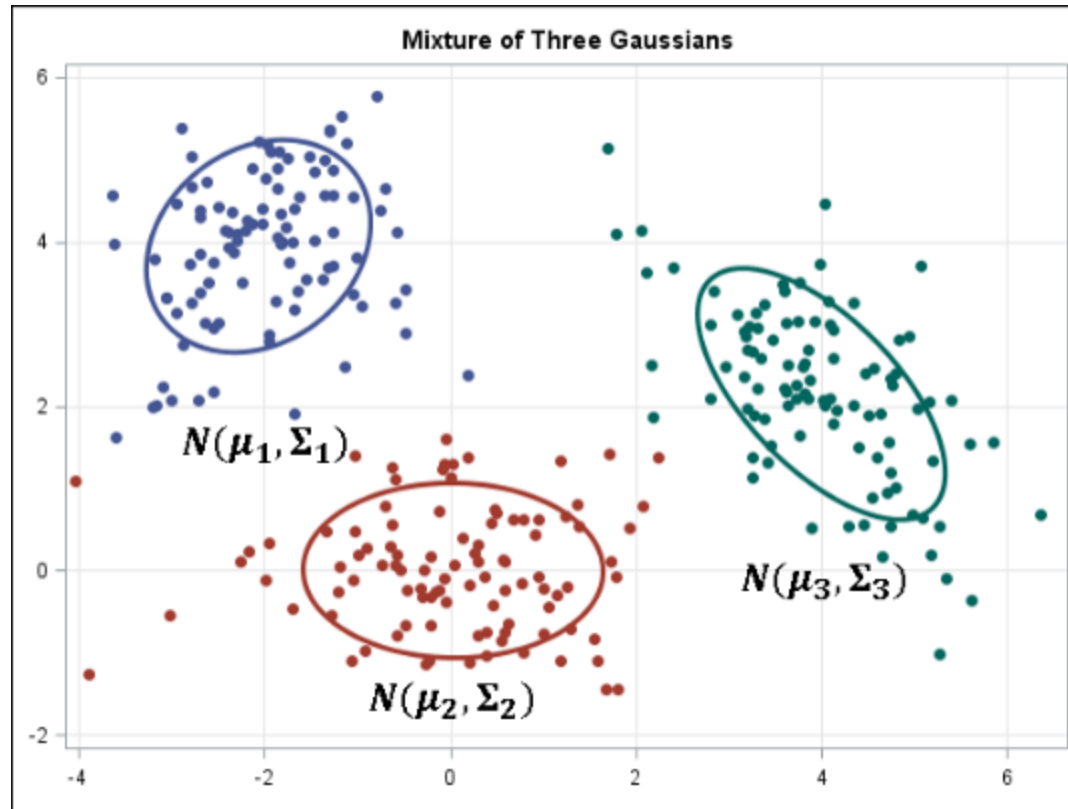


- Here the “bell” of the bell-shaped curve is shaped like a real bell

Gaussian Mixture Models

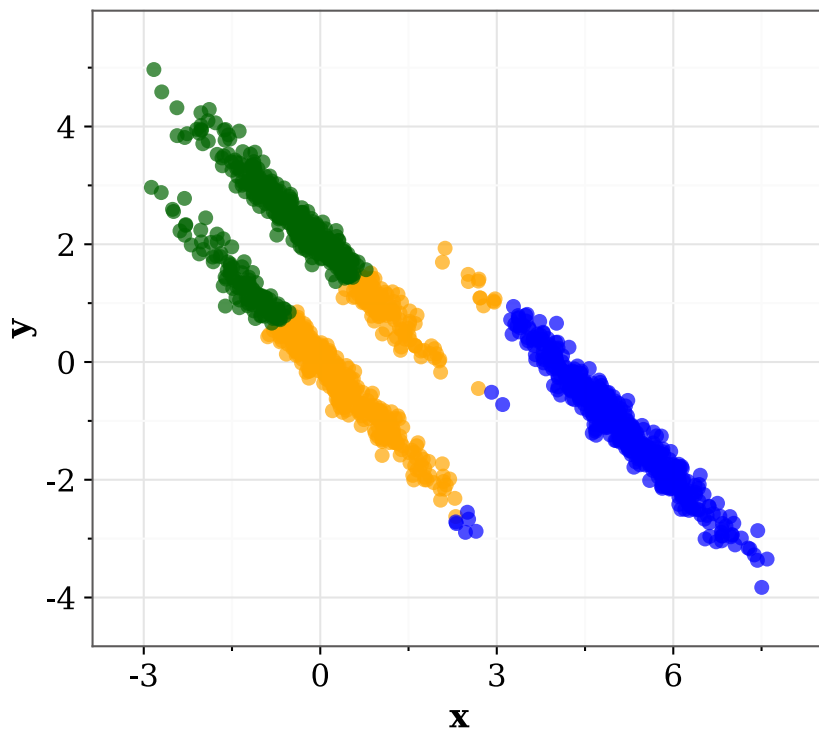
- We imagine that there are k distinct “multivariate” Normal distributions which give rise to the data (with unknown means, standard deviations, and “correlation” structures)
- But we do not know which individuals came from which distribution
- Nonetheless, we can compute estimates of the values for the means, standard deviations, and correlation structures that would fit the data best, using what is known as the Expectation-Maximization (EM) algorithm
- Once we have these estimates, we can actually calculate the **probability** that an individual observation is from a given cluster
- We then assign each individual to the cluster with the highest probability

Gaussian Mixture Model

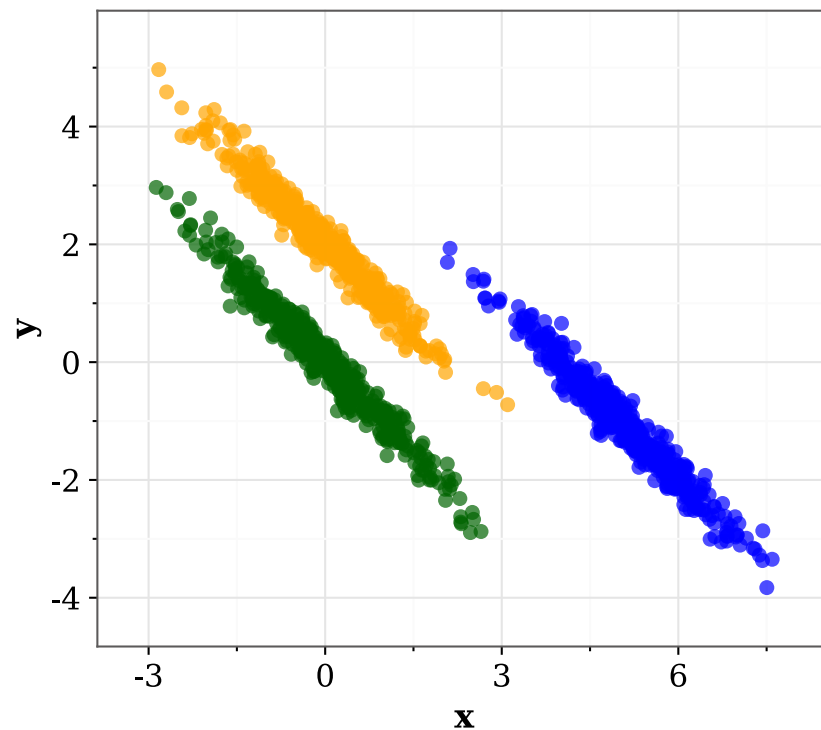


Comparing GMMs and k -means

k -Means

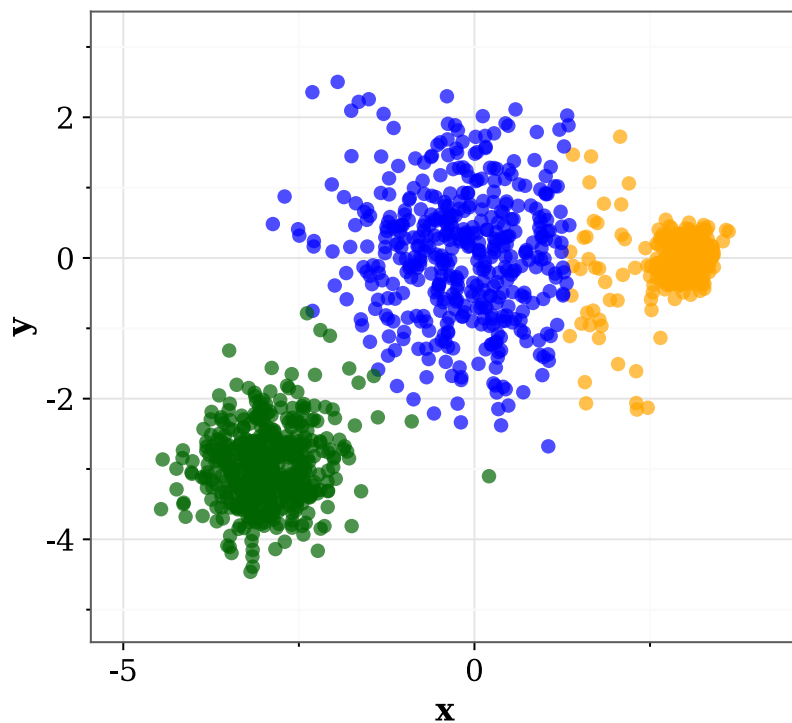


GMM

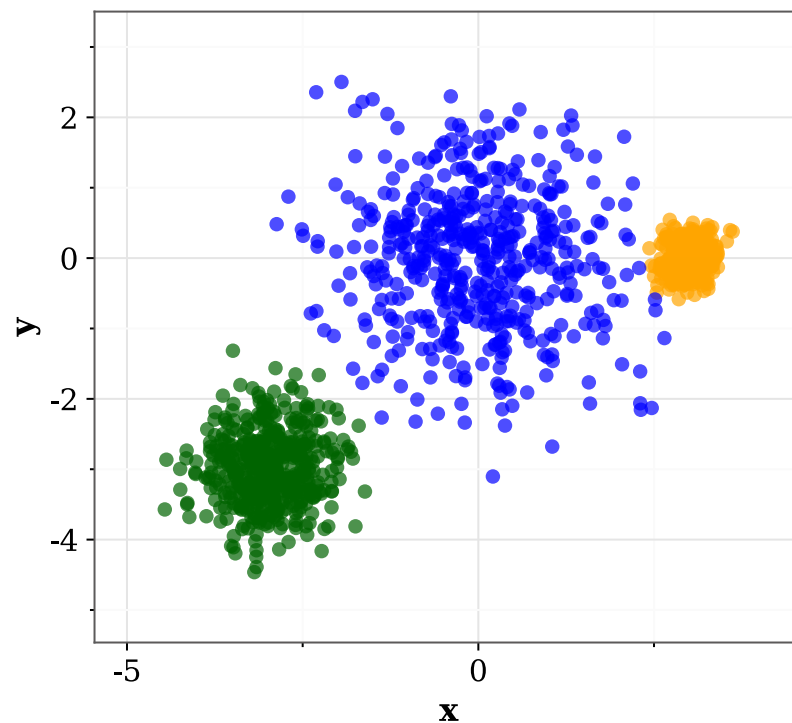


Comparing GMMs and k -means

k -Means



GMM



Summary of Clustering

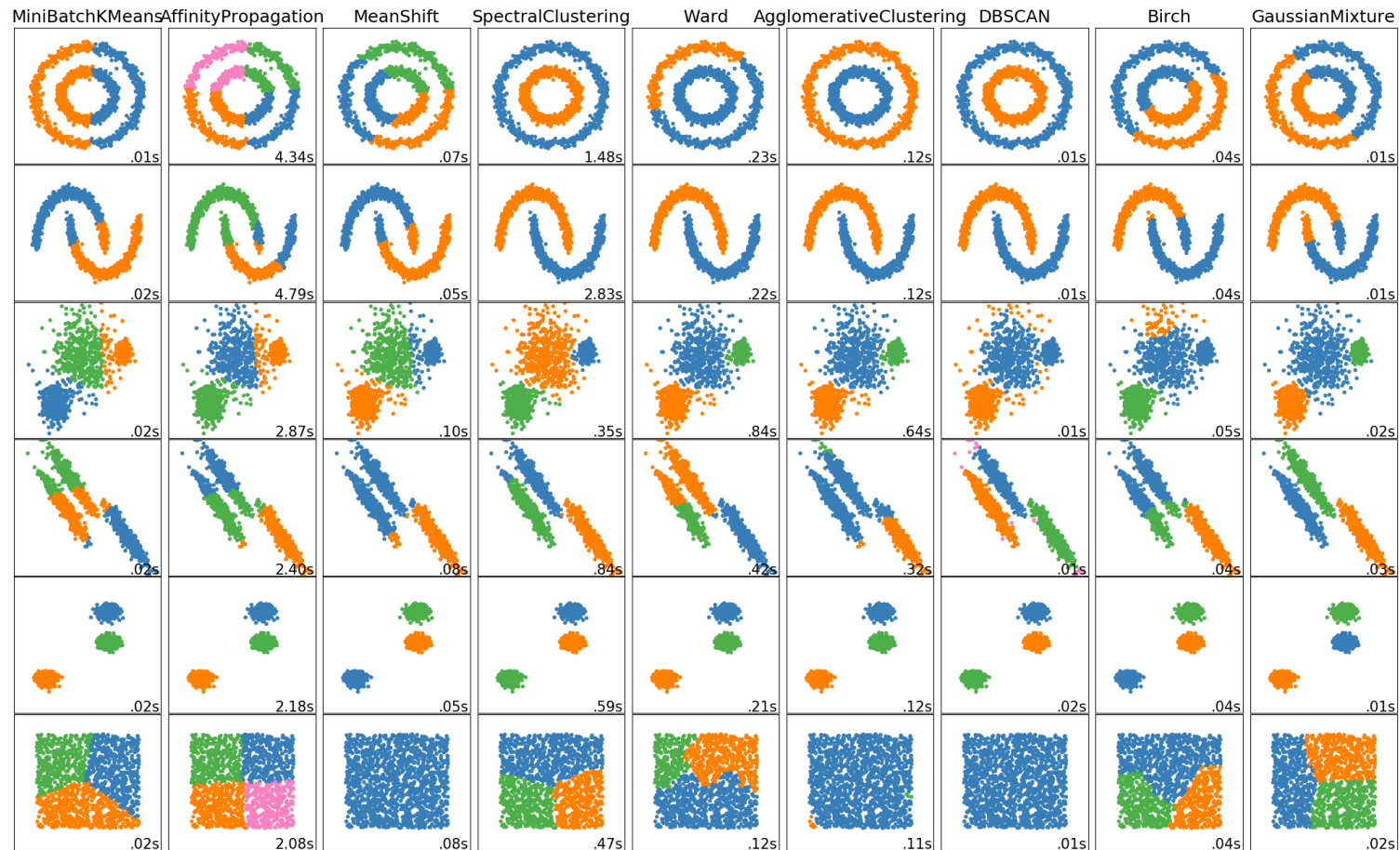
Clustering and Customer Segmentation

- Clustering for customer segmentation is now pervasive
 - Better understanding of key customer types and their prevalence
 - Design of appropriate engagement strategies, tailored to each segment
- Key steps in clustering analytics:
 - Aim for data to be compatible in scale through standardization
 - Identify relevant clusters with k -means, or hierarchical clustering, GMMs, or PAM
 - Analyze cluster centroids (or medoids) to interpret the type of customers in each cluster and determine appropriate implications for management
- Clustering cannot be validated with out-of-sample predictions
 - very important to engage decision-makers throughout model building

Comparison of Clustering Methods

- k -means, hierarchical clustering, and GMMs often yield consistent insights
- Still, the methods have different strengths/weaknesses
- k -means clustering
 - Is more computationally efficient — hence scalable to ever-larger datasets
- Hierarchical clustering
 - Flexibility through various distance and cluster dissimilarity metrics
 - Amenable to iterative decision making — at various levels of granularity
- GMMs
 - Best when clusters are thought to be “elliptical” with potentially different correlations and variances
- Many modeling issues can have significant impacts on results
 - whether to standardize data, how to quantify “distance”, how many clusters to use, etc.

Clustering can become very mathematically deep when the data itself is inherently complex



<https://www.cs.bu.edu/fac/snyder/cs4100/Medium.Clustering.Survey.pdf>